



FINC-3100
Principles of Finance
Student Textbook
Fall 2024



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Topic 1: Financial Statements

This first Topic establishes the foundation for the entire semester of Principles of Finance. We cover the three major financial statements used to measure and report the performance of a firm, and briefly look at a fourth. Throughout the semester, we will refer to the information provided in these documents, especially the three in bold font:

1. **Balance Sheet**
2. **Income Statement (also called the Statement of Operations)**
3. **Statement of Cash Flows**
4. Statement of Shareholders' Equity (we learn a part of this called the Statement of Retained Earnings)

These financial statements are used to examine the three main areas of corporate financial management:

1. Working Capital Management (cash in, cash out)
2. Financial Capital Management (the debt and equity used to finance assets)
3. Fixed Capital Management (capital budgeting decisions for investing in long-term/fixed assets)

Hopefully, the material covered in this first Topic is mostly a review of your previous course: Principles of Financial Accounting (ACCT-2090). In accounting, students learn how financial transactions are recorded and how this system of recording (double-entry bookkeeping) results in the production of financial statements. In Principles of Finance, our task is to learn how managers, analysts, lenders, and investors use these financial statements to make informed decisions about the firm, such as its present financial condition and its future financial prospects. The manager's job is to help the firm achieve and maintain financial health. The job of the analyst, lender, and/or investor is to determine which firms represent good investment opportunities because of their ability to achieve and maintain financial health.

Definition: Financial statements present historical information that allows users (i.e., managers, analysts, lenders, investors, regulators) to both determine the firm's past performance and project its future prospects. The four statements described in this Topic are closely related to one another, and taken together they provide a complete picture of the firm's operations and financial position.

For demonstration purposes, we will use financial information from real companies (e.g., a local Atchison company named **MGP Ingredients, Inc.** which is publicly traded under the symbol **MGPI**) and a fictitious company named Ravencorp. The fictitious financial statements for Ravencorp will be deliberately simple and straightforward. While these fictitious statements will lack some of the line items and complexities that appear on the financial statements of many publicly traded companies (including MGPI), they will contain the most basic and important information students need to learn. Comparing Ravencorp's statements with MGPI's statements will allow the student to first learn fundamental principles and then apply them to a real example in the market. It will also demonstrate an important principle: Finance can be both an art and a science. It is a science to the extent that formal rules must be followed and all firms must operate financially within a certain basic framework; but it is also an art because of the rich variety of terms, formats, and reporting methods that are utilized by companies and practitioners.

The financial statements for publicly traded companies (such as MGPI) are found: **1.)** in the firm's **Annual Report** (usually available on an Investor Relations page in the company's website), **2.)** in the quarterly and annual **SEC filings** that can be retrieved from the EDGAR database (<https://www.sec.gov/edgar/searchedgar/companysearch.html>), and **3.)** on one of many **business websites** (such as Yahoo Finance at www.finance.yahoo.com) which reproduce or summarize the information obtained from the public reports of companies.

The Annual Report is the most important publication that firms produce for their stockholders and the general public. It contains not only detailed accounting and financial information (including the financial statements), but also information about operations, markets, strategies, and management's analysis of both past performance and future operations. The statements we use this semester for MGPI are taken from their 2023 Annual Report.

The Balance Sheet

The **Balance Sheet** reports the monetary value (or balance) of accounts as of a specified date. It is sometimes referred to as a “snapshot” of the firm’s financial condition because most of the account balances are not cumulative over time but are balances at a specific moment in time (such as End-of-Year, End-of-Quarter, etc.). There are three types of accounts on the Balance Sheet: **Assets**, **Liabilities** (also called debt), and **Equity** (or stockholders’ equity).

<u>Account Type</u>	<u>Account Description</u>
Assets	What the firm “owns”
Liabilities (Debt)	What the firm “owes” to lenders and creditors
Stockholders’ Equity	What the owners (stockholders) have contributed to the firm

Under general rules of accounting (i.e., double-entry bookkeeping), a firm’s assets are “financed” either with capital that is owed to parties outside the firm (and listed on the Balance Sheet as a liability¹) or with capital that is provided by the firm’s stockholders (and listed on the Balance Sheet as equity²). The total value of all assets must be equal to the total value of the combined liabilities and equity that is needed to finance those assets. In other words, there must be an exact “balance” between total assets and the combined total liabilities and equity.

In its simplest form (conforming to the classic “T-account” used in double-entry bookkeeping), the Balance Sheet is presented as two equal sides. The “left” side lists all asset accounts and the “right” side lists all liabilities and equity accounts. Exhibit 1 is an example of a “Two-Sided” Balance Sheet containing some of the more common accounts.

Exhibit 1

Balance Sheet "Two Sides" or "T-Account"	
Assets	Liabilities & Equity
<u>Current Assets</u> Cash & Equivalents Accounts Receivable Inventory Other Current Assets	<u>Liabilities</u> Current Maturities of LT Debt Accounts Payable Accrued Expenses Short-Term Debt Long-Term Debt Other Non-Current Liabilities
<u>Long-Term Assets</u> Net Property, Plant & Equipment Equity Investments Goodwill/Intangible Assets Other Non-Current Assets	<u>Stockholders' Equity</u> Preferred Stock Common Stock + Paid-in-Capital Retained Earnings
Total Assets	Total Liabilities & Equity

Companies publish their Balance Sheet in the “over-and-under” style rather than the “side-by-side” style above. The Balance Sheet on the following page is taken from the Annual Report for MGP Ingredients (ticker symbol: MGPI). Notice the line item for “Total Assets” (approximately one-third down the page) has a double underline, as does “Total Liabilities and Stockholders’ Equity” (at the bottom of the page). Look for accounts on MGPI’s Balance Sheet that are among the most common accounts (listed above on Exhibit 1) and look for those that are not among them.

¹ Not all liabilities on the Balance Sheet represent capital that is borrowed from a conventional lender. For example, some liabilities are created through the normal course of purchasing materials on credit terms (which are represented as Accounts Payable), and some by incurring payment obligations that arise through normal business operations (such as accrued payroll, accrued taxes, accrued interest on debt, which are represented as Accrued Expenses).

² Generally speaking, owners provide equity capital either by the direct purchase of stock from the firm (listed on the Balance Sheet as Common Stock and Paid-in Capital), or by allowing the firm to keep some or all of the profits (listed on the Balance Sheet as Retained Earnings). Preferred Stock is discussed in more detail later in this text.

MGP INGREDIENTS, INC.
CONSOLIDATED BALANCE SHEETS
(Dollars in thousands, except share amounts and par value)

	December 31,	
	2023	2022
Current Assets		
Cash and cash equivalents	\$ 18,388	\$ 47,889
Receivables (less allowance for credit loss of \$1,475 at both December 31, 2023 and 2022)	144,286	109,267
Inventory	346,853	289,722
Prepaid expenses	3,580	2,957
Refundable income taxes	1,190	4,327
Total current assets	514,297	454,162
Property, plant, and equipment, net	262,303	235,632
Operating lease right-of-use assets, net	13,975	15,042
Investment in joint venture	5,197	5,534
Intangible assets, net	271,706	216,768
Goodwill	321,544	226,294
Other assets	3,326	4,779
Total assets	\$ 1,392,348	\$ 1,158,211
Current Liabilities		
Current maturities of long-term debt	\$ 6,400	\$ 5,600
Accounts payable	73,594	66,432
Federal and state excise taxes payable	2,251	4,627
Accrued expenses and other	31,861	28,716
Total current liabilities	114,106	105,375
Long-term debt, less current maturities	85,305	29,510
Convertible senior notes	195,544	195,225
Long-term operating lease liabilities	11,292	11,622
Contingent consideration	69,200	—
Other noncurrent liabilities	4,763	3,723
Deferred income taxes	63,071	67,112
Total liabilities	543,281	412,567
Commitments and Contingencies – Note 11		
Stockholders' Equity		
Capital stock		
Preferred, 5% non-cumulative; \$10 par value; authorized 1,000 shares; issued and outstanding 437 shares	4	4
Common stock		
No par value; authorized 40,000,000 shares; issued 23,125,166 shares at December 31, 2023 and 2022; 22,016,113 and 21,994,042 shares outstanding at December 31, 2023 and 2022, respectively	6,715	6,715
Additional paid-in capital	325,453	318,839
Retained earnings	539,883	443,061
Accumulated other comprehensive income	(397)	(304)
Treasury stock, at cost, 1,109,053 and 1,131,124 shares at December 31, 2023 and 2022, respectively	(21,166)	(21,591)
Total MGP Ingredients, Inc. stockholders equity	850,492	746,724
Noncontrolling interest	(1,425)	(1,080)
Total equity	849,067	745,644
Total liabilities and equity	\$ 1,392,348	\$ 1,158,211

See Accompanying Notes to Consolidated Financial Statements

As noted above, the monetary value of each Balance Sheet account is reported as of a certain date that is always specified on the statement (e.g., “December 31” as with MGPI). In general, the firm’s End-of-Year (EOY) Balance Sheet is the most relevant.³ Tracking the changes in Balance Sheet accounts over time (e.g., from year to year) is an important part of corporate financial analysis, as we will learn in Topic 2.

Important Note: The Balance Sheet accounts that appear in these Exhibits do not represent an exhaustive list of accounts, as you have already noted with MGPI. However, they do represent the accounts common to most firms. Students should become familiar with the name, description, and type (i.e., asset, liability, or equity) of the following accounts:

Assets

Cash
Accounts Receivable
Inventory
Net PP&E*

Liabilities

Accounts Payable
Accrued Liabilities
ST Debt
LT Debt

Equity

Preferred Stock
Common Stock and Paid-in-Capital
Retained Earnings

* Another term used for Net Property, Plant and Equipment (Net PPE) is Net Fixed Assets. A “fixed asset” is something considered tangible and permanent (like land, buildings, fixtures, etc.) and is distinguished from assets that are either “paper” (such as marketable securities and accounts receivable), tangible but non-permanent (such as inventory), and intangible (such as patents, goodwill, pre-paid expenses, etc.). While not all LT Assets are Fixed Assets, we will often use the terms “Net Fixed Assets” and “Net PP&E” interchangeably.

The two sides of the Balance Sheet can be further divided into smaller groupings based on shared characteristics (such as short-term accounts, long-term accounts, etc.). This allows a manager or analyst to review a firm’s Balance Sheet based on considerations that are more detailed than the simple “two-sides” of the T-account.

One alternative presentation of Balance Sheet accounts is to sub-divide each of the “two sides” precisely into short-term and long-term. This results in what can be referred to as the **“Four Corners”** of the Balance Sheet in Exhibit 2:

Exhibit 2

Balance Sheet	
"Four Corners"	
Assets	Liabilities & Equity
NW <u>Current Assets</u> Cash & Equivalents Accounts Receivable Inventory Other Current Assets <u>Long-Term Assets</u> Net PP&E Equity Investments Goodwill/Intangible Assets Other Non-Current Assets SW Total Assets	NE <u>Current Liabilities</u> Current Maturities of LT Debt Accounts Payable Accrued Expenses Short-Term Debt <u>LT Liabilities and SHs' Equity</u> Long-Term Debt Other Non-Current Liabilities Preferred Stock Common Stock + Paid-in-Capital Retained Earnings SE Total Liabilities & Equity

³ A firm’s end-of-year will either correspond to the end of the calendar year (i.e. 12-31-2023) or to the end of what is known as their “fiscal year” (i.e., 6-30-2023 for a company whose fiscal years ends on June 30th each year).

The “Four Corners” Balance Sheet highlights an important element of corporate financial analysis: a comparison of the “duration” of assets with the financing accounts (the corresponding liabilities & equity). While it is not an exact rule of corporate finance, some firms try to “match durations” of assets and liabilities as closely as possible. In other words, a firm may strive to have Current Assets roughly equal to Current Liabilities, and Long-Term Assets roughly equal to Long-Term Liabilities and Equity. This is because it can become problematic for a firm to finance short-term assets with long-term liabilities, or to finance long-term assets with short-term liabilities.

Notice that the lower-right corner of the “Four Corners” Balance Sheet on the previous page contains “long-term” financing accounts, and that it includes both debt and equity accounts. The decision to finance long-term assets with either debt or equity is one of strategic significance. Thus, one shortcoming of the “Four Corners” Balance Sheet is that it groups Long-Term Debt and Equity in the same sub-category. For this reason, another alternative presentation of the Balance Sheet will distinguish these accounts by subdividing LT Debt and Equity into separate areas. This can be referred to as the “**Five Accounting Areas**” of the Balance Sheet:

Exhibit 3

Balance Sheet "Five Accounting Areas"	
Assets	Liabilities & Equity
<u>Current Assets</u> Cash & Equivalents Accounts Receivable Inventory Other Current Assets	<u>Current Liabilities</u> Current Maturities of LT Debt Accounts Payable Accrued Expenses Short-Term Debt
<u>Long-Term Assets</u> Net PP&E Equity Investments Goodwill/Intangible Assets Other Non-Current Assets	<u>Long-Term Liabilities</u> Long-Term Debt Other Non-Current Liabilities
	<u>Stockholders' Equity</u> Preferred Stock Common Stock + Paid-in-Capital Retained Earnings
Total Assets	Total Liabilities & Equity

The total amount of Stockholders' Equity is also known as the “Book Value” of the firm. Note the formula:

$$\text{Stockholder's Equity} = \text{Total Assets less Total Liabilities} = \text{"Book Value"}$$

As the “Five Accounting Areas” Balance Sheet makes clear, a typical firm can consider two types of assets (short-term and long-term) and three types of financing for those assets (ST Liabilities, LT Liabilities, and Equity).

Note: Students should become familiar with each of these five areas, with the accounts inside these five areas, and with the relationships between and among the accounts within these five areas.

A third alternative presentation of the Balance Sheet is to group the accounts according to the type of **“capital”** they represent. Students should note that the term “capital” has a very broad use in finance and can be used to identify any of the three main Balance Sheet categories: assets, liabilities, and equity. Note the following examples:

- The term **“Working Capital”** usually means short-term (current) assets such as Cash, Accounts Receivable, and Inventory. Sometimes “working capital” is used synonymously with simply “cash.”
- The term **“Net Working Capital”** is used to identify the combined “net” of both short-term (current) assets and short-term (current) liabilities such as Accounts Payable, Accrued Liabilities, and ST Debt. The term “working capital” is sometimes used to mean “net working capital” although it is not as precise.
- The term **“Net Operating Working Capital”** is used to identify a combination (“net”) of both short-term (current) assets and those short-term (current) liabilities that are generated from normal operations (such as Accounts Payable and Accrued Expenses). **Note:** Short-term liabilities that are generated from outside financing activities (such as Short-Term Debt or Notes Payable) rather than from normal operations (such as Accounts Payable and Accrued Expenses) are excluded from Net Operating Working Capital calculations.
- The term **“Fixed Capital”** is used to identify long-term (fixed) assets such as PP&E, Investments, Other, etc.
- The term **“Financial Capital”** is used to identify all capital that is acquired from sources outside the firm, specifically interest-bearing debt from lenders (whether short or long-term) and equity (from the owners).

The **“Three Capitals”** Balance Sheet groups accounts according to the three major capital categories as shown in Exhibit 4 below. **Note: The 11 account names listed are those we use the most this semester.**

Exhibit 4

Balance Sheet "Three Capitals"	
Assets	Liabilities & Equity
<u>Net Operating Working Capital</u>	
Cash & Equivalents Accounts Receivable Inventory	Accounts Payable Accrued Expenses
<u>Fixed Capital</u> Net PP&E Equity Investments Goodwill/Intangible Assets Other Non-Current Assets	<u>Financial Capital</u> Short-Term Debt Long-Term Debt Preferred Stock Common Stock Retained Earnings
Total Assets	Total Liabilities & Equity

Students should note that the colors used in Exhibit 4 are meant to symbolize each type of capital. For instance:

- Net Operating Working Capital contains “liquid” accounts (i.e., accounts that are quickly converted into cash) that are generated as part of normal operations. This section is colored green to represent cash.
- Fixed Capital includes the firm’s “productive” assets (i.e., the assets needed to generate the goods and services that produce revenue). This section is colored black to represent revenue and profits (“in the black”).
- Financial Capital includes all accounts that represent the “financing” provided by outside sources (i.e., lenders who provide debt financing and stockholders who provide equity financing). This section is colored red to represent obligations that must be met in order to avoid insolvency (“in the red”) and determine value.

Students should note in Exhibit 4 that one of the Three Capitals (“green”) contains both asset and liability accounts. Students must learn the terms Net Working Capital (NWC) and Net Operating Working Capital (NOWC). Each of these terms describes the “net” of accounts from both sides of the Balance Sheet: current assets from the left side and certain current liabilities from the right side. As mentioned above, the only difference between NWC and NOWC is that externally generated liabilities (such as ST Debt or Current Maturities of LT Debt) are NOT included in the calculation of NOWC because they are NOT operating accounts. Note the two formulas:

$$\text{Net Working Capital (NWC)} = \text{Current Assets} - \text{Current Liabilities}$$

$$\text{Net Operating Working Capital (NOWC)} = \text{Current Assets} - \text{Internal Current Liabilities}$$

Important Note: The word “net” often signals the “sum” of offsetting accounts, such as revenue and expenses, or assets and liabilities. On the Balance Sheet, asset accounts are assigned a positive value and liability accounts are assigned a negative value. In the case of NOWC, the current assets (Cash, Accounts Receivable, and Inventory) are given a positive value, and the internal liabilities (Accounts Payable and Accrued Expenses) are given a negative value. The “net” is the “sum” of the positives and negatives. This means that NOWC is negative if the sum of the liabilities is greater than the sum of the assets.

Exhibit 5

Ravencorp - Year-End Balance Sheet 12/31/20--			
ASSETS		LIABILITIES	
Cash	\$5,000,000	Accounts payable	\$10,000,000
Accounts Receivable	\$5,000,000	Accrued expenses	\$8,000,000
Inventory	\$5,000,000	Short-Term Debt	\$2,000,000
Total current assets	\$15,000,000	Total current liabilities	\$20,000,000
Net PP&E	\$18,000,000	LT Debt	\$5,000,000
		Total liabilities	\$25,000,000
		STOCKHOLDERS' EQUITY	
		Preferred Stock	\$500,000
		Common stock	\$1,500,000
		Retained Earnings	\$6,000,000
		Total common equity	\$8,000,000
Total assets	\$33,000,000	Total liabilities and equity	\$33,000,000

NOWC = Current Assets less Internally Generated Current Liabilities
 NOWC = \$15,000,000 less \$18,000,000 = -\$3,000,000

Exhibit 6

Ravencorp - Year-End Balance Sheet 12/31/20--			
ASSETS		LIABILITIES	
Cash	\$5,000,000	Accounts payable	\$10,000,000
Accounts Receivable	\$10,000,000	Accrued expenses	\$10,000,000
Inventory	\$10,000,000	Short-Term Debt	\$4,000,000
Total current assets	\$25,000,000	Total current liabilities	\$24,000,000
Net PP&E	\$20,000,000	LT Debt	\$11,000,000
		Total liabilities	\$35,000,000
		STOCKHOLDERS' EQUITY	
		Preferred Stock	\$500,000
		Common stock	\$1,500,000
		Retained Earnings	\$8,000,000
		Total common equity	\$10,000,000
Total assets	\$45,000,000	Total liabilities and equity	\$45,000,000

NOWC = Current Assets less Internally Generated Current Liabilities
 NOWC = \$25,000,000 less \$20,000,000 = \$5,000,000

Note in Exhibit 5 that NOWC is negative (-\$3,000,000) because the sum of current assets (Cash + A/R + Inventory) equals \$15,000,000 and the sum of current liabilities generated internally from normal operations (A/P + A/E) equals \$18,000,000.⁴ A negative NOWC is rarely the case. It is much more common for a firm to have a positive NOWC, which means the sum of current assets exceeds the sum of current liabilities generated internally from normal operations. Note in Exhibit 6 that NOWC is positive (\$5,000,000) because the sum of current assets (\$25,000,000) is greater than the sum of current liabilities generated internally from normal operations (\$20,000,000). This means that

⁴ The precise NOWC formula in some textbooks is “Total Current Assets minus Total Non-Interest Bearing Current Liabilities.” Here we prefer to refer to NOWC as Total Current Assets minus Total Internally Generated Current Liabilities. Remember that ST Debt is a current liability but is NOT included in the NOWC calculation because ST Debt is interest-bearing and is external to the operations of the firm (and, as a result, is included in Financial Capital).

the firm requires additional financing for its Net Operating Working Capital, which must come from Financial Capital (the “red” area).⁵ The NOWC accounts will be relevant in the weeks ahead, particularly with respect to our study of short-term financial forecasting (Topic 3) and cash management (Topic 4).

Important Note: Students must become familiar with the terms (and the accounts included in) Working Capital (WC), Net Working Capital (NWC), and Net Operating Working Capital (NOWC). Students should know both the similarities and differences of each grouping, and that the word “Net” indicates that the grouping includes both assets and liabilities.

Students should note that the amount of Financial Capital needed by a firm (from lenders and owners) is equal to the combined amount of Net Operating Working Capital and Fixed Capital the firm employs. In other words:

$$\text{Financial Capital} = \text{NOWC} + \text{Fixed Capital}$$

Exhibit 7

Ravencorp - Year-End Balance Sheet 12/31/20--			
ASSETS		LIABILITIES	
Cash	\$5,000,000	Accounts payable	\$10,000,000
Accounts Receivable	\$10,000,000	Accrued expenses	\$10,000,000
Inventory	\$10,000,000	Short-Term Debt	\$4,000,000
Total current assets	\$25,000,000	Total current liabilities	\$24,000,000
Net PP&E	\$20,000,000	LT Debt	\$11,000,000
		Total liabilities	\$35,000,000
		STOCKHOLDERS' EQUITY	
		Preferred Stock	\$500,000
		Common stock	\$1,500,000
		Retained Earnings	\$8,000,000
		Total common equity	\$10,000,000
Total assets	<u>\$45,000,000</u>	Total liabilities and equity	<u>\$45,000,000</u>
NOWC = Current Assets less Internally Generated Current Liabilities NOWC = \$25,000,000 less \$20,000,000 = \$5,000,000 Fixed Capital = Net Fixed Assets Fixed Capital = \$20,000,000 Financing Capital = NOWC + Fixed Capital Financing Capital = NOWC + Fixed Capital = \$5,000,000 + \$20,000,000 = \$25,000,000 Financing Capital Accounts = ST Debt, LT Debt, Pref Stock, Comm Stock, Retained Earnings Financing Capital = \$4,000,000 + \$11,000,000 + \$500,000 + \$1,500,000 + \$8,000,000 = \$25,000,000			

⁵ As noted above, this additional financing will be provided by a Financial Capital account (the “red” area). As previously noted, many firms will seek to match the maturities of assets and liabilities. In such a case, it will likely use ST Debt (a Financial Capital account) to finance NOWC because of the short-term nature of NOWC assets.

Note in Exhibit 7 that Ravencorp's fictitious Balance Sheet is colored according to the Three Capitals. As we have learned, the total for Financial Capital (\$25,000,000) is equal to the Net Operating Working Capital (\$5,000,000) plus Fixed Capital (\$20,000,000).

In addition, as we saw previously in Exhibit 6, the Net Operating Working Capital (\$5,000,000) is equal to the total sum of current assets (\$25,000,000) less current liabilities generated from normal operations (\$20,000,000). Also, students must remember that ST Debt (a current liability) is not considered a liability that is generated internally from normal operations and is therefore not included in the NOWC calculation (it is part of Financial Capital). However, students should remember that ST Debt is included in the calculation for Net Working Capital (NWC), which does not distinguish between current liabilities that are generated by operations and those generated by financing activities.

Some Accounting Rules that affect the Balance Sheet accounts

Before ending our discussion of the Balance Sheet, it is important to recall some important rules that students should have learned in their accounting courses:

- a. All assets on the Balance Sheet are “booked” at their original cost. If a company owns an asset that was acquired at no cost (e.g., land is donated by a city seeking to attract business), its book value will be \$0 and it will contribute no monetary value to the firm's Balance Sheet.
- b. Changes in the Inventory account are subject to specific rules, such as First-In-First-Out (FIFO), Last-In-First-Out (LIFO), Weighted Average Cost, etc. The choice of rules will affect the reported Inventory balance.
- c. Most tangible (fixed) assets are subject to annual depreciation charges that reduce the book value of the asset according to a schedule that is normally based on the productive life of the asset.⁶ The depreciation charge is reflected as an Expense on the Income Statement, **but it does not involve an expenditure of cash.**
- d. Some companies report both the historic value of their Fixed Assets and the Accumulated Depreciation that has been recorded for these assets. Other companies report only the net amount of fixed assets (e.g., Net Fixed Assets or Net PP&E), which is the “net” of their historic values less the accumulated depreciation.
- e. Certain other non-current assets (e.g., Startup Costs, Goodwill, R & D, Patents, etc.) are subject to amortization according to a standard schedule. Amortization works the same as depreciation, the only difference is the nature of the asset (i.e., fixed assets are depreciated and intangible assets are amortized).
- f. If a portion of Long-Term Debt (known as the “principal” balance) is due within one year of the date of the financial statements, it is recorded under Current Liabilities on the Balance Sheet as Current Maturities of Long-Term Debt (CMLTD). For example, assume a company has issued \$50,000,000 of Long-Term Bonds, but \$5,000,000 mature in six months. The Balance Sheet would report \$45,000,000 of Bonds Payable (LT Debt) and \$5,000,000 of CMLTD under Current Liabilities.
- g. As will be discussed later, Retained Earnings is a cumulative account on the Balance Sheet. In other words, all Net Income that has not been paid to the shareholders in the form of a Dividend will accumulate on the Balance Sheet as Retained Earnings. For example, if the firm begins the year with \$50,000,000 in Retained Earnings, reports \$10,000,000 in Net Income during the year, and pays a \$5,000,000 Dividend, the Retained Earnings will increase from \$50,000,000 to \$55,000,000 at the end of the year. However, this does not necessarily mean that the firm retains \$5,000,000 in cash. It is possible for a firm to show an increase in Retained Earnings and a decrease in Cash from beginning to end of year. On the other hand, if the firm reports a loss (negative net income), the Retained Earnings account will decrease. Certain transactions involving Treasury Stock may also affect the Retained Earnings balance (see bottom of pg. 18 below).

⁶ Land is an example of a Fixed Asset that is not subject to a depreciation charge. As a result, land is always reflected on a firm's Balance Sheet at its original value.

MGP INGREDIENTS, INC.
CONSOLIDATED BALANCE SHEETS
(Dollars in thousands, except share amounts and par value)

	December 31,	
	2023	2022
Current Assets		
Cash and cash equivalents	\$ 18,388	\$ 47,889
Receivables (less allowance for credit loss of \$1,475 at both December 31, 2023 and 2022)	144,286	109,267
Inventory	346,853	289,722
Prepaid expenses	3,580	2,957
Refundable income taxes	1,190	4,327
Total current assets	514,297	454,162
Property, plant, and equipment, net	262,303	235,632
Operating lease right-of-use assets, net	13,975	15,042
Investment in joint venture	5,197	5,534
Intangible assets, net	271,706	216,768
Goodwill	321,544	226,294
Other assets	3,326	4,779
Total assets	\$ 1,392,348	\$ 1,158,211
Current Liabilities		
Current maturities of long-term debt	\$ 6,400	\$ 5,600
Accounts payable	73,594	66,432
Federal and state excise taxes payable	2,251	4,627
Accrued expenses and other	31,861	28,716
Total current liabilities	114,106	105,375
Long-term debt, less current maturities	85,305	29,510
Convertible senior notes	195,544	195,225
Long-term operating lease liabilities	11,292	11,622
Contingent consideration	69,200	—
Other noncurrent liabilities	4,763	3,723
Deferred income taxes	63,071	67,112
Total liabilities	543,281	412,567
Commitments and Contingencies – Note 11		
Stockholders' Equity		
Capital stock		
Preferred, 5% non-cumulative; \$10 par value; authorized 1,000 shares; issued and outstanding 437 shares	4	4
Common stock		
No par value; authorized 40,000,000 shares; issued 23,125,166 shares at December 31, 2023 and 2022; 22,016,113 and 21,994,042 shares outstanding at December 31, 2023 and 2022, respectively	6,715	6,715
Additional paid-in capital	325,453	318,839
Retained earnings	539,883	443,061
Accumulated other comprehensive income	(397)	(304)
Treasury stock, at cost, 1,109,053 and 1,131,124 shares at December 31, 2023 and 2022, respectively	(21,166)	(21,591)
Total MGP Ingredients, Inc. stockholders equity	850,492	746,724
Noncontrolling interest	(1,425)	(1,080)
Total equity	849,067	745,644
Total liabilities and equity	\$ 1,392,348	\$ 1,158,211

See Accompanying Notes to Consolidated Financial Statements

Practice-Exercise: MGPI Balance Sheet and the Three Capitals

Instructions: Convert MGPI's Balance Sheet on the opposite page to the Three Capitals Balance Sheet below.

First, find the 11 main accounts listed on the top of pg. 4 and put them in the appropriate box (with their values).

Then add those accounts on MGPI's Balance Sheet that are in addition to the 11 main accounts

Note: If applicable, move Current Maturities of Long-Term Debt to the Long-Term Debt total in Financial Capital.

Does your Total NOWC + Total Fixed Capital = Total Financial Capital?

MGPI Balance Sheet	
"Three Capitals"	
Assets	Liabilities/Equity
<u>Net Operating Working Capital</u> Total Net Operating Working Capital = _____	
<u>Fixed Capital</u> Total Fixed Capital = _____	<u>Financial Capital</u> Total Financial Capital = _____

The Income Statement (may also be called “The Statement of Operations”)

The **Income Statement** (also called the **Statement of Operations**) reports all of the financial activity, as measured in revenue and expenses, of the firm over a period of time.⁷ Activity that represents revenue is given a positive value, while activity that represents expenditures (or “expense”) is given a negative value (although it is not always listed as a negative number on the statement). The “net” of revenue (positive) and expenses (negative) reflects a level of profitability (or loss) that is labeled as either “income” or “earnings” depending on the firm.

The two most important lines on the Income Statement are the **“top line”** (representing the total revenue produced by the business activity of the firm) and the **“bottom line”** (representing the total profits produced by the firm after all expenses have been assessed). Multiple names can be used for the “top line” such as “Gross Sales” or “Gross Revenue” or simply “Sales” or “Revenue” depending on the firm and perhaps the industry.

In between the “top line” and the “bottom line” are various levels of expenses incurred by the firm in the operation of its business. Some of the common categories of expenses are listed below in Exhibit 8.

Exhibit 8

Income Statement	
Gross Revenue ("Top Line")	
less: Cost of Goods Sold or Cost of Sales	
less: Selling, General & Administration (S,G & A)	
less: Depreciation & Amortization (D&A)	
less: Interest Expense	
less: Taxes	
Net Income ("Bottom Line")	

Students should note that many publicly traded firms no longer list Depreciation and Amortization (D&A) as a stand-alone expense on the Income Statement. Depending on the type of business (e.g., manufacturing, retail, service), D&A may be included under Cost of Goods Sold (sometimes called Cost of Sales), or it may be included under Selling, General & Administration (SG&A), which is sometimes listed simply as Operating Expenses. Manufacturing firms may consider D&A to be part of their Cost of Goods Sold, while retail or service firms consider D&A to be part of the overall Operating Expenses. In either case, the basic Income Statement of these firms might look like Exhibit 9, which differs from Exhibit 8 only by the absence of Depreciation as a stand-alone line item.⁸ Unless otherwise noted, we will assume Depreciation & Amortization expenses are included in SG&A.

Exhibit 9

Income Statement	
Gross Revenue ("Top Line")	
less: Cost of Goods Sold or Cost of Sales	
less: Selling, General & Administration (S,G & A)	
less: Interest Expense	
less: Taxes	
Net Income ("Bottom Line")	

⁷ The Income Statement will reflect whether it is a report of Monthly, Quarterly, or Annual activity.

⁸ In cases where D&A is not included as a stand-alone line item on the Income Statement, the actual D&A expense for the period can be determined by looking at the Statement of Cash Flows.

Using an approach that is slightly more sophisticated than the top-line/bottom-line approach, the Income Statement can be used to determine various levels of profitability (also called “earnings”) that the firm achieved during the particular accounting period. Each level of profits (“earnings”) represents the firm’s performance in a different area of its business operations and may reflect the overall effectiveness of management. A typical Income Statement might list five “levels” of earnings that descend below the top line:

- Level 1: Gross Profit
- Level 2: Earnings Before Interest, Taxes, Depreciation and Amortization (EBITDA)
- Level 3: Earnings Before Interest and Taxes (EBIT)
- Level 4: Earnings Before Taxes (EBT)
- Level 5: Net Earnings (Bottom Line)

Beginning with the top line and moving downward, each level of earnings is normally progressively lower because of specific expenses that offset revenue and lower net earnings (or net profits). This is the case until the bottom line, which reflects earnings after ALL expenses have been charged.

The typical Income Statement of a manufacturing firm might look like Exhibit 10-A below in which each expense line-item represents a distinct area of costs (such as production, business operations, assets, debt, etc.) The corresponding “cost” category is listed on Exhibit 10-B.

Exhibit 10-A	Exhibit 10-B
Income Statement Gross Revenue ("Top Line") less: "Cost of Goods Sold" or "Cost of Sales" Gross Profit less: Selling, General & Administration (S,G & A) EBITDA less: Depreciation & Amortization (D&A) EBIT less: Interest Expense EBT less: Taxes Net Income ("Bottom Line")	Income Statement Gross Revenue ("Top Line") less: Cost of Production Gross Profit less: Cost of Business Operations EBITDA less: Cost of Assets EBIT less: Cost of Debt Capital EBT less: Cost of Government Net Income ("Bottom Line")

As we learned above, because most firms no longer include D&A as a specific line item on the Income Statement or Statement of Operations, the typical firm reflected in Exhibit 10 above would be modified in Exhibit 11 below.

Exhibit 11-A	Exhibit 11-B
Income Statement Gross Revenue ("Top Line") less: "Cost of Goods Sold" or "Cost of Sales" Gross Profit less: Selling, General & Administration (S,G & A) EBIT or "Net Income from Operations" less: Interest Expense EBT less: Taxes Net Income ("Bottom Line")	Income Statement Gross Revenue ("Top Line") less: Cost of Production Gross Profit less: Cost of Business Operations EBIT or "Net Income from Operations" less: Cost of Debt Capital EBT less: Cost of Government Net Income ("Bottom Line")

MGP INGREDIENTS, INC.
CONSOLIDATED STATEMENTS OF INCOME
(Dollars in thousands, except share and per share amounts)

	Year Ended December 31,		
	2023	2022	2021
Sales	\$ 836,523	\$ 782,358	\$ 626,720
Cost of sales	531,811	529,052	427,755
Gross profit	304,712	253,306	198,965
Advertising and promotion expenses	38,213	29,714	16,098
Selling, general, and administrative expenses	91,395	74,627	72,829
Impairment of long-lived assets and other	19,391	—	—
Change in fair value of contingent consideration	7,100	—	—
Insurance recoveries	—	—	(16,325)
Operating income	148,613	148,965	126,363
Interest expense, net	(6,647)	(5,451)	(4,037)
Other expense, net	(220)	(3,342)	(1,230)
Income before income taxes	141,746	140,172	121,096
Income tax expense	34,616	31,300	30,279
Net income	107,130	108,872	90,817
Net loss attributable to noncontrolling interest	345	590	490
Net income attributable to MGP Ingredients, Inc.	107,475	109,462	91,307
Income attributable to participating securities	(1,074)	(871)	(712)
Net income used in earnings per common share calculation	\$ 106,401	\$ 108,591	\$ 90,595
Weighted average common shares			
Basic	22,059,816	22,002,990	20,719,663
Diluted	22,173,918	22,053,966	20,719,663
Earnings per common share			
Basic	\$ 4.82	\$ 4.94	\$ 4.37
Diluted	\$ 4.80	\$ 4.92	\$ 4.37

See Accompanying Notes to Consolidated Financial Statements

Practice Exercise: MGPI Statement of Income

On the opposite page is the Statement of Income of MGP for the most recent three years that is found in their Annual Report. Compare the line-items on MGPI's statement below to the line-items that are common to most firms (listed in the discussion on the previous page). Note the similarities and differences and answer the following questions:

1. Identify the various levels of earnings/profitability for MGPI in the most recent year. Which level of earnings/profitability listed on Exhibit 10-A cannot be calculated from MGPI's Statement of Income? Where can you find this missing information (See Statement of Cash Flow below)?
2. Which line-item terms are different from the terms used in Exhibit 11-A?
3. What do you notice about how "Cost of Sales" is reported compared to "Interest Expense" on MGPI's Statement of Income? What does the word "net" mean on a financial statement?
4. Do you notice any two-year sequence when Gross Profit and Net Income moved in opposite directions? What does this suggest about the company's operations?

Below is a reproduction of MGPI's Balance Sheet and Income Statement onto an Excel spreadsheet in preparation for financial statement analysis. Note that additional years have been added so that we can review trends over multiple accounting periods (6 years total) and calculate long-term growth rates (such as a 5-year CAGR – see pg. 32 below).

MGP Ingredients, Inc. (MGPI)			Balance Sheet			
Assets	2023	2022	2021	2020	2019	2018
Cash & Cash Equivalents	\$ 18,388	\$ 47,889	\$ 21,568	\$ 21,662	\$ 3,309	\$ 5,025
Accounts Receivable (less allowances)	144,286	109,267	92,537	56,966	40,931	38,797
Inventories	346,853	289,722	245,944	141,011	136,931	108,769
Prepaid Expenses	3,580	2,957	1,510	2,644	2,048	1,320
Refundable Income Taxes	1,190	4,327	5,539	0	987	712
Total current assets	514,297	454,162	367,098	222,283	184,206	154,623
Property, Plant, Equipment (Net)	262,303	235,632	207,286	131,992	128,419	120,788
Investment in Joint Ventures/Equity Investments	5,197	5,534	4,944	0	0	0
Operating lease right-of-use assets, net	13,975	15,042	9,671	5,151	6,490	0
Intangible assets, net	271,706	216,768	218,838	890	0	0
Goodwill	321,544	226,294	226,294	2,738	0	0
Other LT Assets	3,326	4,779	7,336	3,521	3,482	2,481
Total assets	\$ 1,392,348	\$ 1,158,211	\$ 1,041,467	\$ 366,575	\$ 322,597	\$ 277,892
Liabilities and Shareholders' Equity						
Current Maturity of Long-Term Debt	\$ 6,400	\$ 5,600	\$ 3,227	\$ 1,600	\$ 401	\$ 386
Accounts payable	73,594	66,432	53,712	30,273	29,511	25,363
Income taxes payable	2,251	4,627	6,992	704	0	0
Accrued Expenses	31,861	28,716	24,869	20,752	9,383	11,714
Total current liabilities	114,106	105,375	88,800	53,329	39,295	37,463
Long-term debt	85,305	29,510	35,266	38,271	40,658	21,040
Convertible Senior Notes	195,544	195,225	194,906	0	0	0
Long-term Credit Facility	0	0	0	0	1	10,588
Long-term Operating Leases	11,292	11,622	6,997	3,057	4,267	0
Contingent Liabilities	69,200	0	0	0	0	0
Deferred Credit	0	0	0	2,196	1,233	1,565
Deferred Income Taxes	63,071	67,112	66,101	2,298	1,929	1,677
Other LT Liabilities	4,763	3,723	5,132	4,898	4,170	4,118
Total long-term liabilities	429,175	307,192	308,402	50,720	52,258	38,988
Total Liabilities	\$ 543,281	\$ 412,567	\$ 397,202	\$ 104,049	\$ 91,553	\$ 76,451
Preferred stock	4	4	4	4	4	4
Common stock	6,715	6,715	6,715	6,715	6,715	6,715
Additional Paid-In Capital	325,453	318,839	315,802	15,503	14,029	15,375
Retained Earnings	539,883	443,061	344,237	262,943	230,784	198,914
Accumulated Other Income (Loss)	(397)	(304)	354	486	(246)	(164)
Treasury Stock	(21,166)	(21,591)	(22,357)	(23,125)	(20,242)	(19,403)
Total MGP shareholders' equity	\$ 850,492	\$ 746,724	\$ 644,755	\$ 262,526	\$ 231,044	\$ 201,441
Noncontrolling interest	(490)	0	0	0	0	0
Total equity	850,002	746,724	644,755	262,526	231,044	201,441
Total liabilities and shareholders' equity	\$ 1,392,348	\$ 1,158,211	\$ 1,041,467	\$ 366,575	\$ 322,597	\$ 277,892

MGP Ingredients, Inc. (MGPI)		Statement of Income										
	2023		2022		2021		2020		2019		2018	
Sales	\$	836,523	\$	782,358	\$	626,720	\$	395,521	\$	362,745	\$	376,089
<u>Cost of Sales</u>		<u>531,811</u>		<u>529,052</u>		<u>427,755</u>		<u>296,715</u>		<u>286,213</u>		<u>292,490</u>
Gross Profit		304,712		253,306		198,965		98,806		76,532		83,599
Operating, General & Adm. Expenses		129,608		104,341		88,927		44,565		29,290		33,451
<u>Net Other Costs and (Gains)/Losses</u>		<u>26,491</u>		<u>0</u>		<u>(16,325)</u>		<u>0</u>		<u>0</u>		<u>0</u>
EBIT or Operating Income (Loss)*		148,613		148,965		126,363		54,241		47,242		50,148
Gain on Sale of Equity Investment		0		0		0		0		0		0
Interest Income (Expense)		(6,647)		(8,793)		(5,267)		(1,640)		(1,305)		(1,168)
<u>Equity Method Investment Gain (Loss)</u>		<u>(220)</u>		<u>0</u>		<u>0</u>		<u>0</u>		<u>0</u>		<u>0</u>
EBT (Loss)*		141,746		140,172		121,096		52,601		45,937		48,980
<u>Income Taxes (Provision)</u>		<u>34,616</u>		<u>31,300</u>		<u>30,279</u>		<u>12,256</u>		<u>7,144</u>		<u>11,696</u>
Net Income (Loss)	\$	107,130		108,872		90,817		40,345		38,793		37,284
<u>Net Loss Attributable to Noncontrolling Interest</u>		<u>345</u>		<u>590</u>		<u>490</u>		<u>0</u>		<u>0</u>		<u>0</u>
Net Income Attributable to MGP Ingredients, Inc.	\$	107,475	\$	109,462	\$	91,307	\$	40,345	\$	38,793	\$	37,284
<u>Discontinued Operations and Adj for Securities</u>		<u>(1,074)</u>		<u>(871)</u>		<u>(712)</u>		<u>(261)</u>		<u>(253)</u>		<u>(708)</u>
Net Income used in EPS Calculation	\$	106,401	\$	108,591	\$	90,595	\$	40,084	\$	38,540	\$	36,576

Important Calculations from the Balance Sheet and Income Statement accounts

*Net Working Capital (NWC) = Total Current Assets less Total Current Liabilities

*Net Operating Working Capital (NOWC) = Total Current Assets less “operating” current liabilities**

***Total Operating Capital (TOC) = NOWC + LT Assets (Net PP&E + all non-current assets)

Net Operating Profit after Taxes (NOPAT) = EBIT x (1 minus the Tax Rate)

Operating Cash Flow (OCF) = NOPAT + Depreciation & Amortization****

Free Cash Flow (FCF) = OCF less the Change in TOC (from beginning to end of year)

* Remember the difference between Net Working Capital and Net Operating Working Capital learned in the discussion of Balance Sheet accounts above.

** The main “operating” current liabilities are Accounts Payable and Accrued Expenses. These are sometimes referred to as “internal” current liabilities because they are generated internally through normal operations. They are also sometimes referred to as “non-interest bearing” current liabilities because they are not liabilities that accrue an interest charge like other debts (although an A/P can be assessed an interest charge if the payment is late). However, for this calculation, all current liabilities except ST Debt and Current Maturity of LT Debt should be included.

*** Note that Total Operating Capital (TOC) equals NOWC (green area) + Net Fixed Assets (black area) and remember that the “green” area plus the “black” area equals the “red” area. Thus, TOC (the “green” area plus the “black” area) will also equal Total Financial Capital (the “red” area).

**** If it is not listed on the Income Statement as a specific line-item, the annual expenditure for Depreciation & Amortization can be found on the Statement of Cash Flows (see pg. 21).

Using the five-year summary of financial statements on the opposite page, complete the following calculations for all five years:

	<u>2023</u>	<u>2022</u>
NOWC	_____	_____
TOC	_____	_____
NOPAT	_____	_____
OCF	_____	_____
FCF	_____	_____

What extra step is needed to calculate FCF?

Why are you unable to calculate FCF in 2022?

The Statement of Stockholders' Equity

The Statement of Stockholders' Equity reports changes that occurred in the Equity accounts of the Balance Sheet from the beginning to the end of the accounting period. While not considered a “major” financial statement (the three major financial statements are the Balance Sheet, Income Statement, and Statement of Cash Flows), the Statement of Shareholders' Equity provides additional information that is not necessarily reflected in the Balance Sheet, especially if major changes occurred to any of the main equity accounts: Preferred Stock, Common Stock, Retained Earnings, and Treasury Stock.

On the opposite page is the Statement of Changes in Stockholders' Equity for MGP Ingredients, Inc. for the most recent three years taken from their Annual Report.

Change in Retained Earnings

As we have learned, the “bottom line” of the Income Statement reflects profits (or losses if such is the case) and is stated as Net Income or Net Earnings (or Net Loss). The common stockholders have a claim on all profits generated by the firm after taxes. Ultimately, the stockholders (typically through their representation on the Board of Directors) decide what the firm does with its profits. They can choose to receive profits through the payment of a dividend directly from the company, or they can allow the company to retain (“keep”) some or all of the profits in order to retire debt or finance new assets. Profits that are retained by the firm are reflected on the Balance Sheet as Retained Earnings. As previously noted, Retained Earnings is an Equity account because it represents capital provided by the owners (who allow the firm to keep earned income rather than distribute it to them). It is also a cumulative account in that it is stated as the total amount of earnings retained since the inception of the firm.

A specific Statement of Change in Retained Earnings would reflect the cumulative nature of the account by starting with the balance at the beginning of the accounting period, adding the amount of Net Income or (Loss) during the accounting period, and subtracting the amount paid in dividends during the period. The ending balance would be reflected on the Balance Sheet. Below is the basic schedule of Change in Retained Earnings that is limited to Net Income and payment of Dividends.

Statement of Change in Retained Earnings

Balance of Retained Earnings, Beginning of the Year
Add: Net Income or (Loss)
Less: Dividends paid to stockholders
= Balance of Retained Earnings, End of the Year

Retirement of Treasury Stock

Besides the payment of dividends, another activity during the accounting period that can affect Retained Earnings is the retirement of treasury stock. Treasury Stock is listed on the Balance Sheet under Stockholders' Equity and represents the value of shares of common stock that the firm has repurchased from the market.

If the firm holds onto repurchased shares (listed as Treasury Stock), they are considered “issued” but not “outstanding” and are recorded at the cost of repurchase (i.e., the market price at time of repurchase). The firm may later resell these shares of Treasury Stock on the public market in the same way an individual can sell shares of stock. However, the firm may also “retire” shares held as Treasury Stock. When Treasury Stock is retired, the difference between the amount received by the firm when the stock was issued, and the amount that was paid by the firm when the stock was repurchased, will be charged to Retained Earnings. (Note: This transaction will also affect Par Value and Additional Paid-in-Capital accounts under Stockholders Equity, but our concern here is only with Retained Earnings).

MGP INGREDIENTS, INC.
CONSOLIDATED STATEMENTS OF CHANGES IN STOCKHOLDERS' EQUITY
(Dollars in thousands)

	Capital Stock Preferred	Issued Common	Additional Paid-In Capital	Retained Earnings	Accumulated Other Comprehensive Income (Loss)	Treasury Stock	Non- Controlling Interest	Total
Balance, December 31, 2020	<u>\$ 4</u>	<u>\$ 6,715</u>	<u>\$ 15,503</u>	<u>\$ 262,943</u>	<u>\$ 486</u>	<u>\$ (23,125)</u>	<u>\$ —</u>	<u>\$ 262,526</u>
Comprehensive income (loss):								
Net income (loss)	—	—	—	91,307	—	—	(490)	90,817
Other comprehensive loss	—	—	—	—	(132)	—	—	(132)
Dividends and dividend equivalents of \$0.48 per common share and per restricted stock unit, net of estimated forfeitures	—	—	—	(10,013)	—	—	—	(10,013)
Share-based compensation	—	—	5,555	—	—	—	—	5,555
Stock shares awarded, forfeited or vested	—	—	(1,535)	—	—	1,535	—	—
Stock shares repurchased	—	—	—	—	—	(767)	—	(767)
Equity consideration for Merger	—	—	296,279	—	—	—	—	296,279
Balance, December 31, 2021	<u>4</u>	<u>6,715</u>	<u>315,802</u>	<u>344,237</u>	<u>354</u>	<u>(22,357)</u>	<u>(490)</u>	<u>644,265</u>
Comprehensive income (loss):								
Net income (loss)	—	—	—	109,462	—	—	(590)	108,872
Other comprehensive loss	—	—	—	—	(658)	—	—	(658)
Dividends and dividend equivalents of \$0.48 per common share and per restricted stock unit, net of estimated forfeitures	—	—	—	(10,638)	—	—	—	(10,638)
Share-based compensation	—	—	4,518	—	—	—	—	4,518
Stock shares awarded, forfeited or vested	—	—	(1,481)	—	—	1,481	—	—
Stock shares repurchased	—	—	—	—	—	(715)	—	(715)
Balance, December 31, 2022	<u>4</u>	<u>6,715</u>	<u>318,839</u>	<u>443,061</u>	<u>(304)</u>	<u>(21,591)</u>	<u>(1,080)</u>	<u>745,644</u>
Comprehensive income (loss):								
Net income (loss)	—	—	—	107,475	—	—	(345)	107,130
Other comprehensive loss	—	—	—	—	(93)	—	—	(93)
Dividends and dividend equivalents of \$0.48 per common share and per restricted stock unit, net of estimated forfeitures	—	—	—	(10,653)	—	—	—	(10,653)
Share-based compensation	—	—	7,840	—	—	—	—	7,840
Stock shares awarded, forfeited or vested	—	—	(1,226)	—	—	1,226	—	—
Stock shares repurchased	—	—	—	—	—	(801)	—	(801)
Balance, December 31, 2023	<u>\$ 4</u>	<u>\$ 6,715</u>	<u>\$ 325,453</u>	<u>\$ 539,883</u>	<u>\$ (397)</u>	<u>\$ (21,166)</u>	<u>\$ (1,425)</u>	<u>\$ 849,067</u>

See Accompanying Notes to Consolidated Financial Statements

The Statement of Cash Flows

The **Statement of Cash Flows** (SCF) measures the amount of cash a business generates in a given year. The net effect on cash is reflected as a change (up or down) in the Cash account on the Balance Sheet from the beginning of the year to the end of the year. The SCF includes activity reflected on both the Balance Sheet and the Income Statement, and each activity is recorded on the SCF as having either a positive or negative effect on the Cash Balance.

The SCF allows the analyst to determine which activities produced cash and which activities consumed cash during the accounting period. A firm's cash position is affected by many factors:

1. Accounting Cash Flow (defined as Net Income + Depreciation)
2. Changes in net working capital accounts (increase or decrease in working capital assets and liabilities)
3. Changes in fixed assets (either investing in new assets or liquidating existing assets)
4. Borrowing activity resulting in an increase or decrease in debt accounts (short-term and long-term)
5. Security transactions (such as issuing new stock or purchasing outstanding stock) and dividend payments

It is important to note that the Income Statement does not always reflect “cash” activity. The prevalent use of accrual accounting means income and expenditures are reported when they occur (“accrue”) and not when a cash payment is made or received. Also, some “expenses” which affect the Income Statement do not involve a cash payment (such as Depreciation & Amortization). Thus, the Statement of Cash Flow is an important tool for analysis as it provides information that is not readily available in isolation on either the Balance Sheet or Income Statement.

Note: The SFC is divided into three types of business activities: **Operating, Investing, and Financing.** The values that appear on the SCF are either positive or negative, depending on whether they constitute an increase (positive value) or decrease (negative value) in the Cash balance.

Cash Flows from Operating Activities measures cash inflows and outflows that occur as part of normal operations. Net Income is the first source of cash and thus the first item on the SCF. However, Net Income is NOT equal to cash from operations because not all sales and expenses are made in cash and some expenses do not involve any cash expenditure at all. Thus, adjustments to NI are needed to determine Net Cash Flow from Operations: a) Depreciation and Amortization expense is added back, b) other adjustments as needed to reconcile Net Income, and c) changes to the net operating working capital accounts – both assets and liabilities – are calculated to reflect the effect these changes had on cash. As a general rule, any increase in an asset account (from beginning to end of year) constitutes a decrease in cash, and any decrease in the account constitutes an increase in cash. Liability accounts are the opposite, an increase in the liability account constitutes an increase in cash, while a decrease in the account constitutes a decrease in cash. The sum of all this activity procures net cash provided by (or used in) operating activities.

Cash Flows from Investing Activities measures cash inflows and outflows produced from activities involving the purchase or disposition of Property, Plant & Equipment (PP&E) and long-term investment accounts. The effect on cash from changes in assets under Investing Activities follows the same rule as with assets under Operating Activities: An increase from beginning to end of year constitutes negative cash while a decrease constitutes positive cash. It is important here to note that the change in Net PP&E on the Balance Sheet (from beginning to end of year) is NOT necessarily entirely due to the purchase or disposition of assets because the Depreciation expense during the year will also affect the ending balance. See the Note below Exhibit 12 on pg. 22 for an explanation of this calculation.

Cash Flows from Financing Activities measures cash inflows and outflows based on changes in the financing accounts: mainly ST Debt, LT Debt, Preferred Stock and Common Stock. In addition, any **dividend payment** is noted as an outflow (i.e., the payment is given a negative value). The effect on cash from changes in liability and equity accounts under Financing Activities follows the same rule as with liabilities under Operating Activities: an increase in the account from beginning to end of year constitutes positive cash while a decrease constitutes negative cash.

On the opposite page is the Statement of Cash Flows for MGP Ingredients, Inc. for the most recent three years taken from their Annual Report.

MGP INGREDIENTS, INC.
CONSOLIDATED STATEMENTS OF CASH FLOWS
(Dollars in thousands)

	Year Ended December 31,		
	2023	2022	2021
Cash Flows from Operating Activities			
Net income	\$ 107,130	\$ 108,872	\$ 90,817
Adjustments to reconcile net income to net cash provided by operating activities:			
Depreciation and amortization	22,113	21,455	19,092
Impairment of long-lived assets and other	19,391	—	—
Gain on insurance recoveries	—	—	(16,325)
Share-based compensation	10,635	5,502	5,555
Equity method investment loss	337	2,220	1,611
Deferred income taxes, including change in valuation allowance	(4,041)	1,011	6,772
Change in fair value of contingent consideration	7,100	—	—
Other, net	728	194	145
Changes in operating assets and liabilities, net of effects of acquisitions:			
Receivables, net	(32,397)	(16,786)	(6,031)
Inventory	(46,921)	(44,350)	(14,214)
Prepaid expenses	(481)	(1,468)	2,586
Income taxes payable (refundable)	3,136	1,212	(6,242)
Accounts payable	(2,406)	10,626	5,301
Accrued expenses and other	348	1,984	738
Federal and state excise taxes payable	(2,375)	(2,365)	(1,467)
Other, net	1,486	829	(75)
Net cash provided by operating activities	83,783	88,936	88,263
Cash Flows from Investing Activities			
Additions to property, plant, and equipment	(55,267)	(45,323)	(47,389)
Purchase of business, net of cash acquired	(103,712)	—	(149,005)
Contributions to equity method investments	—	(2,810)	(1,470)
Proceeds from property insurance recoveries	—	—	16,325
Other, net	(263)	320	(1,080)
Net cash used in investing activities	(159,242)	(47,813)	(182,619)
Cash Flows from Financing Activities			
Payment of dividends and dividend equivalents	(10,675)	(10,646)	(10,017)
Purchase of treasury stock	(801)	(715)	(767)
Loan fees incurred with borrowings	—	—	(7,050)
Principal payments on long-term debt	(5,600)	(3,403)	(1,620)
Proceeds from credit agreement - revolver	105,000	—	242,300
Payments on credit agreement - revolver	(42,000)	—	(242,300)
Proceeds from convertible senior notes	—	—	201,250
Payment on assumed debt as part of the Merger	—	—	(87,509)
Net cash provided by (used in) financing activities	45,924	(14,764)	94,287
Effect of exchange rate changes on cash and cash equivalents	34	(38)	(25)
Increase (decrease) in cash and cash equivalents	(29,501)	26,321	(94)
Cash and cash equivalents, beginning of year	47,889	21,568	21,662
Cash and cash equivalents, end of year	\$ 18,388	\$ 47,889	\$ 21,568

See Accompanying Notes to Consolidated Financial Statements

Statement of Cash Flows

<u>Cash from Operating Activities</u>	<u>Effect on Cash + or (-)</u>
Net Income	+ or (-)
Depreciation (always positive effect on cash)	+
<u>Changes in Net Working Capital Accounts</u>	
Δ A/R (↑ = "-" cash, ↓ = "+" cash)	+ or (-)
Δ Inventory (↑ = "-" cash, ↓ = "+" cash)	+ or (-)
Δ A/P (↑ = "+" cash, ↓ = "-" cash)	+ or (-)
Δ A/E (↑ = "+" cash, ↓ = "-" cash)	+ or (-)
Total Cash from Operating Activities	+ or (-)
<u>Cash from Investing Activities</u>	<u>Effect on Cash + or (-)</u>
Δ in Net PP&E (↑ = "-" cash, ↓ = "+" cash)*	+ or (-)
Δ in Other Non-Current Assets (↑ = "-" cash, ↓ = "+" cash)	+ or (-)
Total Cash from Investing Activities	+ or (-)
<u>Cash from Financing Activities</u>	<u>Effect on Cash + or (-)</u>
Δ in ST Debt (↑ = "+" cash, ↓ = "-" cash)	+ or (-)
Δ in LT Debt (↑ = "+" cash, ↓ = "-" cash)	+ or (-)
Δ in Preferred Stock (↑ = "+" cash, ↓ = "-" cash)	+ or (-)
Δ in Common Stock (↑ = "+" cash, ↓ = "-" cash)	+ or (-)
Dividends Paid (always negative effect on cash)	-
Total Cash from Financing Activities	+ or (-)
Total Increase/Decrease in Cash during Year	+ or (-)

Cash at Beginning of Year

Total Change in Cash during Year

Cash at End of Year

From Balance Sheet

From SCF

Check Balance Sheet

*** Note:** Three steps are needed to calculate the effect on cash from a change in Net Property Plant & Equipment:

Step 1: Calculate the change in Net PP&E from the B/S = Ending Balance (EB) less Beginning Balance (BB)

Step 2: Add back Depreciation & Amortization from Income Statement (if listed) or from internal reports

Simple Formula: $\Delta \text{ Net PP\&E} = \text{EB} - \text{BB} + \text{D\&A}$

Step 3: If $\Delta \text{ Net PP\&E}$ is a positive number, that is an increase in Net PP&E (↑) and the effect on cash is negative (-).

If $\Delta \text{ Net PP\&E}$ is a negative number, that is a decrease in Net PP&E (↓) and the effect on cash is positive (+).

Steps for Constructing a Statement of Cash Flows

In constructing a Statement of Cash Flows (SCF) for this class, the student can refer to Exhibit 12 on the previous page and note the following instructions:

1. It should be noted that the SCF reports “changes” that occur from the BEGINNING of the accounting period to the END of the accounting period. In particular, the SCF explains how the “Cash” account on the Balance Sheet changed from its BEGINNING balance to its ENDING balance.
2. The BEGINNING balances for an accounting year are typically the ENDING balances of the previous year. Thus, for example, the END of 2022 would be the BEGINNING of 2023, so the change from beginning to end of 2022 would compare the ENDING Balance Sheet in 2022 with the ENDING Balance Sheet in 2022.
3. While most information on the SCF is generated from a review of the Balance Sheet, it does require information from the Income Statement as well.
 - a. The SCF begins with the Net Income number, and it should be recorded on the SCF as it is reported on the Income Statement (i.e., if Net Income is positive, it is recorded as a positive number on the SCF, likewise, if Net Income is negative, it is recorded as a negative number on the SCF).
 - b. Also, the Depreciation and Amortization expenses from the Income Statement are ALWAYS recorded as positive numbers on the SCF. The reason: Depreciation and Amortization have been used to reduce Net Income, but they are ALWAYS non-cash expenses. (Note: As was previously explained, most Income Statements do not list Depreciation and Amortization as a separate expense, which means these expenses are included in another line item. It is often the case that the only source of Depreciation and Amortization for an external analyst is the SCF.)
4. As stated above, most of the information on the SCF is generated from the Balance Sheet, and, in particular, from “changes” in the Balance Sheet accounts (i.e., the “change” from the beginning of the year to end of the year). Thus, the SCF requires the student to complete the following sequence: a.) calculate the “change” in the Balance Sheet account, b.) determine whether this “change” had the effect of increasing or decreasing cash, and c.) record this on the SCF by using positive (for an increase in cash) and negative (for a decrease in cash) numbers. Two important rules to remember are:
 - a. Assets: when the “change” in an asset account indicates an increase from beginning to end of the year, this has the effect of decreasing cash (assume the firm needed to use cash in order to increase the asset). Likewise, when the “change” in an asset account indicates a decrease from beginning to end of the year, this has the effect of increasing cash (assume the firm liquidated the asset for cash).
 - b. Liabilities and Equity: when the “change” in either a liability or equity account indicates an increase from beginning to end of the year, this has the effect of increasing cash (assume the firm receives cash in return for the “claim” against it). Likewise, when the “change” in either a liability or equity account indicates a decrease from beginning to end of the year, this has the effect of decreasing cash (assume the firm uses cash to extinguish the claim against it).
5. The last item on the SCF recognizes any Dividends paid by the company to the shareholders during the accounting period. This is ALWAYS recorded as a negative number on the SCF.
6. A change in Retained Earnings is NEVER reported on the SCF. Why? Recall that Net Income is added to Retained Earnings at the end of the year but the Dividend payout is subtracted before the Retained Earnings at the end of the year is reported. Note that the SCF has already accounted for Net Income and Dividends. Therefore, the “change” in Retained Earnings does not factor into the SCF.

Topic 2: Financial Statement Analysis

After learning what information financial statements report, the student must next learn how financial statements are used by professionals to assess the financial health of the firm. Managers use financial statements to monitor and evaluate the firm's past performance and project the future. Lenders use the financial statements to determine the firm's creditworthiness. Investors and outside analysts use the financial statements to determine a firm's value, to predict future performance, and to analyze the stock's attractiveness as an investment.

Four common methods used to analyze the financial statements in greater detail are:

1. Ratio Analysis (see list of the most common ratios on pg. 30)
2. Common-Sizing (an extension of Ratio Analysis)
3. Benchmarking
4. Trend Analysis

These techniques are used to (a) assess the firm's current financial health, (b) identify trends in the firm's financial position over time (usually the most recent past), (c) compare the firm's health and performance with other firms, often in the same industry, and (d) project the firm's performance into the future.

Financial Ratios

Most financial ratios are produced by dividing the amount of one account by the amount of a different account. For example, dividing Net Income by Total Revenue (or Total Sales) produces a ratio known as the Net Profit Margin (NPM), which measures the percent of revenue that remains as profits after all expenses have been recorded. Another example would be dividing Net Income by Total Assets to produce a ratio known as Return on Assets (ROA), which measures profits as a percent of the total assets employed by the company. A schedule of frequently used financial ratios is provided on pg. 28 opposite a schedule of the calculated Ratios for MGPI on pg. 29.

We divide the ratios into five categories:

1. **Liquidity Ratios** – reflect the firm's ability to pay liabilities in the near term (that are due within a year) by comparing liquid assets with current liabilities.
2. **Asset Management Ratios** – reflect how efficiently the firm is using its assets
3. **Debt Management Ratios** – reflect how the firm has financed its assets as well as its ability to repay its long-term debt.
4. **Profitability Ratios** – reflect how profitably the firm is operating and utilizing its assets
5. **Market Value Ratios** – reflect how investors value the firm through the stock price

One important note: As we have learned, the Balance Sheet and Income Statement differ in their reflection of time. The Balance Sheet reports the firm's financial position at a specific moment in time (the "snapshot"), while the Income Statement reports on activities over a specific accounting period (such as a year). This can be problematic when accounts from the Balance Sheet (a moment in time) are measured against accounts from the Income Statement (a year's worth of activity). One solution to this problem is to use an average Balance Sheet account (e.g., the average of the beginning and ending balances for the year). An example of this would be **Return on Assets (ROA)**. The formula is simply **Net Income ÷ Total Assets**. However, Net Income is an Income Statement number and Total Assets is a Balance Sheet account. For this ratio formula, the denominator can be adjusted to the average Total Assets balance from beginning to ending year: Thus: $\frac{A/R (2022) + A/R (2021)}{2} = A/R$ used in the Liquidity Ratio calculations. This is a more precise calculation. However, it requires knowing the previous year balances, which limits the analysis. **The relevant ratios for MGPI below use the average Balance Sheet account.**

Liquidity Ratios:

Liquidity Ratios measure a company's ability to pay off its short-term debt obligations. This is done by comparing a company's most liquid assets (cash and those that can be easily converted to cash) to its most immediate (short-term) liabilities that will require cash to satisfy.

In general, when liquid assets exceed short-term liabilities (i.e., the ratio is greater than 1) it is a clear signal that a company can pay those debts that are coming due in the near future and still fund its ongoing operations. On the other hand, a company with a low liquidity ratio (i.e., the ratio is lower than 1) may signal that the company will have difficulty meeting its near-term obligations and funding operations. The biggest difference between each of the three liquidity ratios is the type of current assets used in the calculation. While each ratio includes current assets, the more conservative ratios will exclude some current assets as they aren't as easily converted to cash.

Asset Management Ratios (also called Operating Performance Ratios)

The Asset Management Ratios measure different segments of a company's overall operating performance. These ratios give analysts insight into the company's performance and management during the period being measured.

Included in the Asset Management ratios are those that reveal how efficiently a company turns its assets into revenue (such as Fixed Asset Turnover, Total Asset Turnover, and Inventory Turnover) as well as how efficiently the company converts its sales into cash. Note that two components of the Cash Conversion Cycle involve assets: the Inventory Conversion Period and the Receivables Collection Period. Basically, these ratios help determine whether a company is using its resources to generate revenue and increase financial health. In addition, some of the ratios in this category track the percent of a company's that are tied up in working capital (cash, inventory, receivables).

Debt Management Ratios (also called Leverage Ratios)

The Debt Management Ratios (or sometimes simply called the Debt Ratios and other times referred to as Leverage Ratios) provide a general idea of the company's overall debt load (e.g., as a percent of Total Assets), its mix of debt and equity, and the debt interest burden relative to Operating Income (EBIT). Debt ratios are used to assess the overall level of financial risk a company and its shareholders face. In general, the greater the amount of debt held by a company (i.e., the more it is "leveraged"), the greater is the financial risk of insolvency due to an inability to both meet debt obligations and fund ongoing operations. In other words, while debt can be attractive as a strategic means of increasing returns for shareholders (which is the goal of "leverage" in its strict sense), it can also lead to a rapid loss of financial health if operations are unable to meet debt obligations that have exceeded the firm's capacity.

Students should learn some of the important terminology used with "debt" and financial statement analysis. In general, companies incur two types of liabilities: those that are incurred during the normal part of operations, and those that are incurred by borrowing. The former category (operational liabilities) includes Balance Sheet accounts such as Accounts Payable, Accrued Expenses, Taxes Payable, pension obligations, etc. The latter category includes Notes Payable (often listed simply as ST Debt) and other short-term borrowings (such as a balance due on a Line of Credit), the Current Portion of Long-Term Debt, Long-Term Debt, and Bonds Payable. The word "debt" is often used synonymously with "liabilities" in finance literature and media reports. However, the student should be aware that the word "debt" may refer only to the second category (i.e., a company's indebtedness that does not include operational liabilities).

The debt ratios used in this Supplement are the most common used. However, what companies, financial analysts, and investment research services use as components to calculate these ratios is not standardized and may contain significant difference.

Profitability Ratios

The Profitability Ratios measure how well the company utilizes its resources to generate profits, increase shareholder value, and provide overall financial health. Long-term profitability is vital for both the survivability of the company as well as the benefit received by shareholders. As the name suggests, these ratios provide data on the critical amount of "profit" the company produces relative to different stages of the Income Statement and relative to certain measures from financial structure of the company as found on the Balance Sheet.

Of the five ratios listed under Profitability, three measure profits against the top line of the Income Statement (Total Revenue or Total Sales) and two measure profits against important Balance Sheet accounts (Total Assets and Total Equity). It might be said that the three most common ratios quoted by analysts are the Net Profit Margin, the Return on Assets (ROA), and the Return on Equity (ROE). In fact, these three profitability ratios are so important that a special system of breaking them down into small components is frequently used by analysts (see discussion of DuPont Analysis on the following page). This is not meant to diminish the importance of Gross Profit Margin and Operating Profit Margin, both of which reveal important facts regarding a company's operating efficiency, especially when assessed over time in a trend analysis (see discussion of trend analysis below).

Market Ratios (also called Price Ratios)

The Market Ratios are specific to publicly traded companies and represent a unique category of financial ratios because they produce an estimate of "value" (as determined by the market price of the company's stock) rather than operating or financial health. The term "value" here is reflected in the price of the company's stock as it trades on public markets.

When first learning how to analyze financial statements using ratios, students can suffer from information overload as there are so many different financial values: Current Ratio, Debt Ratio, Revenue, Gross Margin, Operating Profit Margin, Net Profit Margin, EBIT, EBITDA, FATO, TATO, ROA, ROE, the Cash Conversion Cycle, even Pro Forma Projections. The list goes on and on, and there is a real risk that the beginner will simply learn how to use formulas to calculate a ratio and not make a connection as to how this data can determine the "value" of the company. Market Ratios should help students overcome this risk.

For example, perhaps the best known investment ratio used by analysts and the financial media is the P/E Ratio, which compares the current price of a company's stock to the amount of earnings (another word for "profits") it generates. The purpose of this ratio is to give manager and analysts an idea of how much investors are paying for every \$1 of profits. In other words, a P/E ratio of 12 means that buyers and sellers (investors) on the market are willing to pay \$12 for every \$1 of earnings. Likewise, the Price-to-Book Ratio reveals how much investors are willing to pay for every \$1 of Net Assets (Total Assets less Total Liabilities).

An important component of Market Ratio calculations is the number of shares outstanding. This number is usually disclosed on the Balance Sheet in the company's SEC filing or on financial websites (such as Yahoo Finance). The number of shares outstanding is needed to calculate such items as Earnings per Share (EPS) and Book Value per Share (BVS), which are then compared to the stock price.

Students should recall that a firm might have more shares "issued" than "outstanding" if they are holding Treasury Stock. Thus, it is important to note that Market Ratios are calculated against the number of shares "outstanding" and not the total number of shares "issued" by the firm.

DuPont Analysis

The DuPont Company was a forerunner in stressing that ROA (Return on Assets) may be achieved through a high profit margin or through a rapid turnover of assets, or through a combination of both. Thus, ***DuPont Analysis helps the analyst examine the sources of a company's profitability.*** Since the profit margin is an Income Statement ratio, a high profit margin indicates good cost control, whereas a high asset turnover ratio demonstrates efficient use of the assets on the Balance Sheet. Different industries have different operating and financial structures. For example, a heavy capital goods industry (machinery and equipment) emphasizes high profit margins and low asset turnover, while a retailer or a food processor emphasizes low profit margin but a rapid turnover of assets.

ROA = Return on Assets

ROA = Net Profit Margin (NPM) x Total Asset Turnover Ratio (TATO)

$$\text{ROA} = \frac{\text{Net Income}}{\text{Total Assets}} = \frac{\text{Net Income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Total Assets}}$$

Return on Assets = Net Profit Margin (NPM) x Total Asset Turnover (TATO)

DuPont analysis further stresses that the ROE (Return on Equity) stems from the return on assets multiplied by the amount of financial leverage. Financial leverage is the ratio of equity to assets. This ratio is then used as a “multiplier” of ROA. The higher the financial leverage ratio, the more the company relies on debt for financing its assets. Thus, the higher the spread between ROA and ROE, the more effective the company is using financial leverage to increase its ROE. DuPont Analysis allows a deeper comparison of companies based on their ROA and ROE ratios.

ROE = Return on Equity

ROE = Net Profit Margin x Total Asset Turnover Ratio x Leverage Ratio

ROE = ROA x Leverage Ratio

$$\text{ROE} = \frac{\text{Net Income}}{\text{Equity}} = \frac{\text{Net Income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Total Assets}} \times \frac{\text{Total Assets}}{\text{Equity}}$$

Return on Equity = Net Profit Margin x Total Asset Turnover x “Leverage Ratio”

$$\text{ROE} = \frac{\text{Net Income}}{\text{Equity}} = \text{ROA} \times \frac{\text{Total Assets}}{\text{Equity}}$$

Return on Equity = Return on Assets x “Leverage Ratio”

Important Note: Students should become familiar with the components of ROA and ROE using DuPont Analysis, and to understand how changes in the components affect the ratio (both operationally and mathematically). For example, the ROE and the Leverage Ratio will be positively correlated, i.e., as the Leverage increases, so does ROE (all things being equal).

Ratio Analysis

Liquidity Ratios

Current ratio = **Current Assets** ÷ **Current Liabilities**

Quick ratio (“Acid Test”) = **(Cash + A/R)** ÷ **Current Liabilities**

Cash Ratio = **Cash** ÷ **Current Liabilities**

Asset Management Ratios

Fixed Asset Turnover (FATO) = **Total Sales or Total Revenue** ÷ **Net Fixed Assets**

Total Asset Turnover (TATO) = **Total Sales or Total Revenue** ÷ **Total Assets**

Inventory Turnover = **COGS** ÷ **Avg. Inventory** (Note: Some textbooks use Sales in the numerator for this ratio)

Receivables Turnover = **Total Sales** ÷ **Avg. Accounts Receivable**

Debt Management (“Solvency”) Ratios

Debt to Assets (also called the “Debt Ratio”) = **Total Liabilities** ÷ **Total Assets**

state as a %

Debt to Equity Ratio = **Total Liabilities** ÷ **Total Shareholders’ Equity**

state as a %

Long-Term Debt to Equity Ratio = **LT Debt** ÷ **Total Shareholders’ Equity**

state as a %

Times Interest Earned Ratio (TIE) = **EBIT** ÷ **Interest Expense**

Leverage Ratio = **Total Assets** ÷ **Total Shareholders’ Equity**

Note: DO NOT state as a %

Payables Turnover = **COGS** ÷ **Avg. Accounts Payable**

Profitability Ratios (all = %)

Gross Profit Margin = **Gross Profit** ÷ **Total Sales or Total Revenue**

state as a %

Operating Profit Margin = **Operating Income (EBIT)** ÷ **Total Sales or Total Revenue**

state as a %

Net Profit Margin = **Net Income** ÷ **Total Sales or Total Revenue**

state as a %

Return on Assets (ROA) = **Net Income** ÷ **Total Assets**

state as a %

Return on Common Equity (ROE) = **Net Income** ÷ **Common Equity (“Net Worth”)**

state as a %

Market (Price) Ratios

Book Value per Share (BVPS) = **Total Common Equity** ÷ **number of common shares**

Price-to-Book (P/B) Ratio = **Market Price of Stock** ÷ **BVPS**

Earnings per Share (EPS) = **Net Income** ÷ **number of common shares**

Price-to-Earnings (P/E) Ratio = **Market Price of Stock** ÷ **EPS**

Dividend Yield = **Dividend per share** ÷ **Market price of stock at beginning of year**

state as a %

Cash Conversion Cycle Ratios (see Topic 3)

Inventory Conversion Period (ICP) = **Inventory** ÷ **Purchases per Day (COGS ÷ 365)**

Receivables Collection Period (RCP) = **Receivables** ÷ **Sales per Day (Sales ÷ 365)**

Payables Deferral Period (PDP) = **Accounts Payable** ÷ **Purchases per Day (COGS ÷ 365)**

Cash Conversion Cycle = **ICP + RCP - PDP**

DuPont Analysis

ROA = **Net Profit Margin (NPM)** x **Total Asset Turnover Ratio (TATO)**

state as a %

ROE = **Net Profit Margin (NPM)** x **Total Asset Turnover Ratio (TATO)** x **Leverage Ratio**

state as a %

Other Financial Statement Formulas

Net Working Capital (NWC) = **total current assets** – **total current liabilities**

Net Operating Working Capital (NOWC) = **total current assets** – **“internal” current liabilities**

Total Operating Capital (TOC) = **NOWC** + **Net PP&E**

Net Operating Profit after Taxes (NOPAT) = **EBIT** x **(1 – Tax Rate)**

Operating Cash Flow (OCF) = **NOPAT** + **Depreciation**

Free Cash Flow (FCF) = **OCF** - **Change in TOC (from beginning to end of year)**

MGP Ingredients, Inc.			Key Financial Ratios			
	2023	2022	2021	2020	2019	2018
Liquidity Ratios						
Current Ratio (All Current Assets)	4.51	4.31	4.13	4.17	4.69	4.13
Quick Ratio (Cash and A/R only)	1.43	1.49	1.28	1.47	1.13	1.17
Cash Ratio (Cash only)	0.16	0.45	0.24	0.41	0.08	0.13
Asset Management Ratios						
Fixed Assets Turnover (FATO) using avg. of BY	3.4	3.5	3.7	3.0	2.9	3.1
Total Assets Turnover (TATO) using avg. of BY	0.7	0.7	0.9	1.1	1.2	1.4
Inventory Turnover using avg. of BY and EY*	1.7	2.0	2.2	2.1	2.3	2.7
Receivables Turnover using avg. of BY and EY*	6.6	7.8	8.4	8.1	9.1	9.7
Debt Management (Solvency) Ratios						
Debt to Assets (Debt Ratio)	39%	36%	38%	28%	28%	28%
Debt-to-Equity Ratio	64%	55%	62%	40%	40%	38%
LT Debt-to-Equity Ratio	34%	32%	37%	16%	19%	16%
Times-interest-earned ratio (TIE based on EBIT)	22.36	16.94	23.99	33.07	36.20	42.93
Financial Leverage Ratio	1.64	1.55	1.62	1.40	1.40	1.38
Payables Turnover using avg. of BY and EY*	7.60	8.81	10.19	9.93	10.43	11.53
Profitability Ratios						
Gross Profit Margin	36.4%	32.4%	31.7%	25.0%	21.1%	22.2%
Operating Profit Margin	17.8%	19.0%	20.2%	13.7%	13.0%	13.3%
Net Profit Margin	12.7%	13.9%	14.5%	10.1%	10.6%	9.7%
Return on Assets (ROA)	8.3%	9.9%	12.9%	11.6%	12.8%	26.3%
Return on Equity (ROE)	12.5%	14.5%	14.1%	15.3%	16.7%	18.2%
Valuation Calculations using Financial Statements						
Earnings per share (EPS)	\$4.83	\$4.94	\$4.12	\$2.37	\$2.26	\$2.17
Price-to-Earnings ratio (P/E)	20.98	21.55	20.61	19.86	21.41	26.29
Book Value per share (BVPS)	\$38.63	\$33.95	\$29.35	\$15.52	\$13.57	\$11.95
Price-to-book ratio (P/B)	2.63	3.13	2.90	3.03	3.57	4.77
Dividend Yield (based on prior YE price)	0.45%	0.56%	1.02%	0.99%	0.70%	0.42%
Cash Conversion Cycle						
Inventory Conversion Period*	218.5	184.8	165.1	171.0	156.7	135.7
Receivable Collection Period*	55.3	47.1	43.5	45.2	40.1	37.7
Payables Deferral Period*	48.1	41.4	35.8	36.8	35.0	31.7
Cash Conversion Cycle	225.7	190.4	172.8	179.4	161.8	141.7
* All calculations in Red Font use the average of BY/EY for Balance Sheet accounts when used in a formula that includes an Income Statement item (see Text)						
Dupont Analysis						
Net Profit Margin	12.7%	13.9%	14.5%	10.1%	10.6%	9.7%
TATO	0.66	0.71	0.89	1.15	1.21	1.35
ROA	8.3%	9.9%	12.9%	11.6%	12.8%	13.2%
Leverage Ratio	1.64	1.55	1.62	1.40	1.40	1.38
ROE	13.7%	15.3%	20.8%	16.2%	17.9%	18.2%
Other Key Financial Statement Data						
Net Working Capital (NWC)	\$400,191	\$348,787	\$278,298	\$168,954	\$144,911	\$117,160
Net Operating Working Capital (NOWC)	\$406,591	\$354,387	\$281,525	\$170,554	\$145,312	\$117,546
Total Operating Capital (TOC)	\$1,284,642	\$1,058,436	\$955,894	\$314,846	\$283,703	\$240,815
Net Operating Profits after Taxes (NOPAT)	\$112,320	\$115,702	\$94,767	\$41,603	\$39,895	\$38,173
Operating Cash Flow (OCF)	\$134,433	\$137,157	\$113,859	\$54,564	\$51,467	\$49,535
Free Cash Flow (FCF)	-\$91,773	\$34,615	-\$527,189	\$23,421	\$8,579	-\$191,280

Common Sizing

To produce a common-sized Balance Sheet, each asset, liability, and equity account is divided by the amount of total assets. The result (called the quotient) is expressed as a percentage. To produce a common-sized Income Statement, each item on the statement is divided by Total Sales (or Total Revenue), and the result is again expressed as a percentage. To produce a common-sized Balance Sheet, each account is divided by Total Assets. In addition, the Balance Sheet accounts can be common-sized to Total Sales. Likewise, the Income Statement items can be common-sized to Total Assets. These extra steps might be used to give a more complete picture. The percentages produced by common-sizing allow analysts to compare small firms with big firms despite a great disparity in the aggregate values of financial statement accounts and line items.

Note the common-sized Balance Sheet and Income for MGPI on the opposite page.

Benchmarking

Benchmarking compares a firm's current performance against an industry average or against a top competitor (or a subset of top competitors) in their industry. Typically, ratios are used for the analysis rather than aggregate numbers. Benchmarking allows management to see where a firm stands relative to its competitors, which allows management to set strategic goals based on certain key ratios. Below is an example of industry benchmark ratios.

Industry Name	Current Ratio	Inventory Turnover	TATO	LT Debt/ LT Capital	RCP (days)	NPM	ROA	ROE
Aerospace/Defense	1.6	5.9	0.9	94.8%	42.44	6.7%	5.6%	87.0%
Apparel Stores	2.3	4.2	1.8	21.3%	4.64	6.5%	11.9%	33.6%
Auto Manufacturing	1.0	9.0	0.7	62.0%	32.30	4.7%	3.6%	15.6%
Beverage (soft drink)	1.2	7.5	0.8	51.0%	36.87	14.1%	8.2%	27.1%
Electronics	3.9	5.1	0.7	19.4%	24.50	8.8%	4.7%	6.7%
Food Wholesalers	1.7	13.5	3.0	43.2%	25.35	2.3%	6.9%	17.6%
Grocery Stores	1.1	17.2	2.3	84.9%	6.90	2.9%	5.8%	14.7%
Health Services	1.8	21.9	0.8	55.8%	64.04	9.5%	6.9%	21.4%
Lodging	1.1	165.1	0.7	50.7%	13.93	9.6%	5.8%	18.3%
Newspapers	0.5	7.4	0.2	6.5%	214.71	7.0%	1.0%	2.3%
Paper Products	2.1	6.6	0.7	51.5%	52.14	0.7%	1.2%	4.6%
Railroad	1.1	15.0	0.5	41.2%	33.80	15.9%	7.3%	18.7%
Restauranant	0.7	25.9	0.8	61.5%	9.58	13.7%	6.4%	21.3%
Retail (Dept Stores)	1.5	5.8	1.9	49.6%	2.80	5.1%	10.6%	22.9%
Tech Instruments	2.9	8.6	0.7	37.5%	41.01	8.3%	6.2%	14.3%
Sporting Goods	2.3	3.4	2.0	36.7%	27.24	1.3%	4.9%	15.4%
Steel and Iron	2.0	3.9	0.9	42.2%	40.11	5.1%	4.0%	8.0%
Tobacco	1.1	2.7	1.3	57.3%	14.43	16.1%	18.2%	35.8%

Notes:

The ratios are presented as averages for each industry

LT Debt/LT Capital is calculated by $LT\ Debt \div (LT\ Debt + Equity)$

The highest industry ratio in each category is highlighted in blue

The lowest industry ratio in each category is highlighted in yellow

This data was taken from calendar year 2013

MGP Ingredients, Inc.							Common-Size Balance Sheet (against Total Assets)						Common-Size Balance Sheet (against Sales)					
Assets	2023	2022	2021	2020	2019	2018	2023	2022	2021	2020	2019	2018	2023	2022	2021	2020	2019	2018
Cash & Cash Equivalents	1.3%	4.1%	2.1%	5.9%	1.0%	1.8%	2.2%	6.1%	3.4%	5.5%	0.9%	1.3%	17.2%	14.0%	14.8%	14.4%	11.3%	10.3%
Accounts Receivable (less allowances)	10.4%	9.4%	8.9%	15.5%	12.7%	14.0%	17.2%	14.0%	14.8%	14.4%	11.3%	10.3%	41.5%	37.0%	39.2%	35.7%	37.7%	28.9%
Inventories	24.9%	25.0%	23.6%	38.5%	42.4%	39.1%	41.5%	37.0%	39.2%	35.7%	37.7%	28.9%	0.4%	0.4%	0.2%	0.7%	0.6%	0.4%
Prepaid Expenses	0.3%	0.3%	0.1%	0.7%	0.6%	0.5%	0.4%	0.4%	0.2%	0.7%	0.6%	0.4%	0.1%	0.6%	0.9%	0.0%	0.3%	0.2%
Refundable Income Taxes	0.1%	0.4%	0.5%	0.0%	0.3%	0.3%	0.1%	0.6%	0.9%	0.0%	0.3%	0.2%	0.1%	0.6%	0.9%	0.0%	0.3%	0.2%
Total current assets	37%	39%	35%	61%	57%	56%	61%	58%	59%	56%	51%	41%	61%	58%	59%	56%	51%	41%
Property, Plant, Equipment (Net)	18.8%	20.3%	19.9%	36.0%	39.8%	43.5%	31.4%	30.1%	33.1%	33.4%	35.4%	32.1%	0.6%	0.7%	0.8%	0.0%	0.0%	0.0%
Investment in Joint Ventures/Equity Investments	0.4%	0.5%	0.5%	0.0%	0.0%	0.0%	0.6%	0.7%	0.8%	0.0%	0.0%	0.0%	1.7%	1.9%	0.0%	0.0%	0.0%	0.0%
Operating lease right-of-use assets, net	1.0%	1.3%	0.0%	0.0%	0.0%	0.0%	1.7%	1.9%	0.0%	0.0%	0.0%	0.0%	32.5%	27.7%	0.0%	0.0%	0.0%	0.0%
Intangible assets, net	19.5%	18.7%	0.0%	0.0%	0.0%	0.0%	32.5%	27.7%	0.0%	0.0%	0.0%	0.0%	38.4%	28.9%	0.0%	0.0%	0.0%	0.0%
Goodwill	23.1%	19.5%	0.0%	0.0%	0.0%	0.0%	38.4%	28.9%	0.0%	0.0%	0.0%	0.0%	0.4%	0.6%	1.2%	0.9%	1.0%	0.7%
Other LT Assets	0.2%	0.4%	0.7%	1.0%	1.1%	0.9%	0.4%	0.6%	1.2%	0.9%	1.0%	0.7%	0.4%	0.6%	1.2%	0.9%	1.0%	0.7%
Total assets	100%	100%	100%	100%	100%	100%	166%	148%	166%	93%	89%	74%	166%	148%	166%	93%	89%	74%
Liabilities and Shareholders' Equity																		
Current Maturity of Long-Term Debt	0.5%	0.5%	0.3%	0.4%	0.1%	0.1%	0.8%	0.7%	0.5%	0.4%	0.1%	0.1%	8.8%	8.5%	8.6%	7.7%	8.1%	6.7%
Accounts payable	5.3%	5.7%	5.2%	8.3%	9.1%	9.1%	8.8%	8.5%	8.6%	7.7%	8.1%	6.7%	0.3%	0.6%	1.1%	0.2%	0.0%	0.0%
Income taxes payable	0.2%	0.4%	0.7%	0.2%	0.0%	0.0%	0.3%	0.6%	1.1%	0.2%	0.0%	0.0%	0.3%	0.6%	1.1%	0.2%	0.0%	0.0%
Accrued Expenses	2.3%	2.5%	2.4%	5.7%	2.9%	4.2%	3.8%	3.7%	4.0%	5.2%	2.6%	3.1%	2.3%	2.5%	2.4%	5.7%	2.9%	4.2%
Total current liabilities	8%	9%	9%	15%	12%	13%	14%	13%	14%	13%	11%	10%	14%	13%	14%	13%	11%	10%
Long-term debt	6.1%	2.5%	3.4%	10.4%	12.6%	7.6%	10.2%	3.8%	5.6%	9.7%	11.2%	5.6%	23.4%	25.0%	31.1%	0.0%	0.0%	0.0%
Convertible Senior Notes	14.0%	0.0%	0.0%	0.0%	0.0%	0.0%	23.4%	25.0%	31.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2.8%
Long-term Credit Facility	0.0%	0.0%	0.0%	0.0%	0.0%	3.8%	0.0%	0.0%	0.0%	0.0%	0.0%	2.8%	0.0%	0.0%	0.0%	0.0%	0.0%	2.8%
Long-term Operating Leases	0.8%	0.0%	0.0%	0.6%	0.4%	0.6%	1.3%	0.0%	0.0%	0.6%	0.3%	0.4%	1.3%	0.0%	0.0%	0.6%	0.3%	0.4%
Contingent Liabilities	5.0%	5.8%	6.3%	0.6%	0.6%	0.6%	8.3%	8.6%	10.5%	0.6%	0.5%	0.4%	0.0%	0.5%	0.8%	1.2%	1.1%	1.1%
Deferred Credit	0.0%	0.3%	0.5%	1.3%	1.3%	1.5%	0.0%	0.5%	0.8%	1.2%	1.1%	1.1%	0.0%	0.5%	0.8%	1.2%	1.1%	1.1%
Deferred Income Taxes	4.5%	5.8%	6.3%	0.6%	0.6%	0.6%	7.5%	8.6%	10.5%	0.6%	0.5%	0.4%	7.5%	8.6%	10.5%	0.6%	0.5%	0.4%
Other LT Liabilities	0.3%	0.3%	0.5%	1.3%	1.3%	1.5%	0.6%	0.5%	0.8%	1.2%	1.1%	1.1%	0.6%	0.5%	0.8%	1.2%	1.1%	1.1%
Total long-term liabilities	31%	27%	30%	14%	16%	14%	51%	39%	49%	13%	14%	10%	51%	39%	49%	13%	14%	10%
Total Liabilities	39%	36%	38%	28%	28%	28%	65%	53%	63%	26%	25%	20%	65%	53%	63%	26%	25%	20%
Preferred stock	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.9%	1.1%	1.7%	1.9%	1.8%
Common stock	0.5%	0.6%	0.6%	1.8%	2.1%	2.4%	0.8%	0.9%	1.1%	1.7%	1.9%	1.8%	38.9%	40.8%	50.4%	3.9%	3.9%	4.1%
Additional Paid-In Capital	23.4%	27.5%	30.3%	4.2%	4.3%	5.5%	38.9%	40.8%	50.4%	3.9%	3.9%	4.1%	64.5%	56.6%	54.9%	66.5%	63.6%	52.9%
Retained Earnings	38.8%	38.3%	33.1%	71.7%	71.5%	71.6%	64.5%	56.6%	54.9%	66.5%	63.6%	52.9%	0.0%	0.0%	0.1%	0.1%	-0.1%	0.0%
Accumulated Other Income (Loss)	0.0%	0.0%	0.0%	0.1%	-0.1%	-0.1%	0.0%	0.0%	0.1%	0.1%	-0.1%	0.0%	0.0%	0.0%	0.1%	0.1%	-0.1%	0.0%
Treasury Stock	-1.5%	-1.9%	-2.1%	-6.3%	-6.3%	-7.0%	-2.5%	-2.8%	-3.6%	-5.8%	-5.6%	-5.2%	-2.5%	-2.8%	-3.6%	-5.8%	-5.6%	-5.2%
Total MGP shareholders' equity	61%	64%	62%	72%	72%	72%	102%	95%	103%	66%	64%	54%	102%	95%	103%	66%	64%	54%
Total liabilities and shareholders' equity	100%	100%	100%	100%	100%	100%	166%	148%	166%	93%	89%	74%	166%	148%	166%	93%	89%	74%

MGP Ingredients, Inc.							Common-Size Income Statement (against Total Assets)						Common-Size Income Statement (against Sales)					
	2023	2022	2021	2020	2019	2018	2023	2022	2021	2020	2019	2018	2023	2022	2021	2020	2019	2018
Sales	60%	68%	60%	108%	112%	135%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Cost of Sales	38%	46%	41%	81%	89%	105%	64%	68%	68%	47%	79%	78%	64%	68%	68%	47%	79%	78%
Gross Profit	22%	22%	19%	27%	24%	30%	36%	32%	32%	16%	21%	22%	36%	32%	32%	16%	21%	22%
Operating, General & Adm. Expenses	9%	9%	9%	12%	9%	12%	15%	13%	14%	7%	8%	9%	15%	13%	14%	7%	8%	9%
Net Other Costs and (Gains)/Losses	2%	0%	-2%	0%	0%	0%	3%	0%	-3%	0%	0%	0%	3%	0%	-3%	0%	0%	0%
EBIT or Operating Income (Loss)	11%	13%	12%	15%	15%	18%	18%	19%	20%	9%	13%	13%	18%	19%	20%	9%	13%	13%
Gain/Loss on Sale of Equity Investment	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Interest Expense	0%	-1%	-1%	0%	0%	0%	-1%	-1%	-1%	0%	0%	0%	-1%	-1%	-1%	0%	0%	0%
Equity Method Investment Loss	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
EBT (Loss)	10%	12%	12%	14%	14%	18%	17%	18%	19%	8%	13%	13%	17%	18%	19%	8%	13%	13%
Income Taxes (Provision)	2%	3%	3%	3%	2%	4%	4%	4%	5%	2%	2%	3%	4%	4%	5%	2%	2%	3%
Net Income (Loss)	8%	9%	9%	11%	12%	13%	13%	14%	14%	6%	11%	10%	13%	14%	14%	6%	11%	10%
Net Loss Attributable to Noncontrolling Interest	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Net Income Attributable to MGP Ingredients, Inc.	8%	9%	9%	11%	12%	13%	13%	14%	15%	10%	11%	10%	13%	14%	15%	10%	11%	10%
Discontinued Operations and Adj for Securities	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Net Income (Loss)	8%	9%	9%	11%	12%	13%	13%	14%	14%	6%	11%	10%	13%	14%	14%	6%	11%	10%

Trend Analysis

Trend analysis is an evaluation of the firm's financial performance over time. A trend analysis can include the evaluation of ratios (such as NPM, TATO, ROA and ROE) and the aggregate balances of important accounts taken from the financial statements (such as Total Sales, COGS, Net Income, Current Assets, Total Assets, Total Debt, Total Equity, etc.). A trend analysis might typically measure three growth rates using the following formulas:

1.) A year-over-year (**YOY**) rate, which is the simple increase or decrease from the previous year,

$$\text{Year-over-Year Rate (YOY)} = \frac{\text{Ending Balance} - \text{Beginning Balance}}{\text{Beginning Balance}}$$

2.) A "holding period" (**HP**) rate which is the gross increase or decrease from the beginning year (the basis year) to the end of the holding period (i.e., 3 years, 5 years, etc.) and

$$\text{Holding Period Rate (HP)} = \frac{\text{Balance in Last Year of HP} - \text{Balance in First Year of HP}}{\text{Balance in First Year of HP}}$$

3.) A compounded annual growth rate (**CAGR**) which is the annual compounded rate of growth that increases (or decreases) the number from the beginning year (the basis year) to the end year.

$$\text{Compounded Annual Growth Rate (CAGR)} = \left[1 + \frac{\text{Balance in Last Year of HP} - \text{Balance in First Year of HP}}{\text{Balance in First Year of HP}} \right]^{1/N} - 1$$

On the following page you will find an example of a trend analysis on MGPI that evaluates the growth in major accounts from the Balance Sheet and Income Statement from 2018 to 2023.

Note that an annual Year-over-Year (**YOY**) growth rate is calculated for each year from 2019 thru 2023 (the YOY rate in 2018 is not available because it requires data from 2017). Also, two types of 5-year growth rates are calculated (which cover the five years from 2018 to 2023). One is labeled the 5-year Holding Period Growth Rate (**HPGR**), and the other is labeled the 5-year Compounded Annual Growth Rate (**CAGR**). Recall that a spreadsheet exhibit of MGPI's 2023 Balance Sheet and 2023 Statement of Income is found on page 16 of this text.

Note: Students should recall from their course in Statistics that a "five-year" growth rate (regardless of whether it is the HP growth rate or the CAGR) requires six years of data, which is exactly what is contained on the schedule opposite this page (2023, 2022, 2021, 2020, 2019, and 2018). Note that 2018 to 2019 marks one year of growth (YOY), followed by four subsequent periods of one-year (YOY) growth rates. Thus, with 2018 as the base year, there are five subsequent years of growth through 2023 (even though the data set has the end-of-year data from six total calendar years).

MGP Ingredients, Inc. (MGPI)	2023-2018		Balance Sheet																	
	5-Yr	5-Yr																		
	CAGR	HPGR	YOY	2023	%TA	YOY	2022	%TA	YOY	2021	%TA	YOY	2020	%TA	YOY	2019	%TA	2018	%TA	
Assets																				
Cash & Cash Equivalents	30%	266%	-62%	\$ 18,388	1%	122%	\$ 47,889	4%	0%	\$ 21,568	2%	555%	\$ 21,662	6%	-34%	\$ 3,309	1%	\$ 5,025	2%	
Accounts Receivable (less allowances)	30%	272%	32%	144,286	10%	18%	109,267	9%	62%	92,537	9%	39%	56,966	16%	6%	40,931	13%	38,797	14%	
Inventories	26%	219%	20%	346,853	25%	18%	289,722	25%	74%	245,944	24%	3%	141,011	38%	26%	136,931	42%	108,769	39%	
Prepaid Expenses	22%	171%	21%	3,580	0%	96%	2,957	0%	-43%	1,510	0%	29%	2,644	1%	55%	2,048	1%	1,320	0%	
Refundable Income Taxes	n/a	n/a	n/a	1,190	0%	n/a	4,327	0%	n/a	5,539	1%	n/a	0	0%	n/a	987	0%	712	0%	
Total current assets	27%	233%	13%	514,297	37%	24%	454,162	39%	65%	367,098	35%	21%	222,283	61%	19%	184,206	57%	154,623	56%	
Property, Plant, Equipment (Net)	17%	117%	11%	262,303	19%	14%	235,632	20%	57%	207,286	20%	3%	131,992	36%	6%	128,419	40%	120,788	43%	
Investment in Joint Ventures/Equity Investments	n/a	n/a	-6%	5,197	0%	n/a	5,534	0%	n/a	4,944	0%	n/a	0	0%	n/a	0	0%	0	0%	
Operating lease right-of-use assets, net	n/a	n/a	-7%	13,975	1%	56%	15,042	1%	88%	9,671	0%	n/a	5,151	0%	n/a	6,490	0%	0	0%	
Intangible assets, net	n/a	n/a	25%	271,706	20%	n/a	216,768	19%	n/a	218,838	0%	n/a	890	0%	n/a	0	0%	0	0%	
Goodwill	n/a	n/a	42%	321,544	23%	n/a	226,294	20%	n/a	226,294	0%	n/a	2,738	0%	n/a	0	0%	0	0%	
Other LT Assets	6%	34%	-30%	3,326	0%	-35%	4,779	0%	108%	7,336	1%	1%	3,521	1%	40%	3,482	1%	2,481	1%	
Total assets	38%	401%	20%	\$1,392,348	100%	11%	\$1,158,211	100%	184%	\$1,041,467	100%	14%	\$366,575	100%	16%	\$322,597	100%	\$277,892	100%	
Liabilities and Shareholders' Equity																				
Current Maturity of Long-Term Debt	75%	1558%	14%	\$ 6,400	0%	74%	\$ 5,600	0%	102%	\$ 3,227	0%	299%	\$ 1,600	0%	4%	\$ 401	0%	\$ 386	0%	
Accounts payable	24%	190%	11%	73,594	5%	24%	66,432	6%	77%	53,712	5%	3%	30,273	8%	16%	29,511	9%	25,363	9%	
Income taxes payable	n/a	n/a	-51%	2,251	0%	100%	4,627	0%	0%	6,992	1%	0%	704	0%	0%	0	0%	0	0%	
Accrued Expenses	22%	172%	11%	31,861	2%	15%	28,716	2%	20%	24,869	2%	121%	20,752	6%	-20%	9,383	3%	11,714	4%	
Total current liabilities	25%	205%	8%	114,106	8%	19%	105,375	9%	67%	88,800	9%	36%	53,329	15%	5%	39,295	12%	37,463	13%	
Long-term debt	32%	305%	189%	85,305	6%	-16%	29,510	3%	-8%	35,266	3%	-6%	38,271	10%	93%	40,658	13%	21,040	8%	
Convertible Senior Notes	n/a	n/a	0%	195,544	14%	0%	195,225	0%	0%	194,906	0%	0%	0	0%	0%	0	0%	0	0%	
Long-term Credit Facility	n/a	n/a	n/a	0	0%	n/a	0	0%	#####	0	0%	-100%	0	0%	-100%	1	0%	10,588	4%	
Long-term Operating Leases	n/a	n/a	-3%	11,292	1%	0%	11,622	1%	0%	6,997	0%	0%	3,057	0%	0%	4,267	0%	0	0%	
Contingent Liabilities	n/a	n/a	n/a	69,200	5%	n/a	0	0%	n/a	0	0%	n/a	0	0%	n/a	0	0%	0	100%	
Deferred Credit	n/a	n/a	n/a	0	0%	n/a	0	0%	n/a	0	0%	78%	2,196	1%	-21%	1,233	0%	1,565	1%	
Deferred Income Taxes	n/a	n/a	-6%	63,071	5%	2%	67,112	6%	2776%	66,101	6%	19%	2,298	1%	15%	1,929	1%	1,677	1%	
Other LT Liabilities	3%	16%	28%	4,763	0%	-27%	3,723	0%	5%	5,132	0%	17%	4,898	1%	1%	4,170	1%	4,118	1%	
Total long-term liabilities	62%	1001%	40%	429,175	31%	0%	307,192	27%	508%	308,402	30%	-3%	50,720	14%	34%	52,258	16%	38,988	14%	
Total Liabilities	48%	611%	32%	\$ 543,281	39%	4%	\$ 412,567	36%	282%	\$ 397,202	38%	14%	\$104,049	28%	20%	\$ 91,553	28%	\$ 76,451	28%	
Preferred stock	0%	0%	0%	4	0%	0%	4	0%	0%	4	0%	0%	4	0%	0%	4	0%	4	0%	
Common stock	0%	0%	0%	6,715	0%	0%	6,715	1%	0%	6,715	1%	0%	6,715	2%	0%	6,715	2%	6,715	2%	
Additional Paid-In Capital	84%	2017%	2%	325,453	23%	1%	318,839	28%	1937%	315,802	30%	11%	15,503	4%	-9%	14,029	4%	15,375	6%	
Retained Earnings	22%	171%	22%	539,883	39%	29%	443,061	38%	31%	344,237	33%	14%	262,943	72%	16%	230,784	72%	198,914	72%	
Accumulated Other Income (Loss)	n/a	n/a	31%	(397)	0%	-186%	(304)	0%	-27%	354	0%	-298%	486	0%	50%	(246)	0%	(164)	0%	
Treasury Stock	2%	9%	-2%	(21,166)	-2%	-3%	(21,591)	-2%	-3%	(22,357)	-2%	14%	(23,125)	-6%	4%	(20,242)	-6%	(19,403)	-7%	
Total MGP shareholders' equity	33%	322%	14%	\$ 850,492	61%	16%	\$ 746,724	64%	146%	\$ 644,755	62%	14%	\$262,526	72%	15%	\$231,044	72%	\$201,441	72%	
Noncontrolling interest	n/a	n/a	n/a	(1,425)	0%	n/a	(1,080)	0%	n/a	(490)	0%	n/a	0	0%	n/a	0	0%	0	0%	
Total equity	33%	321%		849,067			745,644			644,265			262,526			231,044		201,441		
Total liabilities and shareholders' equity	38%	401%	20%	\$1,392,348	100%	11%	\$1,158,211	100%	184%	\$1,041,467	100%	14%	\$366,575	100%	16%	\$322,597	100%	\$277,892	100%	

MGP Ingredients, Inc. (MGPI)	2023-2018		Statement of Income																
	5-Yr	5-Yr																	
	CAGR	HPGR	YOY	2023	%TS	YOY	2022	%TS	YOY	2021	%TS	YOY	2020	%TS	YOY	2019	%TS	2018	%TS
Sales	17%	122%	7%	\$ 836,523	100%	25%	\$ 782,358	100%	58%	\$ 626,720	100%	9%	\$395,521	100%	-4%	\$362,745	100%	\$376,089	100%
Cost of Sales	13%	82%	1%	531,811	64%	24%	529,052	68%	44%	427,755	68%	4%	296,715	47%	-2%	286,213	78%	292,490	78%
Gross Profit	30%	264%	20%	304,712	36%	27%	253,306	32%	101%	198,965	32%	29%	98,806	16%	-8%	76,532	21%	83,599	22%
Operating, General & Adm. Expenses	31%	287%	24%	129,608	15%	17%	104,341	13%	100%	88,927	14%	52%	44,565	7%	-12%	29,290	8%	33,451	9%
Net Other Costs and (Gains)/Losses	n/a	n/a	n/a	26,491	3%	n/a	0	0%	n/a	(16,325)	-3%	n/a	0	0%	n/a	0	0%	0	0%
EBIT or Operating Income (Loss)*	24%	196%	0%	148,613	18%	18%	148,965	19%	133%	126,363	20%	15%	54,241	9%	-6%	47,242	13%	50,148	13%
Gain on Sale of Equity Investment	n/a	n/a	n/a	0	0%	n/a	0	0%	n/a	0	0%	n/a	0	0%	n/a	0	0%	0	0%
Interest Income (Expense)	42%	469%	-24%	(6,647)	-1%	67%	(8,793)	-1%	221%	(5,267)	-1%	26%	(1,640)	0%	12%	(1,305)	0%	(1,168)	0%
Equity Method Investment Gain (Loss)	n/a	n/a	n/a	(220)	0%	n/a	0	0%	n/a	0	0%	n/a	0	0%	n/a	0	0%	0	0%
EBT (Loss)*	24%	189%	1%	141,746	17%	16%	140,172	18%	130%	121,096	19%	15%	52,601	8%	-6%	45,937	13%	48,980	13%
Income Taxes (Provision)	24%	196%	11%	34,616	4%	3%	31,300	4%	147%	30,279	5%	72%	12,256	2%	-39%	7,144	2%	11,696	3%
Net Income (Loss)	24%	187%	-2%	\$ 107,130	13%	20%	\$ 108,872	14%	125%	\$ 90,817	14%	4%	\$ 40,345	6%	4%	\$ 38,793	11%	\$ 37,284	10%
Net Loss Attributable to Noncontrolling Interest	n/a	n/a	0%	345	0%	0%	590	0%	0%	490	0%	0%	0	0%	0%	0	0%	0	0%
Net Income Attributable to MGP Ingredients, Inc.	n/a	n/a	-2%	\$ 107,475	13%	0%	\$ 109,462	14%	0%	\$ 91,307	15%	0%	\$ 40,345	10%	0%	\$ 38,793	11%	\$ 37,284	10%
Discontinued Operations and Adj for Securities	n/a	n/a	n/a	(1,074)	0%	n/a	(871)	0%	n/a	(712)	0%	n/a	(261)	0%	n/a	(253)	0%	(708)	0%
Net Income used in EPS Calculation	24%	191%	-2%	\$ 106,401	13%	20%	\$ 108,591	14%	126%	\$ 90,595	14%	4%	\$ 40,084	6%	5%	\$ 38,540	11%	\$ 36,576	10%

Topic 3: Financial Forecasting

Using the “% of Sales” method to calculate AFN (“Additional Funds Needed”)

Financial forecasting generally involves managers projecting what future financial statements will look like based on certain assumptions about the near future. The most common activity that triggers the need to project future financial statements is an estimated growth in annual sales. Financial statements that forecast the company’s future financial position are called **Pro Forma Statements**.

Forecasting is an essential part of overall strategic planning, and financial plans generally begin with a sales forecast. If the sales forecast reflects an assumption for growth, this will often trigger corresponding changes across other areas of the firm such as **production costs** (labor, materials, etc.), **operating accounts** (such as Cash, Inventory, A/R, A/P, and A/E), **investing activity** (the Fixed Assets needed to support production), and **financing activity** (the capital accounts needed to finance Fixed Capital and Net Operating Working Capital).

Financial planning for the future involves multiple steps, which include (but may not be limited to):

Step 1: Assumptions are made about future levels of sales, costs, interest rates paid on borrowed funds, production capacity, operating capacity, dividend payouts, etc.

Step 2: A set of projected financial statements is developed (Pro Forma Statements) based on the assumptions generated in Step 1. The projected statements help managers assess changes across all activities of the firm based on certain key assumptions.

Step 3: Projected ratios, such as those discussed in Topic 2 (i.e., Cash Conversion Cycle, factors of DuPont analysis, etc.), are calculated and analyzed for potential changes caused by the growth.

Step 4: The entire plan is examined and re-examined, key assumptions are reviewed and revised, managers assess how additional changes in operations might improve results, and plans are developed and executed.

This process emphasizes the importance of accurate and timely information, and underscores the need for managers to anticipate changes triggered by important future events such as sales growth.

The Percent of Sales Method

A common method used to produce Pro Forma Financial Statements for the purpose of financial forecasting is called the **Percent of Sales Method** which tracks certain Income Statement items and most Balance Sheet accounts as a percentage of Sales. These line items and accounts may remain a constant % of sales over time, or they may change based on deliberate decisions made by managers. For instance, a company might assume that Cash will always be 5% of Sales, A/R will always be 10% of Sales, Inventory will always be 15% of Sales, and so on. On the other hand, a company might decide to specifically reduce Inventory as a % of Sales or to specifically decrease A/R and increase A/P each as a % of Sales, and so on. Recall our discussion of Working Capital Management and the various policies (e.g. payment terms, inventory delivery schedules, etc.) that will affect the Cash Conversion Cycle. Implementing new policies to reduce the Cash Conversion Cycle will affect the level of Working Capital Accounts as a % of Sales, so forecasting must take into account any changes that are the result of deliberate management decisions.

Ultimately, the Percent of Sales Method is used to determine how much new **Financial Capital** is required to “finance” a projected increase in Sales. Recall that Financial Capital (red area) is the external funds needed by a firm to finance the combined total of Net Operating Working Capital (green area) and Fixed Capital (black area). As noted above, an increase in Sales is almost certain to trigger an increase in working capital assets (Cash, A/R, and Inventory). This alone may require additional financing. In addition, sales growth may only be possible with an increase in PP&E (because additional plant and equipment is necessary for the increased production), which will require new capital.

Increase in Sales triggers an Increase in Assets

Financing is required when an increase in Sales triggers an increase in Total Assets (because an increase in Total Assets requires an equal increase in Total Liabilities & Equity). It is important to note that an increase in sales will usually trigger an automatic (**spontaneous**) increase in Working Capital Assets, but not necessarily an automatic increase in PP&E (Fixed Assets). The level of PP&E depends on such issues as the available capacity of existing equipment and the targeted ratio of Fixed Assets to Sales. In any case, the key principle is this: an increase in sales will usually trigger a necessary increase in assets. It is the finance manager's task to anticipate this increase of assets and plan for the corresponding (and necessary) increase in financing accounts (liabilities and equity). Financing will be available both from "internal" and "external" sources, as described next.

First source of Financing (Internal): Spontaneous Liabilities and Retained Earnings

The amount of new "Financial Capital" required for an increase in Sales depends first on how much financing is provided by the two internal sources: Spontaneous Current Liabilities and Retained Earnings.

Spontaneous Accounts are those accounts that increase spontaneously as the firm expands (i.e. as when the firm experience a growth in Sales). Typically, a firm assumes that all Net Operating Working Capital Accounts (Cash, Accounts Receivable, Inventory, Accounts Payable, and Accrued Expenses) are spontaneous. Three of these accounts are Assets (Cash, A/R, and Inventory), but two are liabilities. Thus, an increase in the spontaneous liabilities (A/P and A/E) is the first source of financing for the increase in spontaneous assets.

The next available source of financing available to the firm is **Retained Earnings**. Like spontaneous liabilities, Retained Earnings is considered "internal financing" because it is comprised of **profits** that are "retained" by the firm. Financing available from Retained Earnings is determined by first producing a Pro Forma Income Statement based on the increase in Sales and appropriate assumptions for expenses. Net Income from the Pro Forma Income Statement is then factored by the firm's Retention Ratio (sometimes called the Plowback Ratio) to determine the amount of Retained Earnings that will be available.

Note: The Retention or Plowback Ratio is the % of Net Income retained by the firm (added to Retained Earnings) and is the inverse of the Payout Ratio, which measures the % of Net Income paid to stockholders as a dividend.

Second source of Financing: Financial Capital

Once financing available from spontaneous liabilities and Retained Earnings has been calculated, any capital still needed to finance an increase in assets is referred to as Additional Funds Needed (AFN) and must be provided by one or more the Financial Capital accounts (ST Debt, LT Debt, and Equity/Common Stock). The decision on how to finance AFN is an important strategic matter for the firm and will depend on many factors. Nonetheless, assumptions are needed to round out the forecasting process and balance the accounts.

The term Additional Funds Needed (AFN) was developed to show how much additional capital a firm needs to support its planned growth. However, a firm may be able to grow without any additional outside capital. In fact, this was the case in the U.S. manufacturing industry for many years. A firm may actually generate excess financial capital (primarily through high Profits/Net Income and a high Retention Ratio) that can be used to retire debt, repurchase stock, increase the dividend, expand into different markets, etc. If the increase in Retained Earnings is high enough when forecasting sales growth, the AFN calculated from the formula below will be negative (which translates into excess financial capital).

The AFN Equation:

AFN = (Increase in Total Assets) less (Increase in Spontaneous Liabilities) less (Increase in R.E.)

or

AFN = (↑Cash + ↑A/R + ↑Inventory + ↑Net PP&E if applicable) less (↑A/P + ↑A/L) – (↑R.E.)

Calculating Additional Funds Needed (AFN) due to an increase in Sales – Two Methods:

Assumption #1: Determine a projected increase in sales (e.g., 15%).

Assumption #2: Assume that all Net Operating Working Capital accounts (both assets and liabilities) remain a constant percent of sales. Thus, the Net Operating Working Capital account balances (Cash, A/R, Inventory, A/P and A/E) **spontaneously** increase at the same percentage as sales increases (e.g., 15%).

Assumption #3: Determine whether PP&E will increase or remain unchanged based on the forecasted sales increase. **Note: Recall that Net PP&E (or Net Fixed Assets) do not spontaneously increase as sales increase, but are affected by capacity and/or other factors.**

Method 1: The AFN Equation (see previous page)

$$\text{AFN} = (\text{Increase in Total Assets}) \text{ less } (\text{Increase in Spontaneous Liabilities}) \text{ less } (\text{Increase in R.E.})$$

Method 2: The Two-Pass Pro Forma Balance Sheet

The manual (step-by-step) method of calculating AFN requires the following steps:

Step 1: Produce a Pro Forma Income Statement based on the projected increase in Sales from Assumption #1. (Note: In addition to an assumption on the increase of sales, you need assumptions for all appropriate expenses.)

Step 2: Calculate the projected Retained Earnings available to the firm during the Pro Forma year. This requires multiplying Net Income from Step 1 by the firm's Retention Ratio.

Step 3: Produce the "First Pass" of a Pro Form Balance Sheet:

- a. Adjust all "spontaneous" Net Operating Working Capital accounts to equal the same percentage as sales increased on the Pro Forma Income Statement (Assumption #1),
- b. Adjust Net PP&E based on the decision in Assumption #2. (Note: you could decide that Net Fixed Assets will not increase at all or will decrease through accumulated depreciation).
- c. Increase Retained Earnings by the amount calculated in Step 2.
- d. Calculate the sum of Total Assets and the sum of Total Liabilities & Equity.

If the sum of Total Assets exceeds the sum of Total Liabilities & Equity, the difference equals AFN. This amount should be identical to the amount (a positive number) produced by using the simple AFN Equation at the top of the page. If the sum of Total Liabilities & Equity exceeds the sum of Total Assets, it means the firm has excess financing and no AFN is needed (i.e., the amount calculated in the AFN Equation is negative).

Step 4: Produce the "Second Pass" of the Pro Forma Balance Sheet in which Total Assets and Total Liabilities/Equity balance (are equal to each other):

- a. Calculate the amount of new capital required in each of the Financial Capital accounts (ST Debt, LT Debt, Common Stock) based on the assumed distribution. **Note: The amount of ST Debt, LT Debt, and/or Common Stock that is used to cover AFN is an important strategic decision made by the firm's directors and management, and will be based on factors beyond our scope in this Topic.**
- b. Increase the Financial Capital accounts by the amount needed to cover AFN and recalculate the Balance Sheet so that the deficit from the First Pass is eliminated and the two sides balance.

% of Sales Method to determine AFN

Key Assumptions

Sales growth in 2014	15%
COGS as % of sales in 2014	50%
SG&A as % of sales in 2014	20%
Increase in Net Fixed Assets in 2014	\$3,500,000
Depreciation Expense in 2014	\$4,000,000
Tax rate in 2014	40%
Interest Expense in 2014	\$600,000
Dividend Payout Ratio in 2014	75%
Retention Ratio in 2014	25%

AFN to be raised as:

ST Debt	15%
LT Debt	35%
New Stock	50%

Income Statement	Actual	Pro Forma
	2013	2014
Sales	\$35,000,000	\$40,250,000
<u>Cost of Goods Sold (COGS)</u>	<u>17,500,000</u>	<u>20,125,000</u>
Gross Profit	17,500,000	20,125,000
<u>S,G & A</u>	<u>7,000,000</u>	<u>8,050,000</u>
EBITDA	10,500,000	12,075,000
<u>Depreciation Expense</u>	<u>4,000,000</u>	<u>4,000,000</u>
EBIT	6,500,000	8,075,000
<u>Interest Expense</u>	<u>540,000</u>	<u>600,000</u>
EBT	5,960,000	7,475,000
<u>Taxes</u>	<u>2,384,000</u>	<u>2,990,000</u>
Net Income	<u>3,576,000</u>	<u>4,485,000</u>
Common dividends	2,682,000	3,363,750
Addition to retained earnings	894,000	1,121,250

Balance Sheet	Pro Forma 2014				
	2013	Growth	1st Pass	AFN	2nd Pass
Assets:					
Cash	\$4,950,000	15%	\$5,692,500		\$5,692,500
Accounts Receivable	\$3,850,000	15%	\$4,427,500		\$4,427,500
Inventory	\$7,150,000	15%	\$8,222,500		\$8,222,500
Total current assets	\$15,950,000		\$18,342,500		\$18,342,500
Net PP&E	\$17,000,000		\$20,500,000		\$20,500,000
Total assets	\$32,950,000		\$38,842,500		\$38,842,500
Liabilities and equity:					
Accounts payable	\$3,850,000	15%	\$4,427,500		\$4,427,500
Accrued Expenses	\$1,375,000	15%	\$1,581,250		\$1,581,250
ST Debt	\$1,023,492		\$1,023,492	\$598,125	\$1,621,617
Total current liabilities	\$6,248,492		\$7,032,242		\$7,630,367
LT Debt	\$7,467,460		\$7,467,460	\$1,395,625	\$8,863,085
Total liabilities	\$13,715,952		\$14,499,702		\$16,493,452
Common stock	\$1,093,968		\$1,093,968	\$1,993,750	\$3,087,718
Retained Earnings	\$18,140,080		\$19,261,330		\$19,261,330
Total common equity	\$19,234,048		\$20,355,298		\$22,349,048
Total liabilities and equity	\$32,950,000		\$34,855,000		\$38,842,500
AFN			\$3,987,500		
AFN Equation:					
Increase in Total Assets			\$5,892,500		
Less: Increase in Spontaneous liabilities			\$783,750		
Less: Increase in Retained Earnings			\$1,121,250		
= Additional Funds Needed			\$3,987,500		
ST Debt			\$598,125	15% of AFN	
LT Debt			\$1,395,625	35% of AFN	
Common Stock			\$1,993,750	50% of AFN	

Topic 4: Working Capital Management

Working capital management seeks optimal levels for current assets (Cash, Accounts Receivable, and Inventory) that are critical to operations. Effective working capital management is mainly focused on the management of cash inflows and outflows. It is a continual “balancing act” because it involves the timing of customer payments (Accounts Receivable), the timing of payments to vendors (Accounts Payable) and the relative speed in the turnover of Inventory. It is of strategic importance because it involves choosing appropriate credit terms for both customers (from sales) and vendors (from purchases), and managing Inventory most effectively without any detriment to production or sales. First, a review of terms that the student has already learned:

Current assets = cash and other liquid assets that convert into cash as a matter of course.

Current liabilities = obligations that must be paid within one year (this includes ST Debt)

Current ratio = current assets ÷ current liabilities

Working capital = current assets (although it is also used frequently to imprecisely mean Net Working Capital)

***Net working capital** = (current assets) minus (current liabilities)

***Net operating working capital** = (current assets) minus (current liabilities generated internally from normal operations such as Accounts Payable & Accrued Expenses)

*Note that the only difference between “net working capital” and “net operating working capital” is the inclusion or exclusion of ST Debt in the calculation. Students should note that Current Maturities of Long-Term Debt is treated the same as ST Debt for purposes of calculating the NOWC. See the example below in Exhibit 13:

Exhibit 13

Ravencorp - Year-End Balance Sheet			
<u>ASSETS</u>		<u>LIABILITIES</u>	
Cash	\$1,000,000	Accounts payable	\$3,000,000
Accounts Receivable	\$2,500,000	Accrued expenses	\$1,500,000
Inventory	\$4,000,000	Short-Term Debt	\$1,500,000
Total current assets	\$7,500,000	Total current liabilities	\$6,000,000
Net Fixed Assets	\$16,000,000	LT Debt	\$9,000,000
		Total liabilities	\$15,000,000
		<u>STOCKHOLDERS' EQUITY</u>	
		Preferred Stock	\$500,000
		Common stock	\$1,500,000
		Retained Earnings	\$6,500,000
		Total common equity	\$8,500,000
Total assets	\$23,500,000	Total liabilities and equity	\$23,500,000

Total current assets = \$7,500,000

Total current liabilities = \$6,000,000

Total current liabilities from normal operations = \$4,500,000 (current liabilities less ST Debt)

Question: What is **Working Capital**?

Answer: **\$7,500,000** (current assets)

Question: What is the **Net Working Capital**?

Answer: **\$1,500,000** = \$7,500,000 (current assets) less \$6,000,000 (current liabilities)

Question: What is **Net Operating Working Capital**?

Answer: **\$3,000,000** = \$7,500,000 (current assets) less \$4,500,000 (current liabilities from normal operations)

Cash Conversion Cycle

Cash conversion cycle = the length of time (days) between actual cash expenditures used to pay for resources (i.e. materials used in production) and actual cash receipts from the sale of products or services. The CCC measures the time between cash outflows (payments) and cash inflows (receipts) and can be calculated from balances on the financial statements (see the formula page for Ratio Analysis above).

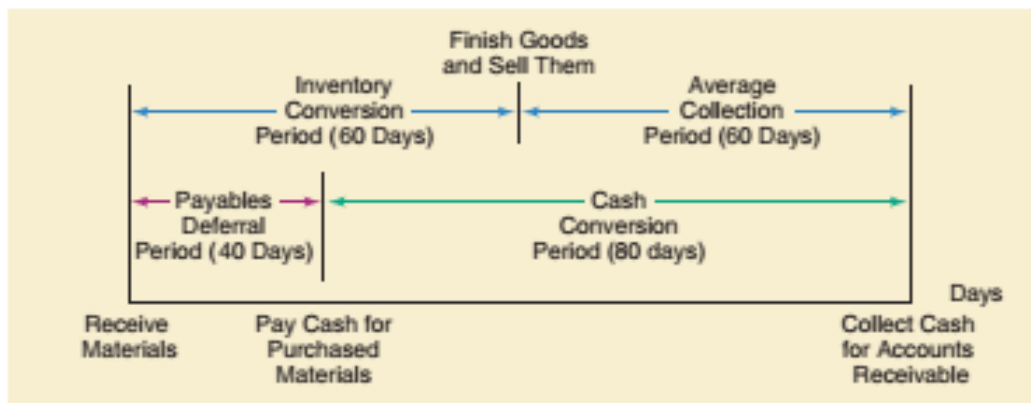
CASH CONVERSION CYCLE	=	INVENTORY CONVERSION PERIOD	+	RECEIVABLES COLLECTION PERIOD	-	PAYABLES DEFERRAL PERIOD
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Inventory Conversion Period (# of days to convert inventory into sales) = $\frac{\text{Inventory}}{\text{Avg. Purchases per Day}}$ or $\frac{\text{Inventory}}{\text{COGS} \div 365}$

Receivables Collection Period (# of days to convert sales into cash) = $\frac{\text{Accounts Receivable}}{\text{Avg. Sales per Day}}$ or $\frac{\text{Accounts Receivable}}{\text{Annual Sales} \div 365}$

Note: This number is also called “Days Sales Outstanding” (DSO)

Payables Deferral Period (# days until purchases must be paid for) = $\frac{\text{Accounts Payable}}{\text{Avg. Purchases per Day}}$ or $\frac{\text{Accounts Payable}}{\text{COGS} \div 365}$



Company	ICP	RCP	PDP	CCC
Apple	10.79	28.10	106.75	-67.86
IBM	20.35	39.44	81.89	-22.10
Microsoft	13.85	83.86	123.29	-25.58
Walmart	44.53	4.74	43.08	6.19
Target	59.92	3.91	59.87	3.96
Costco	28.59	3.44	30.02	2.01
Lowe's	104.38	0.00	65.93	38.45
Home Depot	86.82	7.69	39.92	54.59
Caterpillar	122.24	106.72	81.06	147.90
John Deere	85.12	344.77	58.72	371.17
McDonalds	1.90	33.30	35.87	-0.67
Wendy's	1.89	17.17	10.93	8.13
Chipotle	1.98	4.52	10.25	-3.75
Cracker Barrel	35.40	3.88	43.16	-3.88
Denny's	6.42	20.45	23.07	3.80
Pepsi	46.97	42.94	210.23	-120.32
Coca-Cola	85.84	29.60	343.75	-228.31
American Airlines	20.83	15.94	19.64	17.13
Southwest Airlines	15.13	15.94	38.37	-7.30
Gap	85.01	7.95	48.75	44.21
Abercrombie & Fitch	115.84	10.32	62.98	63.18
American Eagle	65.82	17.73	27.59	55.96

“Making your Working Capital work for you” by Deloitte LLP

Most businesses have formal accounts receivable policies that dictate when to bill, how much to bill and when to collect. Unfortunately, not all businesses enforce those policies effectively – or even adopt the right processes at all. In many cases, it comes down to culture. Businesses that prioritize sales often fall into the trap of extending credit to customers, offering discounts or ignoring payment terms if it means winning new sales. However, if management does not have a focus on working capital, no one will. The upshot? You end up unintentionally providing customers with free financing.

Some may argue this is no big deal, but the truth isn't so simple. If a company needs to borrow money to meet its obligations because customers are paying late, it could incur losses on the financing charges alone. Even if that's not the case, carrying overdue accounts receivable still has a cost. It puts you on a cash flow tightrope. Rather than having free capital to invest in growth opportunities, increase shareholder payouts, buy new equipment or introduce new products, your money is tied up on your balance sheet.

Common risks

While no company intends to adopt weak accounts receivable policies, lack of planning, poor enforcement or a failure to focus on the function can result in unintended consequences. These often arise when companies:

- Fail to follow up with customers in a timely manner when payments are past due
- Allow sales reps to override credit limits – and end up suffering losses from bad credit risks
- Neglect to provide staff with appropriate training on how to deal with late paying customers
- Don't pay sufficient attention to the accuracy of their bills, invoices or credit terms
- Allocate cash payments incorrectly, making it harder to figure out which payments are outstanding

Adopting best practices

Gaining access to this cash takes more than introducing new processes. It also requires the finance and sales departments to work together to develop payment terms that make sense for both the business and its customers. While there are no hard-and-fast rules for achieving this end-state, certain best practices do exist. For example, businesses can realize cost efficiencies by centralizing accounts receivable processing, perhaps through a center of excellence which develops and enforces common practices and standards. Similarly, by automating processes, you can eliminate manual data entry errors and reduce transaction times.

Adopting key performance indicators (KPI) and defined metrics is also important. For instance, by adding working capital metrics to your standard revenue and profit tracking reports, you'll get a clear picture on items like days sales outstanding (DSO), the percentage of customers who pay late, the number of invoices or customers passed through your system, unreconciled items or accounts, the monthly percentage of write-offs, collection rates on bad receivables and even collection efforts made. This will position you to track performance on an ongoing basis.

A note on choosing KPIs Rather than relying simply on a high-level metric like DSO, try to identify the metrics that impact DSO. After all, even if you know daily sales are 45 days outstanding, do you know if that's appropriate for your industry? To delve deeper, consider some more granular metrics, such as:

- How many invoices are past due?
- How many have unapproved discounts?
- How often does the sales team override standard terms? • If overrides are frequent, should you change your terms?

Common Invoice Payment Terms

Accounts Receivable and Accounts Payable

Net monthly account	Payment due on last day of the month following the one in which the invoice is dated
PIA	Payment in advance
Net 7	Payment seven days after invoice date
Net 10	Payment ten days after invoice date
Net 30	Payment 30 days after invoice date
Net 60	Payment 60 days after invoice date
Net 90	Payment 90 days after invoice date
EOM	End of month
21 MFI	21st of the month following invoice date
1% 10 Net 30	1% discount if payment received within ten days otherwise payment 30 days after invoice date
COD	Cash on delivery
Cash account	Account conducted on a cash basis, no credit
Letter of credit	A documentary credit confirmed by a bank, often used for export
Bill of exchange	A promise to pay at a later date, usually supported by a bank
CND	Cash next delivery
CBS	Cash before shipment
CIA	Cash in advance
CWO	Cash with order
1MD	Monthly credit payment of a full month's supply
2MD	As above plus an extra calendar month
Contra	Payment from the customer offset against the value of supplies purchased from the customer
Stage payment	Payment of agreed amounts at stage

Cash Budget

The Cash Conversion Cycle is one of the most important components of cash budgeting. Companies need to forecast their cash flows (in and out) in order to determine when they might face deficits that require additional cash (most likely from short-term borrowing) to pay bills, and when they might enjoy a large surplus of cash that can be put to productive use (short-term investing). The primary forecasting tool for this management task is the Cash Budget.

Cash Budget = a schedule showing projected cash inflows (receipts) and outflows (expenditures) over some period in order to predict cash surpluses or deficits relative to the Target Cash Balance. See opposite page.

Target Cash Balance = the desired cash balance that a firm maintains in order to conduct business.

Cash Budgets can be of any length (daily, weekly, monthly), and the typical time-frame depends on the company's complexity and priorities. Daily and weekly cash budgets are more precise while the monthly budget is an effective tool for annual and longer-term planning. The cash budget requires forecasts (assumptions) for such key variables as:

- Monthly Sales
- Collections from Sales (e.g., % collected during the month of sale and % collected in subsequent months, etc.)
- Monthly Purchases of Materials
- Payments for Purchases (e.g., % paid during the month of purchase and % paid in subsequent months, etc.)
- Payments for Other Expenses such as:
 - Labor
 - Leases
 - S, G & A Expenditures
 - Miscellaneous Expenditures
 - Taxes
 - Extraordinary Purchases paid for with Cash (such as New Equipment)

The forecasted monthly payments (NOT purchases) are subtracted from the forecasted monthly collections (NOT sales) to calculate the expected "cash" gain or loss for each month. These monthly gains and losses accumulate over time to produce monthly cash surpluses and deficits. The surpluses signal an opportunity for a productive use of cash (e.g., short-term investments), while the deficits signal a need for short-term borrowing in order to pay for expenses as they come due. The Cash Budget on the opposite pages is based on the following assumptions:

Sales collections schedule

Payments received during month of sale	25%
Payments received one month following sale	50%
Payments received two months following sale	25%

Payment schedule for Purchases of Materials

Payments made during month of purchase	40%
Payments made one month following purchase	60%

Schedule of Other Expenditures

Monthly Labor (Wages)	\$1,000,000
Monthly Lease Payments	\$100,000
Monthly S, G, & A expenditures	\$3,000,000
Monthly Miscellaneous Expenses	\$500,000
Income Tax Payments (end of each quarter)	\$1,500,000
Payment for New Equipment (October only)	\$1,000,000
Cash at start of month of July	\$4,000,000
Minimum monthly cash balance requirement	\$4,000,000

Forecasted Schedule of Sales and Purchases

	<u>Sales</u>	<u>Purchases</u>
May	\$15,000,000	\$4,000,000
June	\$15,000,000	\$7,000,000
July	\$20,000,000	\$20,000,000
August	\$25,000,000	\$15,000,000
September	\$35,000,000	\$15,000,000
October	\$25,000,000	\$5,000,000
November	\$10,000,000	\$1,000,000
December	\$5,000,000	\$1,000,000

Part I - Schedule of Collections and Payments

	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sales	\$15,000,000	\$15,000,000	\$20,000,000	\$25,000,000	\$35,000,000	\$25,000,000	\$10,000,000	\$5,000,000
<u>Collections on Sales</u>								
Payments received during month of sale			\$5,000,000	\$6,250,000	\$8,750,000	\$6,250,000	\$2,500,000	\$1,250,000
Payments received one month following sale			\$7,500,000	\$10,000,000	\$12,500,000	\$17,500,000	\$12,500,000	\$5,000,000
Payments received two months following sale			\$3,750,000	\$3,750,000	\$5,000,000	\$6,250,000	\$8,750,000	\$6,250,000
Total collections			\$16,250,000	\$20,000,000	\$26,250,000	\$30,000,000	\$23,750,000	\$12,500,000
	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Purchases of Materials	\$4,000,000	\$7,000,000	\$20,000,000	\$15,000,000	\$15,000,000	\$5,000,000	\$1,000,000	\$1,000,000
<u>Payments on Purchases of Material</u>								
Payments made during month of purchase			\$8,000,000	\$6,000,000	\$6,000,000	\$2,000,000	\$400,000	\$400,000
Payments made one month following purchase			\$4,200,000	\$12,000,000	\$9,000,000	\$9,000,000	\$3,000,000	\$600,000
Total payments for purchases (Labor and Materials)			\$12,200,000	\$18,000,000	\$15,000,000	\$11,000,000	\$3,400,000	\$1,000,000

Part II - Schedule Inflows and Outflows: Net cash gain or loss for month

<u>Inflows:</u>	Jul	Aug	Sep	Oct	Nov	Dec
Collections	\$16,250,000	\$20,000,000	\$26,250,000	\$30,000,000	\$23,750,000	\$12,500,000
<u>Outflows:</u>						
Payments for Purchases of Materials	\$12,200,000	\$18,000,000	\$15,000,000	\$11,000,000	\$3,400,000	\$1,000,000
Monthly Labor (Wages)	\$1,000,000	\$1,000,000	\$1,000,000	\$1,000,000	\$1,000,000	\$1,000,000
Monthly Lease Payments	\$100,000	\$100,000	\$100,000	\$100,000	\$100,000	\$100,000
Monthly S, G, & A expenditures	\$3,000,000	\$3,000,000	\$3,000,000	\$3,000,000	\$3,000,000	\$3,000,000
Monthly Miscellaneous expenses	\$500,000	\$500,000	\$500,000	\$500,000	\$500,000	\$500,000
Income tax payments	\$0	\$0	\$1,500,000	\$0	\$0	\$1,500,000
Payment for New Equipment	\$0	\$0	\$0	\$1,000,000	\$0	\$0
Total cash expenditures	\$16,800,000	\$22,600,000	\$21,100,000	\$16,600,000	\$8,000,000	\$7,100,000
Net cash gain (loss) during month	-\$550,000	-\$2,600,000	\$5,150,000	\$13,400,000	\$15,750,000	\$5,400,000

Part III - Surplus/deficit schedule

	0	0	0	0	0	0
Cash at start of month	\$4,000,000	\$3,450,000	\$850,000	\$6,000,000	\$19,400,000	\$35,150,000
Net cash gain (loss) during month	-\$550,000	-\$2,600,000	\$5,150,000	\$13,400,000	\$15,750,000	\$5,400,000
Cumulative cash balance	\$3,450,000	\$850,000	\$6,000,000	\$19,400,000	\$35,150,000	\$40,550,000
Minimum cash balance required	\$4,000,000	\$4,000,000	\$4,000,000	\$4,000,000	\$4,000,000	\$4,000,000
Cash surplus or deficit	-\$550,000	-\$3,150,000	\$2,000,000	\$15,400,000	\$31,150,000	\$36,550,000
Maximum Deficit		-\$3,150,000				
Maximum Surplus						\$36,550,000

Note that if the amount of cash on hand does not meet the target cash balance at any given time during a budget cycle (a deficit) the firm may need to acquire short-term financing⁹ or reduce its Cash Conversion Cycle¹⁰ in order to improve the cash position. If the amount of cash on hand exceeds the target cash balance at any given time during a budget cycle (surplus), the firm may want to invest in short-term liquid securities (reported as “cash equivalents”).

⁹ Short-term financing options (listed as ST Debt) for most firms include a Line of Credit (LOC), a Revolving Credit Agreement, and short-term Bank Loans. These may be all grouped under ST Debt on the Balance Sheet.

¹⁰ The cash conversion cycle can be reduced by one, or any combination of, the following:

- reducing the Inventory Conversion Period (e.g. “just-in-time” inventory policy)
- reducing the Receivables Collection Period (e.g. “tighten” credit terms with customers – collect sooner)
- increasing the Payables Deferral Period (e.g. “stretching” terms with suppliers – pay later)

Topics 5: Time Value of Money

Understanding the time value of money (TVM) is not only crucial to effective financial management, it is essential for anyone planning an investment strategy or even a simple household budget. The modern economic world is characterized by (among other things) the unlimited creation of credit (i.e., money borrowed at interest) and the inexorable loss in the purchasing power of money (PPM) over time. Thus, anyone who is involved with money should have some comprehension of the time value of money. This subject is defined by the term itself, and can be summarized as follows: the **Value of Money** is affected by **Time**.

Consider the following:

- A banker who makes loans and other investments from the money deposited in the bank by its customers
- A corporate finance manager whose job includes assessing various alternative sources of funds in terms of their cost (which is determined by the required return of those persons or entities providing the funds)
- A corporate manager who must choose among various alternative investment projects based on their predicted generation of future cash flow
- An investment analyst who evaluates the financial securities that a firm sells to investors and may trade on public capital markets (both debt and equity)
- An individual who is confronted with a host of daily personal and/or household financial problems ranging from personal credit account management to deciding how to finance a car or home purchase

Each of these individuals makes important decisions based on the time value of money. Many people fear that a working knowledge of TVM concepts might be difficult to master. However, the availability of financial calculators and spreadsheet programs (Excel) makes the subject readily accessible and easy to master the basic principles.

Although an understanding of the time value of money is useful in and of itself, it is also necessary for upper division courses in Finance, Accounting, and even Law. For example, some of the topics that require knowledge of TVM principles include:

- Valuation of securities and other assets
- Capital budgeting (the analysis of investment projects)
- The cost of capital
- Working capital (short-term asset and liability) management
- Lease analysis
- Merger and acquisition decisions
- Damages assessed by courts

The following pages introduce the concepts necessary to understand the time value of money and its applications. The analysis assumes that students will use a financial calculator, an Excel spreadsheet, or an application for solving the TVM problems. However, it is important that the student learn not just how to solve a TVM problem, but learns the principles of the time value of money.

Why Does Money Have Time Value?

Let us first discuss why money has time value? Time value of money is the economic principle that **a dollar received today has greater value than a dollar received tomorrow, a week from today, a month from today, a year from today, ten years from today, and so on.** In fact, the disparity between the value of a dollar today and one in the future increases as the future timeline increases. The intuition behind this concept is easy to see with a simple example. Suppose you were given the choice between receiving \$100,000 today or \$100,000 in 10 years. Which option would you rather take? Clearly the first option is more valuable for the following reasons:

Risk – Time raises the specter of uncertainty (i.e., not receiving the money that you have chosen to defer into the future), and uncertainty is risk. Thus, time adds risk, and the longer the time into the future, the higher the risk.

Purchasing Power – Because of the inexorable loss in the purchasing power of money over time, it is all but certain that \$100,000 can be exchanged for more goods and services today than \$100,000 in 10 years.

Opportunity Cost – Money in hand today can be invested now in a productive manner, resulting in a higher value when the investment matures or is liquidated at some future date. In contrast, a dollar received in the future cannot begin becoming productive until it is received. This lost opportunity to use money productively is considered (and aptly named) the opportunity cost of deferring money until a future date.

For these reasons, we can reduce time value of money to a fundamental principle: ***Sooner is better than later!*** In fact, this axiom can be amended to: ***Much sooner is much better than much later!*** As we will see, the value today (known as the “present value”) of a future dollar decreases exponentially as the future becomes more remote.

We can verify this principle with a simple calculation of the value in today’s terms (Present Value or PV) of \$100,000 that is not available until some future date. We will use an interest rate of 10% to calculate the PV.

<u>Future Value</u>	<u>PV if 1 year removed</u>	<u>PV if 2 years removed</u>	<u>PV if 5 years removed</u>	<u>PV if 10 years removed</u>
\$100,000	\$90,909	\$82,645	\$62,092	\$38,554

Let us turn to the two basic techniques used in all time value of money calculations: compounding and discounting.

Compounding and Discounting: The Foundation for All Time Value of Money Problems

All time value of money problems involve one of two fundamental techniques: either compounding or discounting. Compounding and discounting are processes used to compare dollars in our pocket or bank account (or even a hypothetical sum) today versus those same dollars at some time in the future.

Compounding is about moving money forward in time. It is the process of mathematically calculating the future value of an investment made today, or the future value of a series of payments made over time (periodic payments). The most critical feature of future value calculations is the assumption that interest is “compounded” over time. Students should learn the difference between simple and compounded interest. The basic mathematical function used in compounding is multiplication (and simplified by using exponents for time). The variables of a compounding calculation will be discussed below.

Discounting is about moving money backwards in time. It is the process of determining the present value of money to be received in the future (either as a lump sum or in installments often referred to as periodic payments). Present value is determined by applying an annual interest rate (which has many names, e.g., discount rate, opportunity cost, required return, cost of capital, and many others) to the sums of money to be received in the future. The basic mathematical function used in discounting is division (and simplified by using exponents for time). The variables of a discounting calculation will be discussed below. **(Note: The example above of the PV of \$100,000 1 year removed, 2 years removed, 5 years removed, and 10 years removed is an example of discounting)**

Time Value of Money (TVM) calculations have five variables. The standard symbols for these are:

FV	=	Future Value (a \$ amount)	
PV	=	Present Value (a \$ amount)	
PMT*	=	Periodic Cash Payment (a \$ amount)	
N*	=	# of periods (a number)	Note: a “period” can be a year, a half-year, a quarter, a month, etc.
I%* ¹¹	=	interest rate per period (a %)	Note: w/calculators the integer (5%), w/Excel the decimal (.05)

*** These three variables must all be adjusted to the same “periodic value” when doing the TVM calculation.**

Note: TVM problems typically involve solving for one of the five variables based on values for the other four variables. One of the variables may have a value of zero. For example, when solving for the **PMT** of a loan for \$25,000 (**PV = \$25,000**), with an interest rate of 7% (**I% = 7%**) and a term of 5 years (**N = 5**), the future value is equal to zero (**FV = \$0**) because the payments are intended to retire the loan.

Note: Three TVM variables represent a cash amount (FV, PV, PMT). However, each cash amount does not “move” in the same direction in the problem (i.e. some cash flows “out” and some flows “in”). Solving the problem requires distinguishing cash that flows in opposite directions by using a “negative” sign. For example, the PV value must often be input as a negative number (think of it as the amount “given”) while the FV and PMT must be input as positive (think of it as the amount “received”).

FV and PV equations (compounding and discounting):

Future Value of a present value amount: **$FV = PV \times (1 + I\%)^N$** = (“compounding”)

Present Value of a future value amount: **$PV = \frac{FV}{(1 + I\%)^N}$** = (“discounting”)

Perpetuity: A perpetuity is a stream of fixed future cash payments, at fixed periodic intervals (e.g., annual), that is expected to continue forever. The two formulas for the Present Value¹² of a perpetuity are:

$PV = \frac{PMT}{I\%}$ and **$PV = \frac{PMT_1}{I\% - g}$** when the PMT grows at a constant annual growth rate (g). Note the PMT is the “next” payment (PMT₁)

Annuity: An annuity is a stream of fixed future cash payments, at fixed periodic intervals (e.g., annual) for a finite period of time. The three main features of an annuity payment are fixed: the amount, the interval, and the date of termination. It may be referred to as an example of **even future cash flow** as opposed to an example of **uneven future cash flow**.¹³

The Net Present Value of an annuity = the sum of the present values of all payments:¹⁴ **$NPV = \sum \frac{PMT}{(1 + I\%)^N}$**

Year:	0	1	2	3	4	5						
		I-----	I-----	I-----	I-----	I-----	I-----					
Equation:		$\frac{PMT_1}{(1 + I\%)^1}$	$+$	$\frac{PMT_2}{(1 + I\%)^2}$	$+$	$\frac{PMT_3}{(1 + I\%)^3}$	$+$	$\frac{PMT_4}{(1 + I\%)^4}$	$+$	$\frac{PMT_5}{(1 + I\%)^5}$	$\longrightarrow$	Sum of PVs of each PMT = NPV

This “timeline” of future cash flow (payments) is an important tool that students should learn to utilize. Note that the numerator is each annual PMT (e.g., PMT₁, PMT₂, PMT₃, etc.), the denominator is the discount division (1+I%)^N that produces the PV, and the NPV is simply the sum of the PVs.

¹¹ In finance, the I% in TVM problems is often called the “discount factor” or the “required return”

¹² Note that, by definition, there is no time limit in a perpetuity (i.e., the future payment stream “never ends”). Thus, there are no values for FV and N. The PV is simply the PMT divided by the I% (and can be adjusted for growth).

¹³ Calculating the PV of “uneven future cash flow” cannot be performed using the TVM inputs on your calculator because the PMT is not fixed. The PV of uneven cash flow is the sum of the PVs of all payments (see equation and time line above), except that each PMT may be a different amount.

¹⁴ The sum of the PVs of future cash flows (PMTs) is often called the Net Present Value (NPV).

Annual and other Compounding Periods

Recall that three of the TVM variables (PMT, N, I%) must all reflect the same “periodic” value. Thus, if the PMT is made at semi-annual intervals, N must reflect the number of total semi-annual periods and I% must reflect the semi-annual rate of interest (called the periodic rate). Likewise, if the PMT is monthly, N must reflect the number of months in the payment stream and I% must reflect the monthly rate of interest. This requires making adjustments to the known variables prior to solving for the missing variable.

Adjusting PMT, N and I% for multiple periods per year: When the TVM equation involves multiple periods per year rather than one annual payment, the letter “m” is used to represent the # periods per year. Solving for PV when there are multiple periods per year is simple. Assume you are solving for the PV of a regular annuity with an annual payment of \$1,000, but the payment is made in semi-annual installments (i.e., twice per year) for 20 years at an annual interest rate of 10%. The PV would be solved as follows:

m =	2	(semi-annual intervals means the problem involves 2 periods per year)
PMT =	-\$500	(annual payment ÷ m = \$1,000 ÷ 2)
N =	40	(# years x m = 20 x 2)
I% =	5%	(annual interest rate ÷ m = 10% ÷ 2)
Solve: PV =	\$8,579.54	

Periodic Rates and Annual Rates (EAR and APR): Solving for I% in a periodic situation is more complicated. The general rule in finance is that, unless otherwise specifically noted, all interest rates are quoted at their annual rate. For example, when a bank offers a six-month CD at 5.00%, the interest rate quoted (5%) is the annual rate and not the rate an investor would receive for the six-month period of the investment (the six-month rate is called the “periodic rate” of interest and would be 2.5% in this example). Assume you are solving for the interest rate (I%) of a regular annuity that makes semi-annual payments of \$500 for 20 years with a PV of \$10,000. The I% would be solved as follows:

m =	2	(semi-annual intervals means the problem involves 2 periods per year)
PMT =	-\$500	(annual payment ÷ m = \$1,000 ÷ 2)
N =	40	(# years x m = 20 x 2)
PV =	\$10,000	(PV and FV are not adjusted for periodic TVM problems because they are a fixed amount)
Solve: I% =	3.93%	

The solution (I%) in this problem (3.93%) represents the periodic rate (semi-annual) and not the annual rate. An additional step is required to convert the periodic rate into an annual rate. The annual rate of return on future cash flow can be presented in two forms that differ based on whether the mathematical formula factors compounded interest: The Annual Percentage Rate (APR) does not compound the annual interest rate (and can be referred to as the “nominal” rate), while the Effective Annual Rate (EAR) does compound the annual interest rate (and can be referred to as the “effective” rate). The two mathematical methods for calculating these two rates are easy to learn.

Converting a periodic rate to an annual rate: APR and EAR

Note: The periodic rate is designated by the symbol: k_p

Nominal Rate = APR (Annual Percentage Rate) Effective Rate = EAR (Effective Annual Rate)

$$\begin{aligned} \text{APR} &= k_p \times m \\ \text{APR} &= .0393 \times 2 \\ \text{APR} &= .0786 = 7.86\% \end{aligned}$$

$$\begin{aligned} \text{EAR} &= [(1 + k_p)^m] - 1 \\ \text{EAR} &= [(1 + .0393)^2] - 1 \\ \text{EAR} &= 1.0393^2 - 1 \\ \text{EAR} &= 1.0801 - 1 = .0801 = 8.01\% \end{aligned}$$

Topic 6: Interest Rates, Risk, and Rates of Return

I. The relationship between Financial Securities, Financial Capital, and Investments

1. **FINANCIAL SECURITIES** are legal documents (literally “pieces of paper”) which describe the rights and obligations of two counterparties (the **owner** of the security vs. the **issuer** of the security).
2. Financial securities are **FINANCIAL ASSETS** to one party (the owner/investor/lender – often an individual but also corporation, banks, and governments), and **FINANCIAL LIABILITIES** to the other party (the issuer/borrower – often corporations, but also individuals and governments).
3. The owner/investor of a stock share, or the owner/lender of a bond, exchanges cash for the financial security. This is then referred to as an **INVESTMENT**.
4. The issuer of a stock share, or borrower of a bond, exchanges the financial security (the piece of paper) for the investor’s cash, which is needed by the business. The cash is called **FINANCIAL CAPITAL**.
5. All financial assets (stocks, bonds, etc.) are expected to produce Future Cash Flow.
(**Note:** The riskiness of a financial asset is judged in terms of the riskiness of its Future Cash Flow.)
6. The Future Cash Flow produced by a financial asset comes in two forms:
 - a. An increase in the price of the financial asset itself (called a **CAPITAL GAIN**).
 - b. A direct cash payment to the owner of the financial security which is called **INCOME** or **CASH FLOW**. With stock certificates, this payment is called a **DIVIDEND**. With bonds, this payment is called an **INTEREST PAYMENT** or a **COUPON PAYMENT**.
7. The two forms of Future Cash produced by an investment (capital gain and income) are used to measure the performance of the investment. Investment performance is stated as a **RATE OF RETURN (%)**.

***SUMMARY:** An **INVESTMENT** is the spending of cash today with the expectation of receiving Future Cash in excess of the investment. This may be accomplished through the purchase of a **FINANCIAL SECURITY** issued by an entity in need of cash today (**FINANCIAL CAPITAL**) that is willing and able to meet the obligation of paying Future Cash to the owner of the security.*

II. Calculating of a rate of return (k) on a financial asset (this is an **actual** or **historical return**)

Rate of return = (Amount of Cash Produced) ÷ (Amount of Cash Invested)

Amount of Cash Produced = **Capital Gain + Income**

Capital Gain = Ending Price – Beginning Price

Income = Dividend (stock) or Interest (bond) or Rental Income (real estate), etc.

Amount of Cash Invested = **Beginning Price**

Rate of return = (**Capital Gain + Income**) ÷ (**Beginning Price**)

Rate of return (**k**) = (**Ending Price – Beginning Price + Income**) ÷ (**Beginning Price**)

$$k = \frac{EP - BP + \text{Income}}{BP}$$

III. The relationship between Interest Rates and the “Prices” of Financial Securities

General Rules:

1. Investors purchase financial securities because they expect a future cash flow (**FCF**).
2. The cost of the investment (the “price” of the security) must be priced today (i.e. as a “present value”).
3. The expected benefits of the investment (capital gain and income) are in the future (i.e. “future values”).
4. The future cash flow (FCF) expected from the investment must be converted into present values.
5. The present value (PV) of future cash flow (FCF) can be calculated by using TVM functions.
6. Calculating present values from future values requires “discounting” the FCF using the interest rate $I\%$.

Note: The interest rate needed to calculate the PV of the FCF produced by a financial security is variously referred to as the “required return” or the “expected return” or the “opportunity cost” or other names. Thus, once the “required return” (or “expected return”) of a financial security is determined, the expected FCF produced by the security can be “discounted” to produce a present value of the financial security.

7. The sum of the present values of all future cash flows can be called the “value” of the financial security.
8. The “price” of the financial security is equal to its value.
9. Thus, **Price = PV of all FCF**

General Rule of Efficient Financial Markets:

Financial securities are **PRICED** in the market in order to produce the return ($I\%$) that is required or expected by investors. This is achieved by using the required return ($I\%$) to discount all **Future Cash Flow (FCF)** to a **Present Value (PV)**, which is the **PRICE** of the security.

Example of Pricing a Bond:

Question #1: What is the appropriate “price” for a bond issued by General Motors today?

Answer #1: The price is the present value (PV) of all future cash flow (FCF) paid to the buyer of the GM bond.

Question #2: In order to calculate the Price of the GM bond, what information do you need?

Answer #2: The **expected future cash flow** (interest payments and redemption value of the bond upon maturity), the timing of future interest payments and the **date** of maturity, and the **required return** of an investor in General Motor bonds.

Question #3: Once you have the information (Answer #2), how do you calculate the Price of the bond?

Answer #3: Calculate the PV of the expected future cash flow (called “discounting”) and sum them. The sum equals the NPV, or the price, of the bond.

Result of the General Rule of Efficient Financial Markets:

Because financial securities are “priced” by calculating the PV of all FCF, and because PV calculations require an interest rate ($I\%$) as the discount factor, interest rates have the most significant effect on daily activity in financial markets.

Note: The lower-case letter “ k ” will be used throughout the rest of this Supplement to represent the rate that will be variously referred to as the interest rate, the required rate of return, the expected rate of return, the discount factor, the opportunity cost of capital, etc. In TVM variables the “ k ” is the same thing as the $I\%$.

IV. Components of an interest rate (or required rate of return)

1. risk-free rate = k_{rf} (Note: this rate can be loosely considered the expected rate of inflation and will be the same for all securities)
2. risk premium = k_{rp} (Note: this rate is determined by the risk associated with the security)

The “required” or “expected” rate of return (k) = risk-free rate + risk premium $k = k_{rf} + k_{rp}$

Note: All financial securities share the same risk-free rate (k_{rf}). Thus, the single most important factor in determining the required return of a financial asset is its risk premium. Riskier securities have higher required returns than less risky securities because they have higher risk premiums.

It is important to note that the required return of a financial security may change over time for one of two reasons: a change in the risk-free rate of return (which affects ALL financial securities), or a change in the risk premium associated with that particular security (and affects ONLY that security).

V. How to determine the expected rate of return $E(k)$ of a financial security: 4 common methods

1. Using historical returns: $E(k)$ = average (“mean”) of annual returns over a given sample period (n).
2. Using probability ratios with the expected return under different future scenarios to produce a simple weighted average return as the expected return:

	<u>$E(k)$</u>	<u>Probability</u>	<u>Product</u>
Most Likely Scenario (i.e., Normal Market Conditions)	10%	50%	.0500
Best-Case Scenario (i.e., Boom Market Conditions)	30%	25%	.0750
Worst-Case Scenario (i.e., Bust Market Conditions)	-10%	25%	-.0250
$E(k)$ = Weighted Average = Sum of the Products =			.1000 = 10%

3. Using the Capital Asset Pricing Model (CAPM) – beta (β) used as a measure of risk.

Note: The Greek letter *Beta* is used to represent the variance of the financial security against the overall market. It is only used as a factor against the market risk premium because all financial securities share the same risk-free rate. *Beta* is produced from a regression analysis. The CAPM formula is as follows:

$$E(k) = k_{rf} + (k_m - k_{rf})\beta$$

k_m = market return (e.g. S&P 500)

$(k_m - k_{rf})$ = market risk premium (k_{mrp})

Because $(k_m - k_{rf})$ equals the market risk premium (k_{mrp}), the equation can also be written:

$$E(k) = k_{rf} + (k_{mrp} \times \beta)$$

4. Using the Discounted Cash Flow (DCF) formula for a perpetuity. When the security is a stock, the DCF formula is often called the Dividend Discount Model with Constant Growth (DDM-CG) for Stocks

$$E(k) = \frac{D_1}{P_0} + g$$

D_1 = amount of the next dividend,

P_0 = Price of the stock today

g = expected growth rate of the dividend

VII. How to determine the risk of a financial asset (i.e. stock)

Because all financial assets (“investments”) are expected to produce cash flows, the riskiness of the financial asset is judged in terms of the riskiness of its future cash flows (FCF).

In other words, the “uncertainty” of receiving the expected future cash flows is called risk.

Thus, the risk of a financial security can be measured by calculating the variability (i.e. uncertainty) of its historical returns (recall that returns are calculated by using the cash flow produced by the investment).

The variance (σ^2) is equal to the average of the squared deviations from the mean.

$$\sigma^2 = \sum_{i=1}^n (k_i - k_e)^2 \div n$$

The standard deviation (σ) = the square root of the Variance

$$\sqrt{\sigma^2} = \sigma$$

VIII. The relationship between risk and return

A financial asset with a high degree of risk must also provide a relatively high expected rate of return in order to attract investors. Investors in general are averse to risk, so they will not buy risky assets unless those assets have high expected returns.

IX. A measure of the trade-off between risk and return

The Coefficient of Variation (“C of V”) is a standard measure of the trade-off between the risk and the expected return of a financial security.

C of V = (standard deviation of historical returns) $\div$ (mean = “average return” = “expected return”)

$$CV = \frac{\sigma}{E(k)}$$

The C of V measures the “trade-off” of risk and return. In other words, it measures how much risk (as measured by the standard deviation) is necessary for each 1% of expected return. The higher the CV, the more “risk” is required for the expected return. The lower the CV, the less “risk” is required for the expected return. Thus, a lower CV suggests a better risk-return tradeoff, all things being equal.

Topic 7: Bonds and Bond Pricing

Everything You Need to Know About Bonds

Source: Pacific Investment Management Company LLC, 840 Newport Center Drive, Newport Beach, CA 92660
<http://www.pimco.com/EN/Education/Pages/Everythingyouneedtoknowaboutbonds.aspx#>

What is a bond?

The bond market is by far the largest securities market in the world, providing investors with virtually limitless investment options. Many investors are familiar with aspects of the market, but as the number of new products grows, even a bond expert is challenged to keep pace. Once viewed as a means of earning interest while preserving capital, bonds have evolved into a \$90 trillion global marketplace that can offer many potential benefits to investment portfolios, including attractive returns.

Before tackling the complexities of this huge and diverse market, it is important to understand the basics: What is a bond and how can bonds help meet your investment goals?

What makes a bond a bond?

A bond is a loan that the bond purchaser, or bondholder, makes to the bond issuer. Governments, corporations and municipalities issue bonds when they need capital. An investor who buys a government bond is lending the government money. If an investor buys a corporate bond, the investor is lending the corporation money. Like a loan, a bond pays interest periodically and repays the principal at a stated time, known as maturity.

Suppose a corporation wants to build a new manufacturing plant for \$1 million and decides to issue a bond offering to help pay for the plant. The corporation might decide to sell 1,000 bonds to investors for \$1,000 each. In this case, the “face value” of each bond is \$1,000. The corporation – now referred to as the bond issuer – determines an annual interest rate, known as the coupon, and a timeframe within which it will repay the principal, or the \$1 million. To set the coupon, the issuer takes into account the prevailing interest-rate environment to ensure that the coupon is competitive with those on comparable bonds and attractive to investors. The issuer may decide to sell five-year bonds with an annual coupon of 5%. At the end of five years, the bond reaches maturity and the corporation repays the \$1,000 face value to each bondholder.

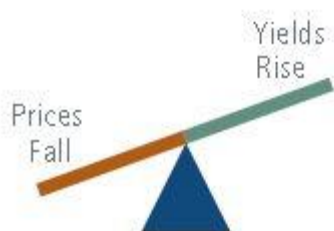
How long it takes for a bond to reach maturity can play an important role in the amount of risk as well as the potential return an investor can expect. A \$1 million bond repaid in five years is typically regarded as less risky than the same bond repaid over 30 years because many more factors can have a negative impact on the issuer’s ability to pay bondholders over a 30-year period relative to a 5-year period. The additional risk incurred by a longer maturity bond has a direct relation to the interest rate, or coupon, the issuer must pay on the bond. In other words, an issuer will pay a higher interest rate for a long-term bond. An investor therefore will potentially earn greater returns on longer-term bonds, but in exchange for that return, the investor incurs additional risk.

Every bond also carries some risk that the issuer will “default,” or fail to fully repay the loan. Independent credit rating services assess the default risk, or credit risk, of bond issuers and publish credit ratings that not only help investors evaluate risk but also help determine the interest rates on individual bonds. An issuer with a high credit rating will pay a lower interest rate than one with a low credit rating. Again, investors who purchase bonds with low credit ratings can potentially earn higher returns, but they must bear the additional risk of default by the bond issuer.

What determines the price of a bond in the open market?

Bonds can be bought and sold in the “secondary market” after they are issued. While some bonds are traded publicly through exchanges, most trade over-the-counter (OTC) between large broker-dealers acting on their clients’ or their own behalf.

If interest rates rise:



If interest rates fall:



A bond's price and yield determine its value in the secondary market. Obviously, a bond must have a price at which it can be bought and sold (see "Understanding Bond Market Prices" below for more), and a bond's yield is the actual annual return an investor can expect if the bond is held to maturity. Yield is therefore based on the purchase price of the bond as well as the coupon.

A bond's price always moves in the opposite direction of its yield, as illustrated above. The key to understanding this critical feature of the bond market is to recognize that a bond's price reflects the value of the income that it provides through its regular coupon interest payments. When prevailing interest rates fall – notably rates on government bonds – older bonds of all types become more valuable because they were sold in a higher interest-rate environment and therefore have higher coupons. Investors holding older bonds can charge a "premium" to sell them in the secondary market. On the other hand, if interest rates rise, older bonds may become less valuable because their coupons are relatively low, and older bonds therefore trade at a "discount."

Understanding bond market prices

In the market, bond prices are quoted as a percent of the bond's face value. The easiest way to understand bond prices is to add a zero to the price quoted in the market. For example, if a bond is quoted at 99 in the market, the price is \$990 for every \$1,000 of face value and the bond is said to be trading at a discount. If the bond is trading at 101, it costs \$1,010 for every \$1,000 of face value and the bond is said to be trading at a premium. If the bond is trading at 100, it costs \$1,000 for every \$1,000 of face value and is said to be trading at par. Another common term is "par value," which is simply another way of saying face value. Most bonds are issued slightly below par and can then trade in the secondary market above or below par, depending on interest rate, credit or other factors.

Face Value:	Price Quoted As:	Market Price:	The Bond is Trading at:
\$1,000	100	\$1,000	Par
\$1,000	102	\$1,020	A premium to par
\$1,000	97	\$970	A discount to par
\$5,000	99	\$4,950	A discount to par

Put simply, when interest rates are rising, new bonds will pay investors higher interest rates than old ones, so old bonds tend to drop in price. Falling interest rates, however, mean that older bonds are paying higher interest rates than new bonds, and therefore older bonds tend to sell at premiums in the market.

On a short-term basis, falling interest rates can boost the value of bonds in a portfolio and rising rates may hurt their value. However, over the long term, rising interest rates can actually increase a bond portfolio's return as the money from maturing bonds is reinvested in bonds with higher yields. Conversely, in a falling interest rate environment, money from maturing bonds may need to be reinvested in new bonds that pay lower rates, potentially lowering longer-term returns.

The role of bonds in a portfolio

Since governments began to issue bonds more frequently in the early twentieth century and gave rise to the modern bond market, investors have purchased bonds for several reasons: capital preservation, income, diversification, and as a potential hedge against economic weakness or deflation. When the bond market became larger and more diverse in the 1970s and 1980s, bonds began to undergo greater and more frequent price changes and many investors began to trade bonds, taking advantage of another potential benefit: price, or capital, appreciation. Today, investors may choose to buy bonds for any or all of these reasons.

Capital preservation: Unlike equities, bonds should repay principal at a specified date, or maturity. This makes bonds appealing to investors who do not want to risk losing capital and to those who must meet a liability at a particular time in the future. Bonds have the added benefit of offering interest at a set rate that is often higher than short-term savings rates.

Income: Most bonds provide the investor with “fixed” income. On a set schedule, whether quarterly, twice a year or annually, the bond issuer sends the bondholder an interest payment, which can be spent or reinvested in other bonds. Stocks can also provide income through dividend payments, but dividends tend to be smaller than bond coupon payments, and companies make dividend payments at their discretion, while bond issuers are obligated to make coupon payments.

Capital appreciation: Bond prices can rise for several reasons, including a drop in interest rates and an improvement in the credit standing of the issuer. If a bond is held to maturity, any price gains over the life of the bond are not realized; instead, the bond’s price typically reverts to par (100) as it nears maturity and repayment of the principal. However, by selling bonds after they have risen in price – and before maturity – investors can realize price appreciation, also known as capital appreciation, on bonds. Capturing the capital appreciation on bonds increases their total return, which is the combination of income and capital appreciation. Investing for total return has become one of the most widely used bond strategies over the past 40 years. (For more, see “Bond Investment Strategies”.)

Diversification: Including bonds in an investment portfolio can help diversify the portfolio. Many investors diversify among a wide variety of assets, from equities and bonds to commodities and alternative investments, in an effort to reduce the risk of low, or even negative, returns on their portfolios.

Potential hedge against economic slowdown or deflation: Bonds can help protect investors against an economic slowdown for several reasons. The price of a bond depends on how much investors value the income the bond provides. Most bonds pay a fixed income that doesn’t change. When the prices of goods and services are rising, an economic condition known as inflation, a bond’s fixed income becomes less attractive because that income buys fewer goods and services. Inflation usually coincides with faster economic growth, which increases demand for goods and services. On the other hand, slower economic growth usually leads to lower inflation, which makes bond income more attractive. An economic slowdown is also typically bad for corporate profits and stock returns, adding to the attractiveness of bond income as a source of return.

If the slowdown becomes bad enough that consumers stop buying things and prices in the economy begin to fall – a dire economic condition known as deflation – then bond income becomes even more attractive because bondholders can buy more goods and services (due to their deflated prices) with the same bond income. As demand for bonds increases, so do bond prices and bondholder returns.

The Many Different Kinds of Bonds

In the 1970s, the modern bond market began to evolve. Supply increased and investors learned there was money to be made by buying and selling bonds in the secondary market and realizing price gains.

Until then, however, the bond market was primarily a place for governments and large companies to borrow money. The main investors in bonds were insurance companies, pension funds and individual investors seeking a high-quality investment for money that would be needed for some specific future purpose.

As investor interest in bonds grew in the 1970s and 1980s, (and faster computers made bond math easier), finance professionals created innovative ways for borrowers to tap the bond market for funding and new ways for investors to tailor their exposure to risk and return potential. The U.S. has historically offered the deepest bond market, but

Europe has expanded greatly since the introduction of the euro in 1999, and developing countries undergoing strong economic growth in the 2000s have become integrated into what is now a global bond marketplace.

Broadly speaking, government bonds and corporate bonds remain the largest sectors of the bond market, but other types of bonds, including mortgage-backed securities, play crucial roles in funding certain sectors, such as housing, and meeting specific investment needs.

Government bonds

The government bond sector is a broad category that includes “sovereign” debt, which is issued and generally backed by a central government. Government of Canada Bonds (GoCs), U.K. Gilts, U.S. Treasuries, German Bunds, Japanese Government Bonds (JGBs) and Brazilian Government Bonds are all examples of sovereign government bonds. The U.S., Japan and Europe have historically been the biggest issuers in the government bond market.

A number of governments also issue sovereign bonds that are linked to inflation, known as inflation-linked bonds or in the U.S., Treasury Inflation Protected Securities (TIPS). On an inflation-linked bond, the interest and/or principal is adjusted on a regular basis to reflect changes in the rate of inflation, thus providing a “real,” or inflation-adjusted, return. But, unlike other bonds, inflation-linked bonds could experience greater losses when real interest rates are moving faster than nominal interest rates.

Corporate bonds

After the government sector, corporate bonds have historically been the largest segment of the bond market. Corporations borrow money in the bond market to expand operations or fund new business ventures. The corporate sector is evolving rapidly, particularly in Europe and many developing countries.

Corporate bonds fall into two broad categories: investment-grade and speculative-grade (also known as high-yield or “junk”) bonds. Speculative-grade bonds are issued by companies perceived to have lower credit quality and higher default risk than more highly rated, investment-grade companies. Within these two broad categories, corporate bonds have a wide range of ratings, reflecting the fact that the financial health of issuers can vary significantly.

Speculative-grade bonds tend to be issued by newer companies, companies in particularly competitive or volatile sectors, or companies with troubling fundamentals. While a speculative-grade credit rating indicates a higher default probability, higher coupons on these bonds aim to compensate investors for the higher risk. Ratings can be downgraded if the credit quality of the issuer deteriorates or upgraded if fundamentals improve.

Mortgage-backed and asset-backed securities

Another major area of the global bond market comes from a process known as “securitization,” in which the cash flows from various types of loans (mortgage payments, car payments or credit card payments, for example) are bundled together and resold to investors as bonds. Mortgage-backed securities and asset-backed securities are the largest sectors involving securitization.

Mortgage-Backed Securities (MBS): These bonds are created from the mortgage payments of residential homeowners. Mortgage lenders, typically banks and finance companies, sell individual mortgage loans to another entity that bundles those loans into a security that pays an interest rate similar to the mortgage rate being paid by the homeowners. As with other bonds, mortgage-backed securities are sensitive to changes in prevailing interest rates and can decline in value when interest rates rise. Securities backed by fixed-rate mortgages, in particular, are sensitive to interest rates because borrowers may prepay and refinance their mortgages when rates drop, causing the securities backed by these loans to pay earlier than expected also. In part for this reason and also to appeal to different types of investors, mortgage-backed securities can be structured into bonds with specific payment dates and characteristics, known as collateralized mortgage obligations (CMOs).

Asset-Backed Securities (ABS): These bonds are securities created from car payments, credit card payments or other loans. As with mortgage-backed securities, similar loans are bundled together and packaged as a security that is then sold to investors. Special entities are created to administer asset-backed securities, allowing credit card companies and other lenders to move loans off of their balance sheets. Asset-backed securities are usually “trunched,” meaning that loans are bundled together into high quality and lower quality classes of securities. Asset-backed securities contain risks, including credit risk.

Bond Rating Agencies

We learned in Topic 6 above that a financial securities Price is equal to the Present Value of All Future Cash Flow. Thus, the Price of the bond will be equal to the PV of the Future Cash payments received by the bond holder. We also learned that the PV calculation requires a discount rate (%) that is the required return on the security. The required return is that return (%) which is required by investors to compensate them for the risk of the investment as compared to other investment opportunities. This begs a question: How is the required return on a bond determined?

Bond Rating Agencies

Companies that assess the creditworthiness of both debt securities and their issuers are known as Bond Rating Agencies. In the United States, the three primary bond rating agencies are Standard and Poor's, Moody's and Fitch. Each uses a unique letter-based rating system to quickly convey to investors whether a bond carries a low or high default risk and whether the issuer is financially stable.

Bonds are “rated” at the time they are issued, and both bonds and their issuers are periodically reevaluated to see if a ratings change is warranted. Bond ratings are important not only for their role in informing investors, but also because they affect the interest rate that companies and government agencies pay on their issued bonds.

What is a 'Bond Rating'

A bond rating is a grade given to bonds that indicates their credit quality. Private independent rating services such as Standard & Poor's, Moody's Investors Service and Fitch Ratings Inc. provide these evaluations of a bond issuer's financial strength, or its the ability to pay a bond's principal and interest in a timely fashion.

From U.S. Treasuries to international corporations, these agencies conduct a thorough financial analysis of the entity that is issuing the bond. Based on each rating agency's individually set criteria, analysts determine the entity's ability to pay their bills and remain liquid and assign a credit rating to the bond. Standard and Poor's highest rating is AAA, and ratings are lowered accordingly (AA, A, BBB, BB, etc.). Once a bond falls to BB+ status, it is no longer considered investment grade, and the lowest rating, D, indicates that the bond is in default (the issuer is delinquent in making interest and principal payments to bondholders).

	Moody's	S&P	Fitch	Meaning
Investment Grade	Aaa	AAA	AAA	Prime
	Aa1	AA+	AA+	High Grade
	Aa2	AA	AA	
	Aa3	AA-	AA-	
	A1	A+	A+	Upper Medium Grade
	A2	A	A	
	A3	A-	A-	
	Baa1	BBB+	BBB+	Lower Medium Grade
	Baa2	BBB	BBB	
	Baa3	BBB-	BBB-	
Junk	Ba1	BB+	BB+	Non Investment Grade Speculative
	Ba2	BB	BB	
	Ba3	BB-	BB-	
	B1	B+	B+	Highly Speculative
	B2	B	B	
	B3	B-	B-	
	Caa1	CCC+	CCC+	Substantial Risks
	Caa2	CCC	CCC	Extremely Speculative
	Caa3	CCC-	CCC-	In Default w/ Little Prospect for Recovery
	Ca	CC	CC+	
		C	CC	
			CC-	In Default
	D	D	DDD	

Bond Ratings Affect Pricing, Yield, and a Reflection of Long-Term Outlook

The bond rating is an important process because the rating alerts investors to the quality and stability of the bond, which in turn influences interest rates. Higher rated bonds (known as investment grade) are seen as safer and more stable investments, which produces a lower risk premium and leads to a corresponding lower required return (and a higher price for the bond). Thus, interest rates on Corporate Bonds at any given time will be determined by a combination of the bond's rating and the yields (another term for interest rates) on US Treasury Bonds because US Treasuries are generally considered to represent the risk-free rate of investment. Compare the Yields on US Treasury Bonds below with the Yields on Corporate Bonds. Note the Yield increases as the "term" on the bond increases, and that the Yield increases as the rating decreases.

Composite Bond Rates

US Treasury Bonds				
Maturity	Yield	Yesterday	Last Week	Last Month
3 Month	0.47	0.45	0.48	0.44
6 Month	0.59	0.58	0.61	0.56
2 Year	1.15	1.12	1.18	1.10
3 Year	1.43	1.45	1.52	1.38
5 Year	1.89	1.91	2.01	1.85
10 Year	2.42	2.45	2.52	2.37
30 Year	3.05	3.06	3.10	3.02

Corporate Bonds				
Maturity	Yield	Yesterday	Last Week	Last Month
2yr AA	1.32	1.34	1.45	1.34
2yr A	1.39	1.40	1.15	1.39
5yr AAA	2.06	2.08	2.27	2.05
5yr AA	2.12	2.16	2.31	2.15
5yr A	2.28	2.32	2.42	2.29
10yr AAA	3.03	3.04	2.97	2.99
10yr AA	2.97	2.92	2.99	2.95
10yr A	3.17	3.18	3.12	3.13
20yr AAA	3.69	3.67	3.71	3.63
20yr AA	3.74	3.86	3.90	4.05
20yr A	4.03	4.06	4.29	4.20

Bond Pricing and TVM

TVM Variables used for pricing Bonds:

FV = Par Value (“Face Value”)

PV = Price (“Worth”)

PMT* = Coupon Payment (“Interest Payment”) = Coupon Rate (“Interest Rate”) x Par Value

N* = Years to Maturity (must be adjusted if bond makes multiple payments per year)

I%** = Yield to Maturity (YTM), or the rate of return (k) for the owner of the bond.

*** As with all TVM calculations, these variables must be same “periodic” interval for bond calculations.**

*** The YTM is always quoted as an APR even though investors may use the EAR for investment decisions.**

Maturity Date

The maturity date is a specified date in the future when the bond matures or expires. On this date, the Par Value of the bond is paid to the owner and Coupon Payments cease.

Bond Prices (PV)

Recall from the General Rules for Financial Securities listed above: The value (“price”) of a financial security is equal to the PV of all FCF.

Thus, the factors needed to determine the price (PV) of a bond are the same factors used in TVM calculations:

1. The amount of the Coupon Payment (PMT)
2. The amount of the par value (FV)
3. The number of years to maturity (N) the coupon payment will be received
4. The required return of an investor purchasing the bond (I% or YTM)

Using TVM functions, the Price/Value of a bond can be solved quickly by using a financial calculator or Excel.

Calculating the value/price of a bond manually would require “discounting” (using the YTM as the I%) and adding together the PV of all future Coupon Payments plus the PV of the Par Value (which is FV of the bond). A manual time-line of a bond with 5 years to maturity would look like this:

$$\begin{array}{ccccccc} \text{Year} & 0 & \text{-----} & 1 & \text{-----} & 2 & \text{-----} & 3 & \text{-----} & 4 & \text{-----} & 5 \\ & & & \frac{PMT}{(1+I\%)^1} & + & \frac{PMT}{(1+I\%)^2} & + & \frac{PMT}{(1+I\%)^3} & + & \frac{PMT}{(1+I\%)^4} & + & \frac{PMT}{(1+I\%)^5} + \frac{FV}{(1+I\%)^5} \end{array}$$

Note: In the bond’s year of maturity (the terminal year), the cash flow generated comes from both the coupon/interest payment and the face/par value of the bond, and both are “discounted” with the same “N” value.

Yield to Maturity, Current Yield, and Capital Gain

The return of an investment is produced by cash in the form of income and capital gains. For bond investors, income comes by way of Coupon Payments and the capital gain comes if the FV of the bond is greater than the PV on the date of purchase. The YTM represents the expected return on the entire investment, and it can be broken down into the “current yield” and the “capital gain.” The formulas are as follows:

$$YTM = \text{Current Yield (CY)} + \text{Capital Gain}$$

$$CY = \frac{\text{Coupon Payment (PMT)}}{\text{Price (PV)}}$$

$$\text{Capital Gain} = YTM - CY$$

Zero Coupon Bonds

A bond that makes no Coupon Payment is able to compensate an investor only in the form of a capital gain. This type of bond is called a “zero coupon bond” and is always priced at a discount (i.e. the Price (PV) of the bond is below the Par Value (FV) of the bond). Many bonds issued by the U.S. Treasury are zero coupon bonds, including all Treasury Bills (T-Bills).

Callable Bond

Some bonds contain a “call provision” which gives the issuer the right to redeem the bond under specified terms prior to the Maturity Date. Bonds with a call provision usually have a “call premium” - which is an amount added to the par value (FV) of the bond (and is typically equal to one year’s coupon payment). The YTM of a callable bond can be adjusted to produce a Yield to Call (YTC). Calculating a YTC requires the following:

FV = The FV is increased to reflect the par value + the call premium
N = N is reduced to the years remaining before the earliest call date on the bond

Note that the PMT and the PV are not affected by calculating a YTC.

Multiple Coupon Payments per year

Recall from the TVM discussion that the PMT is often made at multiple intervals per year. This is frequently the case with bonds because semi-annual payments are common. When doing calculations on bonds that make multiple payments per year, adjustments must be made to the TVM equations. When solving for the Price of the bond, note the following:

m = # payments per year
FV = Par Value (not affected by multiple payments per year)
PV = Price (not affected by multiple payments per year when solving for YTM)
PMT = Coupon Payment per Period = (annual Coupon Payment ÷ m)
N = Periods to Maturity (Years to Maturity x m)
I% = YTM ÷ m = periodic rate

When solving for the YTM of a bond that makes more than one payment per year, adjusting the PMT and the N to reflect periodic intervals will produce an I% that is the periodic rate of return (k_p) and not an annual rate. The formulas for converting the periodic rate into an annual YTM are identical to those listed in the TVM discussion:

Nominal Rate = APR (Annual Percentage Rate)	Effective Rate = EAR (Effective Annual Rate)
= $k_p \times m$	= $[(1 + k_p)^m] - 1$

General Rules of Bonds and the relationship of Price to Yield:

Bonds are often referred to as “fixed income” securities because both the interest/coupon payment and the face/par value of the investment are fixed and do not change or vary over time. Recall that the value/price of a bond is determined by calculating the PV of the FCF. These facts create a predictable relationship between bond prices and interest rates. Interest rates are reflected in the bond’s Yield to Maturity, and recall that interest rates include both the risk-free rate and the risk premium of investing in the bond. Five general rules of bond and interest rates (YTM) are:

1. As YTM increases, PV (Price) decreases
2. As YTM decreases, PV (Price) increases
3. If bond’s YTM = Coupon Rate, then Price = Par
4. If bond’s YTM > Coupon Rate, then Price < Par (the bond is said to be priced at a “discount”)
5. If bond’s YTM < Coupon Rate, then Price > Par (the bond is said to be priced at a “premium”)

Note: Bonds that sell at a premium ($PV > FV$) produce a negative capital gain. This results when the bond’s YTM (its “required return”) falls below the Coupon Rate. This anomaly requires a negative capital gain in order to offset a Current Yield that is too high.

Topic 8: Stock (Equity) Valuation

Stock Market Basics

Source: Scottrade, Inc.

<http://research.scottrade.com/knowledgecenter/Public/Education/Overview>

The stock market may appear daunting at times, but it is a system that has proven to be effective and accessible for all types of investors. Think of it like an auction. Buyers and sellers are matched by the auctioneer to arrive at transactions that are agreeable to both parties. It operates on the basic premise of supply and demand - there are no fixed prices; rather, prices are set based on what buyers are willing to pay and sellers are willing to take. The same is true of the stock market, but instead of physical objects, buyers and sellers trade shares of stock.

Stocks

So what is stock? When you own a share of stock in a company, you own a small fraction of the corporation. When the company profits, you may profit as well, and when the company struggles financially, your investment may struggle too. In addition, shares of common stock come with voting rights, meaning that in addition to owning part of the company, you have a say in some decisions made by the company. The more shares you own, the stronger your voice in company matters. Holders of preferred stock do not have voting rights.

When you buy a share of stock, there is a paper certificate issued to you by the company. Investors used to physically keep these certificates, and some still do, but in today's electronic age, most brokerage firms hold the certificates in their own name on your behalf so that they can execute your trading orders more efficiently.

Stock Exchanges

Stocks in publicly traded companies are bought and sold at a stock market (or stock exchange).

Stock exchanges operate like auction houses in the above example. They are marketplaces in which buyers and sellers come together to trade securities. The NASDAQ and New York Stock Exchange (NYSE) are examples of stock exchanges.

The exchange makes buying and selling easy. You use a stock broker (like Scottrade) who does business with the exchange. You can buy and sell shares because your broker represents you on the exchange and facilitates your transactions.

Because most stock buying and selling occurs in one place (the exchange), it allows the price of a stock to be known every second of the day. Therefore, you can watch as a stock's price fluctuates based on news, economic events, media reports, etc.

Historically, stock trading was done in person at a physical exchange. Now, some exchanges like the NYSE still offer this option, but also allow trades to be made electronically. NASDAQ, on the other hand, is a completely electronic system where trades are placed via an extensive computerized network.

Publicly Traded Companies

Corporations issue shares of their stock to the public to raise capital for growth and operating costs.

In a simplified example, say a company sells 1 million shares of stock at \$50 a share. This raises \$50 million in capital very quickly. The company then invests the \$50 million back into the company (equipment, assets, employees, etc.). The investors (or people that bought the shares) hope that the company will make a profit, meaning the stock price will increase.

When a corporation is publicly traded, all of its financial information is available to the public. The Securities and Exchange Commission (SEC) collects this information and makes it available to investors. When investors are trying to decide whether to invest in a particular company, they use a number of indicators (company financial information, historical performance, expectations for the future, etc.) to determine how much a stock is worth and whether it is something they want to buy.

Broker-Dealer

A broker-dealer is a brokerage firm with a license from the SEC that enables its employees to execute trades on behalf of its clients and the firm. Brokers handle buy and sell orders placed by individual and institutional clients and make money by charging a commission, or sales charge, for doing business on the client's behalf. Dealers execute orders for their own accounts or the firm's account. The spread, or difference, between what dealers pay for a security and what they receive when they sell it is how they make money.

Clearance & Settlement

The process of completing an order is called clearance and settlement. By law, the final transfer of stock ownership must be completed within three business days of the trade. The transfer happens in three steps. First, the number of shares bought and sold is confirmed to be the same. Second, the seller must be credited with payment from money that is transferred out of the buyer's account. Third, the shares must be transferred from seller to buyer.

Some trades settle in less than 3 days from the transaction date, but the majority settle in three days. For example, a municipal bond purchased on Monday, March 16, would settle on Thursday, March 19.

Stock Ownership

Originally, investors received stock certificates when they bought shares of stock. The paper certificate would state the investor's name as well as the number of shares and any other important information. Because approximately three billion shares are traded every day, paper certificates have become impractical. Today, the vast majority of stock purchases are recorded electronically in book entry form. This way, when the stock changes hands, one investor's name can easily be changed for another on the electronic record.

You can register your securities in either street name or through direct registration when recording your investments in book entry form.

About 75% of all electronic registrations are done in street name, meaning the securities are registered in the name of your brokerage firm. Though you don't receive a paper certificate, you do receive periodic statements, interest and dividend payments (if the company pays dividends), annual reports and proxy statements. Furthermore, your securities are insured up to \$500,000 total equities and cash, including \$250,000 for most cash claims, against the possible bankruptcy of your firm through the Securities Investor Protection Corporation (SIPC). Keep in mind that only cash used for investment purposes is covered. For example, customers who use their money market as it is intended, to move money in and out of investments, would be covered by SIPC, but those who use the account like a checking account would not be covered.

With direct registration, securities are registered directly on the books of the issuer or its designated transfer agent. Very few listed securities are eligible for direct registration.

Stock Pricing

Bond Pricing – A Review:

Recall that the features of a typical bond are **1.)** an even cash flow at fixed periodic intervals (Coupon Payments) that terminate at a future date certain (maturity date) and **2.)** a lump-sum cash payment (Par Value) that is paid upon the bond's maturity. These future payments (Future Cash Flow) combine to produce a rate of return (YTM) for the investor. Thus, the value (Price or PV) of a bond is determined by its coupon payment (PMT), its par value (FV), its yield to maturity (I%), and the number of years remaining before the maturity date (N). Recall also that zero coupon bonds make no Coupon Payments, but only a lump-sum payment (FV) at the maturity date.

These features allow simple TVM formulas to price a bond. In addition, recall that bonds can be priced manually by using time lines and discounting formulas to calculate the present value of all future cash flow.

Bond Pricing Example 1: The price (PV) of a “zero coupon” bond is simply the present value of the bond's par value (FV). The pricing formula is the same as calculating the present value of a future cash value:

$$\text{Present Value of a future cash value: } PV = \frac{FV}{(1+I\%)^N}$$

Note: In this case, “N” represents the number of years into the future until the bond matures, at which time the par value is paid to the bond owner.

Bond Pricing Example 2: With a “coupon” bond, the bond's fixed future cash payments resemble an **annuity**, with the added feature of a lump sum payment at maturity. These bonds can easily be priced on a financial calculator using the same TVM functions that are used for an annuity (PMT, N, FV, I%). However, the calculator is simply calculating the present value (PV) of all future cash flow (including the par value), and adding them up. This can be performed manually with the help of a time line. It simply requires calculating the present value of all future values, with the price (PV) being the sum of the present values of all future cash flow (PV of all FCF):

Year	0	-----	1	-----	2	-----	3	-----	4	-----	5
			$\frac{PMT}{(1+I\%)^1}$	+	$\frac{PMT}{(1+I\%)^2}$	+	$\frac{PMT}{(1+I\%)^3}$	+	$\frac{PMT}{(1+I\%)^4}$	+	$\frac{PMT}{(1+I\%)^5} + \frac{FV}{(1+I\%)^5}$

Note: In the bond's year of maturity (the terminal year), the cash flow generated comes from both the final coupon/interest payment and the face/par value of the bond, and both are “discounted” with the same “N” value.

Stock Pricing Equations - Perpetuities:

The features of a typical equity security (stock) differ from the features of a typical debt security (bond) in two important ways:

1. The periodic cash payments generated by owning a stock (called a “dividend”) are neither certain nor fixed. At best, they tend to grow at a rate commensurate with the growth of the company.
2. The periodic cash payments are not expected to terminate at any future date certain. Instead, the periodic cash payments are considered to extend indefinitely (i.e. into perpetuity).

Perpetuities can be priced by using the following two formulas:

1. Present Value of a perpetuity payment that remains constant (no growth): $PV = \frac{PMT}{I\%}$
2. Present Value of a perpetuity payment that grows at a constant annual rate (g): $PV = \frac{PMT_1}{I\% - g}$

Calculating the Price (P_0) of a Stock

When calculating the price (PV or P_0) of common stock using the formulas above for perpetuity, the variables for the “no-growth” formula are as follows:

$$PMT = D_0$$

D_0 = the dividend payment

$I\%$ = the required return for that particular common stock (this must be given or known)

The variables for the “constant growth” formula are as follows:

$$PMT = D_1$$

D_1 = the next expected dividend (Note that $D_1 = D_0 \times (1 + g)$)

g = the rate of constant growth

$I\%$ = the required return for that particular common stock (this must be given or known)

Thus, the price (P_0) of a common stock that grows at a known **constant** annual rate (g) is the next dividend (D_1) divided by the difference between required return ($I\%$) and the growth (g). The formula is identical to the TVM formula for a perpetuity that grows at a constant rate:

$$P_0 = \frac{D_1}{I\% - g}$$

Note that if the stock is not expected to grow, then $g = 0$ and the formula is simply the same as perpetuity: $\frac{D_0}{I\%}$

Note: This formula is referred to as the Dividend Discount Model (DDM) with Constant Growth (DDM-CG).

It is important to understand that the perpetuity formula allows us to calculate the **value today** (PV or P_0) of all future cash flows. In theory, the future cash flow (FCF) from perpetuity continues forever, but the TVM elements are still present: i.e. ***all of these cash flows into perpetuity can be reduced to one present cash value - today.***

Using the TVM timeline, the DDM and DDM-CG formulas allow us to calculate the present value:

$$\begin{array}{l} \text{Year} \quad 0 \text{-----} 1 \text{-----} 2 \text{-----} 3 \text{-----} 4 \text{-----} 5 \text{-----} \text{(into perpetuity)} \\ P_0 = \text{PV of } D_1 + \text{PV of } D_2 + \text{PV of } D_3 + \text{PV of } D_4 + \text{PV of } D_5 + \dots \end{array}$$

Growth that is not constant

Because many firms go through early growth cycles that are not constant, it is sometimes necessary to distinguish between periods of “super” growth (g_s) and periods of constant (“normal”) growth (g_c). When this is the case, calculating the price (P_0) requires making use of both the DDM-CG formula and the TVM timeline.

The DDM-CG formula only applies at that point in time when all future dividends will grow at a constant rate. However, it is important to understand that the DDM-CG formula gives us one cash value for all future dividends from that point forward. Any dividends that are paid prior to that future point in time must be added to the value of all constant dividends calculated by using the perpetuity equation. Below is an example of a TVM timeline which calculates the price of a stock which grows at a “super” rate for three years before settling into a normal/constant growth rate beyond that.

$$\begin{array}{l} \text{Year} \quad 0 \text{-----} 1 \text{-----} 2 \text{-----} 3 \text{-----} [4 \text{-----} 5 \text{-----} 6 \text{-----} 7 \text{-----} 8 \text{-----} (\dots \text{into perpetuity} \dots)] \\ P_0 = \quad \text{PV of } D_1 + \text{PV of } D_2 + \text{PV of } D_3 + \text{PV of } P_3 \leftarrow \end{array}$$

Note: The key to this timeline is calculating P_3 because P_3 captures every dividend from D_4 to eternity.

Note also that P_3 is not a Present Value (today) of all dividends from D_4 forward, it is the value in Year 3 of all dividends from D_4 forward.

Steps for calculating the price (P_0) of a stock that experiences super growth followed by constant growth:

1. Calculate each dividend during the period of super growth (g_s):

$$D_1 = D_0 \times (1+g_s) \quad D_2 = D_1 \times (1+g_s) \quad D_3 = D_2 \times (1+g_s)$$

2. Calculate the *first future* dividend that marks the beginning of the period of constant growth:

$$D_4 = D_3 \times (1+g_c) \text{ Note that the growth rate used here is } g_c \text{ and not } g_s$$

3. Use the DDM-CG formula (and the first future dividend with constant growth from Step #2 above) to calculate a value of all future dividends with constant growth:

$$P_3 = \frac{D_4}{I\% - g_c}$$

4. Calculate the present values (PV) of all dividends in Step 1 and the price in Step #3 (see timeline above).

$$\text{PV of } D_1 = \frac{D_1}{(1+I\%)^1}$$

$$\text{PV of } D_2 = \frac{D_2}{(1+I\%)^2}$$

$$\text{PV of } D_3 = \frac{D_3}{(1+I\%)^3}$$

$$\text{PV of } P_3 = \frac{P_3}{(1+I\%)^3}$$

5. Sum the present values calculated in Step #4 to determine today's price (P_0).

Note on determining the appropriate required return (I%) on equities (stock)

At this point in the semester, students should know that 1.) the required return on a financial security is one of the most important factors in determining its price, and 2.) that the RISK associated with owning a particular financial security is the most important factor in determining its required return. **If you are still uncertain about this, it would be helpful to review Topic 6 once more until it makes sense.**

At this point, a big question in the student's mind should be: How is the required return on a particular stock determined? There is no single (and simple) formula for the answer to this question. Recall that the required return on a bond is reflected in the Yield to Maturity (YTM), which is often determined by the bond's rating (see Topic 7). There is no such rating process for equities (stocks), so analysts are left to their own assessments for determining the required return on any particular stock at any particular moment in time. The list on pp. 62-63 provides the four most common methods used to determine the required return on a stock.

Topic 9: Cost of Capital

The Cost of Capital is the rate used to discount a company's future cash flows back to the present. It is alternatively referred to as the company's required return.

A company's cost of capital is exactly as its name implies. Whether a company raises capital from lenders or from investors (stockholders), both require a return on their investment. Lenders expect to be paid interest on their loans, while owners expect a return through a combination of dividends and capital gain. Thus, the Cost of Capital depends on the mode of financing used – it refers to the cost of equity if the business is financed solely through equity, or to the cost of debt if it is financed solely through debt. Many companies use a combination of debt and equity to finance their entity, and for such companies, their overall cost of capital is derived from a weighted average of all capital sources, widely known as the weighted average cost of capital (WACC). Since the cost of capital represents a **hurdle rate** that a company must overcome before it can generate value, it is extensively used in the capital budgeting process to determine whether the company should proceed with a project.

The cost of the various sources of capital differs from company to company, and depends on factors such as its operating history, profitability, credit worthiness, etc. In general, newer enterprises with limited operating histories will have higher costs of capital than established companies with a solid track record, since lenders and investors will demand a higher risk premium for the former. A stable, predictable company will have a low cost of capital, while a risky company with unpredictable cash flows will have a higher cost of capital. That means, all else equal, that the riskier company's future cash flows are worth less in present value terms, which is why stocks of stable companies often look more expensive on the surface. The cost of capital used in a Discounted Cash Flow (DCF) model can have a significant impact on the fair value, so it's important to pay attention to this estimated figure.

Every company has to determine a strategy for its long-term capital structure. (Note: The most common method for calculating the long-term capital structure is discussed below.) Thus, the cost of capital becomes a critical factor in deciding how to finance project (especially capital budgeting projects, which will be our next topic). Firms must choose between debt, equity, or a combination of the two. Many start-ups do not have sizable assets to pledge as collateral for debt financing, so equity financing becomes the default mode of funding for most of them.

The cost of debt is merely the interest rate paid by the company on such debt. However, since interest expense is tax-deductible, the after-tax cost of debt is calculated as: $\text{Yield to maturity of debt} \times (1 - T)$ where T is the company's marginal tax rate.

The cost of equity is more complicated, since the rate of return demanded by equity investors is not as clearly defined as it is by lenders. Theoretically, the cost of equity is approximated by the Capital Asset Pricing Model (CAPM) = $\text{Risk-free rate} + (\text{Company's Beta} \times \text{Risk Premium})$. There are other models used to calculate the cost of equity that will be discussed on the following pages.

The firm's overall cost of capital is based on the weighted average of these costs. For example, consider an enterprise with a capital structure consisting of 70% equity and 30% debt. If its cost of equity is 10% and its after-tax cost of debt is 7%, its WACC would be $(0.7 \times 10\%) + (0.3 \times 7\%) = 9.1\%$. This is the cost of capital that would be used to discount future cash flows from potential projects and other opportunities to estimate their Net Present Value (NPV) and ability to generate value. If Deferred Stock is added to the mix of capital, the weights would be adjusted accordingly, as will be demonstrated on the next page.

Companies strive to attain the optimal financing mix, based on the cost of capital for various funding sources. Debt financing has the advantage of being more tax-efficient than equity financing, since interest expenses are tax-deductible and dividends on common shares have to be paid with after-tax dollars. However, too much debt can result in dangerously high leverage, resulting in higher interest rates sought by lenders to offset the higher default risk.

General Forms of Financial Capital:

Debt (Liabilities)
Preferred Equity (the “hybrid” capital)*
Common Equity

Specific Sources of Financial Capital:

Short Term Debt
Long Term Debt
Preferred Stock*
Common Stock (Common Equity)
Retained Earnings (Common Equity)

* Note: Preferred stock is a form of capital sometimes referred to as a “hybrid” between debt and equity. Typically preferred stock is a component of financial capital only for large firms, and even then it is not normally a large component of financial capital

A firm’s “Capital Structure” is generally considered to be limited to “long-term” capital accounts:

1. Debt (Long-Term)
2. Preferred Equity (Preferred Stock)
3. Common Equity: (Common Stock + Retained Earnings)

Sources of “NEW” capital are generally considered to come from some or all of 4 sources

1. New Long-Term Debt
2. New Preferred Stock Issues
3. New Common Stock Issues
4. Addition to Retained Earnings (a form of “internal financing”)

Formulas for Calculating the Component Cost of Capital (k):

Cost of Debt (k_d) = $BC \times (1 - T)$ where BC = borrowing costs and T = firm’s tax rate

Cost of Preferred Equity (k_p) = $\frac{\text{Dividend}}{\text{Net Price}}$ where Net Price = Price less Flotation Costs ($NP = P_0 - F$)

Cost of Common Equity (k_e) = 2 types: cost of retained earnings and cost of new common stock

1. Dividend Discount Model for Constant Growth (DDM-CG):

Cost of Equity from Retained Earnings (k_{re}) = $\frac{D_1}{P_0} + g$ where g = constant growth rate

Cost of Equity from New Common Stock (k_{ncs}) = $\frac{D_1}{\text{Net Price}} + g$ where Net Price = $P_0 - F$

Note: The difference between k_{re} and k_{ncs} is called the “new stock premium” and is the result of the fees associated with selling and distributing new stock as compared to no fees associated with using internal profits as retained earnings.

2. Capital Asset Pricing Model:

CAPM cost of Equity from Retained Earnings $k_{re} = k_{rf} + k_{mrp}\beta$

Cost of Equity from New Common Stock $k_{ncs} = \text{CAPM } k_{re} + \text{new stock premium (\%)}$

k_{mrp} = market risk premium, k_m = expected market return, k_{rf} = risk-free rate of return

Weighted Average Cost of Capital - WACC

Step 1: Determine “cost” (k) for each capital component (new long-term debt, new preferred stock, retained earnings, new common stock)

Step 2: Determine “weights” (w) of each capital component (Note: $w = \text{component capital amount} \div \text{total capital amount}$)

Step 3: Multiply component cost (k) times the component weight (w)

Step 4: Sum the products of Step 3

Example Step 1: Calculate the “cost” of each capital component using the assumptions given

A. The firm’s Borrowing Cost (BC) is 8.00% and its tax rate (T) is 40.00%.

The cost of debt is 4.80%: $k_d = .08 \times (1 - .40) = .08 \times .60 = .0480 = 4.80\%$

B. The firm’s preferred stock has a Dividend of \$2.00, a Price of \$20.00, and Flotation costs of \$1.00 per share

The cost of Preferred Equity = 10.53%: $k_p = \frac{2.00}{20.00 - 1.00} = \frac{2.00}{19.00} = .1053 = 10.53\%$

C. The firm’s next dividend (D_1) is expected to be \$1.00, its stock price (P_0) is currently \$10.00, it is expected to grow at a constant rate (g) of 3.00%, and the Flotation costs (F) of issuing new stock is \$0.50 per share.

The cost of Retained Earnings using DDM-CG is 13.00%: $k_{re} = \frac{1.00}{10.00} + .03 = .10 + .03 = .1300 = 13.00\%$

The cost of New Common Stock using DDM-CG is 8.50%: $k_{ncs} = \frac{1.00}{10.00 - 0.50} + .03 = .1053 + .03 = .1353 = 13.53\%$

Example Step 2: Calculate the “weight” of each capital component for a project that is **\$2,500,000**

- **\$500,000** will come from **LT Debt** ($w = 500,000 \div 2,500,000 = .20 = 20\%$)
- **\$250,000** will come from **Preferred Stock** ($w = 250,000 \div 2,500,000 = .10 = 10\%$)
- **\$750,000** will come from **Retained Earnings** (i.e. internal profits) ($w = 750,000 \div 2,500,000 = .30 = 30\%$)
- **\$1,000,000** will come from the issuance of **New Common Stock** ($w = 1,000,000 \div 2,500,000 = .50 = 40\%$)

Example Step 3: Multiply the component cost by the component weight

Long-Term Debt	4.80%	x	20%	=	0.96%
Preferred Stock	10.53%	x	10%	=	1.05%
Retained Earnings	13.00%	x	30%	=	3.90%
New Common Stock	13.53%	x	40%	=	5.41%

Example Step 4: Sum the products of Step 3 = 11.33% is the WACC

Calculating a firm’s Long-Term Capital Structure

A firm’s LT Capital Structure is the ratio of LT Debt and Equity to total long-term capital. Thus, only LT capital accounts on the Liabilities/Equity side of the Balance Sheet are considered: LT Debt, Preferred Stock, Retained Earnings, and Common Stock (where RE and Common Stock combine for Common Equity). See below:

Liabilities & Equity

Accounts Payable	100,000
Accrued Liabilities	100,000
ST Debt	200,000
Total Current Liabilities	400,000
LT Debt	1,000,000
Total Liabilities	1,400,000

Preferred Stock	500,000
Retained Earnings	1,000,000
Common Stock	1,500,000
Total Equity	3,000,000

Total Liabilities & Equity **4,400,000**

Long-Term Capital Structure

LT Debt	1,000,000
Preferred Stock	500,000
Common Stock	1,500,000

Retained Earnings 1,000,000

Total LT Capital **4,000,000** Note that Total LT Capital is less than Total L&E

Long-Term Capital Structure Distribution (%)

LT Debt = $1,000,000 \div 4,000,000 = 25.00\%$

Preferred Equity = $500,000 \div 4,000,000 = 12.50\%$

Common Equity = $(1,000,000 + 1,500,000) \div 4,000,000 = 62.50\%$

Topic 10: Capital Budgeting

Capital budgeting is concerned with long-term decision making and is perhaps the most important function financial managers and staff must perform.

1. Because the results of capital budgeting decisions continue for many years, the firm loses some of its flexibility. For example, the purchase of an asset with an economic life span of 10 years locks in the firm for a 10-year period. A mistake under these conditions is very costly and may put a firm out of business.
2. Because an investment decision is based on projected future sales, the decision to buy an asset that is expected to last 10 years (for example) requires a 10-year sales forecast. Such a process is full of potential mistakes and miscalculations that can be very costly.
3. A firm's capital budgeting decisions define its strategic direction because a move into new products, services, or markets must be preceded by substantial capital expenditures.

We can make three key observations about capital budgeting:

1. A capital budgeting project is typically a go or a no-go decision on a project, i.e., the project is either accepted or rejected. Thus, it is a process that is ordered toward making a decision. The choice of accepting or rejecting proposed projects is the cornerstone of financial management at all levels of the business.
2. A capital budgeting decision requires sound estimates of the amount and timing of cash flow (both inflows and outflows). All of the decision models available to financial managers depend upon a reliable estimation of both initial costs and future cash flow.
3. The capital budgeting model has a predetermined criterion for making decisions that lead to the acceptance or rejection of a project. These models are discussed in greater detail below.

First, capital budgeting projects can be classified based on what they are designed to accomplish:

1. **Replacement: Maintenance of Business** = expenditures to replace worn-out or damaged equipment used in the production of profitable products. Replacement projects are usually necessary if the firm is to continue in a particular line of business.

The analysis of these projects is usually less complex than other capital projects because it is often considered part of a normal cycle of capital replacement.

2. **Replacement: Productivity Gains** = expenditures to replace serviceable but obsolete equipment with the goal to increase productivity. The new equipment might be designed to increase variable output without an increase in variable cost per unit, or it might be designed to lower the cost of labor, materials and other inputs (i.e. electricity, fuel, etc.) at the same level of output, or both. In either case, the productivity gain translates into an increase in after-tax cash flow.

The analysis of these projects is more complex than for maintenance-of-business projects because this might be considered discretionary spending. Thus, a more detailed analysis is usually warranted. The focus of this analysis is likely to be on the expected performance of the capital/equipment itself and does not usually require sales forecasts.

Note: Projects for Replacement-Maintenance and projects for Replacement-Productivity Gains will often occur simultaneously.

3. **Expansion of existing products or markets** = investments to increase the production of existing goods and services (or perhaps to expand retail outlets/distributions facilities).

The analysis of these projects is usually complex because it requires an explicit forecast of growth in demand and sales. Decisions are usually made at high levels as part of a firm's strategic plan.

4. **Expansion into new products or markets** = investments to produce a new product or to expand into a geographic area not currently being served. The analysis of these projects is usually long and can be extremely complex. These projects involve strategic decisions that could change the fundamental nature of the business, and they normally require the expenditure of large sums of money with delayed paybacks. Decisions are almost invariably made at the highest levels of the firm.
5. **Non-Revenue Producing Projects** = capital expenditures necessary to comply with government orders, labor agreements, labor agreements, insurance policies, etc. (these are often mandatory expenditures) or expenditures for parking lots, office space, aircraft, etc. These are often non-revenue producing projects.

The analysis for these projects depends on the nature of the expenditure.

The Capital Budgeting Process

The typical capital budgeting process requires asking (and ultimately answering) four questions:

1. What is the **cost of the assets required for the project**? This cost is expressed as a dollar amount and is referred to as the *Initial Investment* (which includes both hard and soft costs).
2. What is the appropriate **cost of capital** for the funds that will be needed to finance the Initial Investment? This cost is expressed as an interest rate (%) and is often referred to as the "*discount rate*" (because it is used to calculate present values) or the "*hurdle rate*" (because it is the rate that the project must outperform in order to be viable). The cost of capital can be determined by many different methods, the three most common being:
 - a. The firm's Weighted Average Cost of Capital (WACC)
 - b. The "required return" of the firm's owners
 - c. The "opportunity cost" associated with an investment of this particular type.
3. What is the **net annual cash flow** the investment is expected to produce? This is the most detailed and most important stage of the project because it involves forecasts of both sales and expenses. The requirement of "net" cash flow means taxes must be factored, which means depreciation expenses (income tax shielding) must be determined. It also requires calculating net liquidation values of the assets (*Initial Investment*) in the terminal year of the project.

Annual net cash flows from operations (NCF) must be estimated, including:

- a. Input Data for Sales and Expenses
 - b. Depreciation Expenses for each depreciable asset in the Initial Investment
 - c. Liquidation values (estimates of the market values of the assets in the project's terminal year)
4. Should the project be **accepted or rejected**? This is the **Yes/No** decision that requires using one or more of the methods available to analyze capital budgeting projects:
 - a. **Payback period (PB)** = the length of time required for an investment's net cash flows to recover the cost of the assets (*Initial Investment*).

- i. Regular payback period – method used when future cash flows (FV) are not adjusted for time and opportunity cost (i.e. future cash flows are not discounted)
 - ii. Discounted payback – method used when future cash flows are discounted into present values (PV) at the appropriate cost of capital or opportunity cost
- b. **Net Present Value (NPV):** the present value (PV) of all net cash flows is compared to the present value of cash outlays (*Initial Investment*). The sum of all discounted future values (the *Initial Investment* is posted in Year 0 as a negative number) is called the NPV. A positive NPV means the discounted cash flow exceeds the costs of the project.
- c. **Internal Rate of Return (IRR):** the compounded rate of return that measures future cash inflows against the project's costs. The IRR is the discount rate that forces the NPV to equal 0. The hurdle rate is the cost of capital that the IRR must exceed for the project to be viable. This number (IRR) is calculated using Excel (see below pg. 79).
- d. **The Modified Internal Rate of Return (MIRR):** As noted, the IRR is the result of compounding the future cash flow at the internal rate of return of the project itself. However, it is possible to calculate the return of a project that is NOT based on this internal self-compounding factor but is based instead on a compounding rate that is external to the project (which could be thought of as a “reinvestment” rate). This “modified” compound rate can be the firm's cost of capital, or it can be an arbitrary reinvestment rate (such as 0% which eliminates the effect of compounding altogether). If the IRR is higher than the cost of capital, then the MIRR will be lower than the IRR. Once again, this number (MIRR) is calculated using Excel and requires distinguishing between the finance rate (the cost of capital) and the reinvestment rate (the rate at which the future cash flow will be reinvested). See below pg. 79.
- e. **Profitability Index (PI)** is a modification of the NPV to produce the ratio of the present value of benefits (future cash inflows) to the present value of costs (initial investment).

$$\text{Profitability Index} = \frac{\text{Present Value of Benefits}}{\text{Present Value of Costs}}$$

Another way of describing the equation for calculating the Profitability Index is:

$$\text{Profitability Index} = \frac{\text{Present Value of all Future Cash Flow}}{\text{Initial Investment}}$$

Note that the numerator is the Sum of the PV of all FCF (i.e., beginning in Year 1), while the denominator is Initial Investment (often stated in Year 0 as a negative number).

See page 78 – 79 for instructions on formulas and calculations for Payback, NPV, IRR, MIRR, and PI

Cash Flow Estimation

Cash Flow v. Accounting Income

The present value of Cash Flows, not Accounting Income, is the basis of a firm's value. Remember the basic rule of finance: **Price (P_0) = PV of all FCF**. Note that "price" = "value" and it is not considered the PV of all future net income but of future **cash flow**. Thus, only the Operating Cash Flow (OCF) generated by an investment is used in the capital budgeting analysis.

$$\text{OCF} = \text{EBIT} - \text{Taxes} + \text{Depreciation}$$

Timing of Cash Flows

In a capital budgeting analysis, the cash flow is generally considered to occur at the end of the year. A typical exception to this rule occurs when calculating regular Payback and discounted Payback which, unless otherwise known for certain (such as terminal year liquidation values – which always occur at the end of the year) assumes annual net cash flow occurs evenly throughout the year.

Depreciation Schedules

The decision to use accelerated (such as the Modified Cost Recovery System – MACRS) or straight-line depreciation methods will often have an effect on project analysis because accelerated depreciation schedules increase cash flow in the early years, and early year cash flow has a greater effect in all of the analytical tools (Payback, NPV, IRR).

Incremental Cash Flows

Incremental cash flows are those that will occur **ONLY** if some specific event occurs. In capital budgeting, this event is usually the project itself, which is being analyzed.

- a. Any revenue that would occur even without the project is not "incremental" and should not be included in the project analysis.
- b. A **Sunk Cost** is an outlay (cost) that was incurred in the past and cannot be recovered in the future regardless of whether the project is accepted. The analysis only concerns *future* incremental cash.
- c. Any cash inflow (such as from the sale of an unused asset) that is lost or foregone because of the project must be factored into the project, usually as offsetting ("negative") cash flow. This is known as an **Opportunity Cost**.
- d. Whenever a new product competes against a company's already existing products, **Erosion Costs** occur (also referred to as cannibalization). Although difficult to estimate, Erosion Costs must be accounted for in the capital budgeting analysis. The analysis should only include additional revenue net of lost sales due to the effects of cannibalization.
- e. In contrast to erosion costs, sometimes a new product will complement an already existing product and lead to increased sales. Once again, although this is difficult to estimate, it must be included in the capital budgeting analysis.

Terminal Year Cash Flow from the Disposal (Liquidation) of Assets

Most capital budgeting projects are analyzed over a fixed period of time. The final year of the project is called the terminal year. A net cash value (net of taxes) must be estimated for the liquidation of all assets, including working capital. These amounts are then added to the final year (the terminal year) and included in the calculations for Payback, NPV, IRR and PI. A comparison of each asset's Market Value (MV) and Book Value (BV) in the terminal year will determine the tax effect.

Risk Analysis

Risk Analysis

Projects differ in risk, and this should be reflected in the capital budgeting analysis. However, it is difficult to measure risk, especially for a new project with no operational or performance history. Three separate and distinct types of risk are involved:

1. **Stand-alone risk** ignores diversification. The project risk is the variability of the expected return of the project itself. Thus, stand-alone risk assumes:
 - a. The project being considered is the only asset the firm owns
 - b. The firm is the only stock in an investor's portfolio
2. **Corporate (within-firm) risk** ignores the investor's risk (or the risk of an investor's portfolio) and instead takes into account the company's portfolio of assets. This type of risk is measured by the project's impact on uncertainty about the firm's future returns.
3. **Market (beta) risk** takes into account an investor's well-diversified portfolio. The project's market risk is measured by its effect on the firm's beta. This approach assumes:
 - a. The project being analyzed is only one of the firm's assets, and
 - b. The firm's stock is only one part of an investor's portfolio

With the firm's overall WACC as a starting point, a risk-adjusted cost of capital can be used as the hurdle rate for the capital budgeting analysis. For example, a firm might establish three risk classes:

1. Use the WACC for average-risk projects
2. Add a risk premium (e.g. 5%) to the WACC for higher-risk projects
3. Subtract a risk discount (e.g. 2%) from the WACC for low-risk projects

Measuring Stand-Alone Risk: The Sensitivity Analysis

Changes in key input variables will cause changes in each of the decision models used for capital budgeting (Payback, NPV, IRR). A **sensitivity analysis** is the process of measuring a percentage change in NPV that results from a given percentage change in an input (assuming other variables are held at their original values).

Scenario Analysis

A scenario analysis combines various stages of sensitivity analysis into classic cases:

1. Base-Case Scenario – all input variables are set at their most likely values
2. Worst-Case Scenario – all input variables are set at their worst reasonably forecasted variables
3. Best-Case Scenario – all input variables are set at their best reasonably forecasted variables

Capital Budgeting Case: Walton Medical Laboratory

Martin Walton spent 10 years working in the lab at County Hospital. During that time, he advanced to the position of director of the lab and completed an MBA degree. He was looking for a new challenge and became attracted to the idea of starting his own business. Walton decided that he would open an independent lab to provide medical tests for independent medical practices. He believed that his business could help physicians reduce both their capital requirements and administrative chores.

Walton began assembling information. He discovered a piece of land available near a number of independent medical practices. The land could be purchased for \$100,000, and a suitable building would cost approximately \$400,000. The building would have a depreciable life of approximately 40 years. Lab equipment would cost \$700,000. The equipment would have a depreciable life of 7 years for tax purposes. Although the business could continue indefinitely with future purchases of replacement equipment, Walton wanted to do the business analysis based on an assumption that he would liquidate the business in 5 years.

In addition to the hard costs mentioned above, the project would also require soft costs prior to the commencement of operations. Walton wanted to begin with (and maintain until the terminal year) a minimum cash balance of \$50,000 and \$100,000 worth of supplies in inventory. In addition, Walton estimated that total professional fees would be \$60,000. This would include costs for an attorney to draft and file all necessary legal documents, for a CPA to review and certify the financial information, and a finance consultant to help produce the business plan and raise the capital. These professional fees would be capitalized as start-up costs and amortized over three years.

With regard to actual operations, Walton predicted revenue of \$750,000 during the first year and \$1,500,000 each year thereafter. He estimated that labor expenses, including his salary, would be \$300,000 in the first year and \$450,000 each year thereafter. He estimated that supplies expense would be \$100,000 in the first year and \$200,000 each year thereafter. Finally, he estimated overhead expenses, other than depreciation, would be \$100,000 per year for all 5 years, and that total taxes (federal, state, local) would be 40%.

Looking ahead to the terminal year (Year 5), Walton estimated that the equipment would have a no market value, the land and building would increase in market value by 50% from its original cost, the supplies inventory would still be \$100,000 but would be liquidated for 50% of its value net of taxes, and the cash balance would be liquidated at its book value of \$50,000 net of taxes.

He assumed the potential investors would require a return of 12% on their equity, and so decided to use that number as the cost of capital for the capital budgeting model.

The student is required to produce a capital budgeting spreadsheet that will allow Mr. Walton to calculate the Regular Payback, Discounted Payback, NPV, IRR, MIRR, and Profitability Index for this project. In addition, the spreadsheet must allow Mr. Walton to perform a sensitivity analysis based on changes in certain key assumptions to determine which have the largest impact on NPV and IRR as follows:

- Scenario 1: 25% drop in Revenue across all years (all other factors remain at original)
- Scenario 2: 25% increase in Labor expenses in all years (all other factors remain at original)
- Scenario 3: 25% increase in Supplies expenses in all years (all other factors remain at original)
- Scenario 4: 25% increase in Fixed Overhead in all years (all other factors remain at original)
- Scenario 5: A combination of all three Scenarios 2 thru 4
- Scenario 6: The terminal year cash flow from all assets (including working capital) is \$0

Walton Medical Laboratory Steps 1 thru 3

1. **Cost of the project:** List a schedule of the Initial Investment

Note: Include both hard costs (land, buildings, equipment) and soft costs (working capital, fees, etc.)

Land	\$100,000
Building	\$400,000
<u>Equipment</u>	<u>\$700,000</u>
Total Hard Costs	\$1,200,000

Cash	\$50,000
Supplies Inventory	\$100,000
<u>Professional Fees</u>	<u>\$60,000</u>
Total Soft Costs	\$210,000

Total Cost of Project \$1,410,000 → Note: this is the Initial Investment

2. **Cost of capital:** “They viewed this as an investment of moderate risk and indicated that they would want a 12% after-tax return from an investment of this type.” Thus, WACC is not an issue in this case because equity is the only source of capital.

Required return = 12%

Discount Factor = 12%

Question: What WACC would be used if Walton decided to finance 50% of the capital with debt from a bank loan (6% borrowing cost) and 50% with equity from the doctors (12% required return)?

3. **Project Net (annual) Cash Flow from the project:** This step involves creating a spreadsheet that includes all key assumptions, all annual depreciation and amortization expenses, all net cash liquidation values of the assets in the project's terminal year, and all projected annual revenue and expenses (including taxes).

a. Key Assumptions:

<u>Key Assumptions</u>	<u>Cost</u>	<u>BV</u>	<u>Est. MV</u>
	<u>Year 0</u>	<u>Year 5</u>	<u>Year 5</u>
Land	\$100,000	\$100,000	\$150,000
Building	\$400,000	\$350,000	\$600,000
<u>Equipment</u>	<u>\$700,000</u>	<u>\$200,000</u>	<u>\$0</u>
Total Hard Costs	\$1,200,000	\$500,000	\$750,000
Cash	\$50,000	\$50,000	\$50,000
Supplies	\$100,000	\$100,000	\$50,000
<u>Start-Up Fees</u>	<u>\$60,000</u>	<u>\$0</u>	<u>\$0</u>
Total Soft Costs	\$210,000	\$150,000	\$100,000
Total Initial Investment	\$1,410,000		
<u>Terminal Year (Year 5) Liquidation of Working Capital</u>			
Cash	100%		
Supplies	50%		
	<u>Year 1</u>	<u>Years 2-5</u>	
Revenue	\$750,000		\$1,500,000
Labor	\$300,000		\$450,000
Supplies	\$100,000		\$200,000
Fixed Overhead	\$100,000		\$100,000
Tax Rate	40.00%		
Cost of Capital	12.00%		

- b. Depreciation and Amortization – Complete depreciation and amortization tables for each of the depreciable fixed assets (assume for this example that the equipment has a depreciable life of 7 years, and building a depreciable life of 40 years) and for start-up costs (amortized over 3 years). The annual depreciation amount for each asset is listed as an expense on the Net Cash Flow schedule.

Note: Remember that Land is not a depreciable asset. So in this example, there is not a Depreciation Table for Land. The Book Value of Land remains fixed at its historic (original) cost.

<u>Depreciation Table: Building</u>						
Year	0	1	2	3	4	5
Annual Rate (S/L @ 40 years)		2.50%	2.50%	2.50%	2.50%	2.50%
<u>Annual Depreciation Expense (1/40th of Original Cost)</u>		\$10,000	\$10,000	\$10,000	\$10,000	\$10,000
Ending Book Value	\$400,000	\$390,000	\$380,000	\$370,000	\$360,000	\$350,000

<u>Depreciation Table: Equipment</u>						
Year	0	1	2	3	4	5
Annual Rate (S/L @ 7 years)		14.29%	14.29%	14.29%	14.29%	14.29%
<u>Annual Depreciation Expense (1/7th of Original Cost)</u>		\$100,000	\$100,000	\$100,000	\$100,000	\$100,000
Ending Book Value	\$700,000	\$600,000	\$500,000	\$400,000	\$300,000	\$200,000

<u>Amortization Table: Start-Up Fees</u>				
Year	0	1	2	3
Annual Rate (S/L @ 3 years)		33.33%	33.33%	33.33%
<u>Annual Amortization Expense (1/3rd of Original Cost)</u>		\$20,000	\$20,000	\$20,000
Ending Book Value	\$60,000	\$40,000	\$20,000	\$0

- c. Terminal Year Cash Flow from the liquidation of assets – complete a Net Cash Liquidation Value (NCLV) schedule for each of the Fixed Assets and for each Working Capital account (Cash and Supplies).

Note: The NCLV (shaded in yellow below) equals the Market Value of the assets being liquidated (based on an assumption) minus Taxes paid (or credited). Taxes are only paid (or credited) on gains (or losses) of the sale. The gain or loss is determined by subtracting the Book Value of the asset in the terminal year (taken from the Depreciation Tables – shaded in green above and below) from the Market Value. The combined sum of the NVLC amounts from these schedules is included in NCF in the terminal year.

Note: Despite not being a depreciable asset, and thus not being included in the Depreciation Tables above, the NCLV of Land must be included in this calculation because it is an asset that will be liquidated in the terminal year of the project. Its Book Value does not change from its historic cost.

Terminal Year Net Cash Liquidation Value (NCLV) Tables

<u>Land</u>		<u>Building</u>		<u>Equipment</u>	
Est. Market Value in Yr. 5	\$150,000	Est. Market Value in Yr. 5	\$600,000	Est. Market Value in Yr. 5	\$0
Book Value in Yr. 5	\$100,000	Book Value in Yr. 5	\$350,000	Book Value in Yr. 5	\$200,000
Gain/Loss	\$50,000	Gain/Loss	\$250,000	Gain/Loss	-\$200,000
Taxes Paid/Credit	\$20,000	Taxes Paid/Credit	\$100,000	Taxes Paid/Credit	-\$80,000
NCLV	\$130,000	NCLV	\$500,000	NCLV	\$80,000
<u>Working Capital</u>					
	Year 0	Year 5	Year 5		
	<u>Book Value</u>	<u>Book Value</u>	<u>NCLV</u>		
Cash	\$50,000	\$50,000	\$50,000		
Supplies	<u>\$100,000</u>	<u>\$100,000</u>	<u>\$50,000</u>		
Total	\$150,000	\$150,000	\$100,000		

- d. Annual Net Cash Flow - the schedule of net annual net cash flow (NCF) from operations after taxes for each year of the project life represents the future cash flow (Future Value) produced by the investment. These annual amounts are discounted at the Cost of Capital to determine the Present Values (also labeled DCF for Discounted Cash Flow).

The schedule for Net Cash From Operations (see the opposite page) is the most detailed part of the Capital Budgeting Model because it includes inputs from all of the previous steps: Key Assumptions, Depreciation Tables, and Net Cash Liquidation Value tables.

4. **Project appraisal:** Determine the feasibility of the project using one or all of the appraisal methods:
- Regular Payback – using the actual (Future Value) of projected net cash flows
 - Discounted Payback – using the discounted (Present Value) of projected net cash flows
 - NPV – the sum (“net”) of all discounted cash flows (Present Values) including the initial investment
 - IRR – an Excel calculation based on actual cash flows (Future Values) including the initial investment
 - MIRR – based on an adjustment to the IRR to account for differences in returns and cost of capital
 - Profitability Index – a calculation based on discounted cash flow (Present Values)
5. **Sensitivity Analysis:** Create formulas in the spreadsheet so that the five scenarios can be assessed to determine how changes affect NPV and IRR. A table is created to track and compare the results from the different scenarios. Note that for each Scenario, all variables remain at their original assumption except that which is changed in the Description of Scenario column.

Scenario	NPV	IRR	Description of Scenario
Original	\$581,733	23.91%	
1	-\$128,896	9.22%	25% drop in Revenue for all years
2	\$358,500	19.42%	25% increase in Labor expense for all years
3	\$486,982	22.03%	25% increase in Supplies expense for all years
4	\$527,661	22.81%	25% increase in Fixed Overhead for all years
5	\$209,678	16.38%	Scenarios 2 thru 4 combined
6	\$122,117	15.09%	NCLV of all assets = \$0

Questions to consider when examining the results of each Scenario:

- Is the project more sensitive to changes in revenue or expenses?
- Is the project dependent on the Terminal Year Liquidation Values for its viability? Explain?
- What other scenarios might be worth examining?

Year	0	1	2	3	4	5
Initial Investment (Hard Costs)	-\$1,200,000					
Initial Investment (Soft Costs)	-\$210,000					
Revenue		\$750,000	\$1,500,000	\$1,500,000	\$1,500,000	\$1,500,000
Labor		\$300,000	\$450,000	\$450,000	\$450,000	\$450,000
Supplies		\$100,000	\$200,000	\$200,000	\$200,000	\$200,000
Fixed Overhead		\$100,000	\$100,000	\$100,000	\$100,000	\$100,000
Depreciation: Building		\$10,000	\$10,000	\$10,000	\$10,000	\$10,000
Depreciation: Equipment		\$100,000	\$100,000	\$100,000	\$100,000	\$100,000
Amortization: Start-Up Fees		\$20,000	\$20,000	\$20,000	\$0	\$0
Total Operating Expenses		\$630,000	\$880,000	\$880,000	\$860,000	\$860,000
Earnings Before Taxes		\$120,000	\$620,000	\$620,000	\$640,000	\$640,000
Taxes		\$48,000	\$248,000	\$248,000	\$256,000	\$256,000
Net Income		\$72,000	\$372,000	\$372,000	\$384,000	\$384,000
Add Depreciation and Amortization		\$130,000	\$130,000	\$130,000	\$110,000	\$110,000
Net Cash from Operations		\$202,000	\$502,000	\$502,000	\$494,000	\$494,000
Terminal Year Cash Flow (from fixed assets)						\$710,000
Terminal Year Cash Flow (from working capital)						\$100,000
Project Cash Flow Summary						
Year	0	1	2	3	4	5
Total Project Cash Flow - FV	-\$1,410,000	\$202,000	\$502,000	\$502,000	\$494,000	\$1,304,000
Discounted Cash Flow - PV	-\$1,410,000	\$180,357	\$400,191	\$357,314	\$313,946	\$739,925
Payback Schedules						
Cumulative Cash Flow (FV)	-\$1,410,000	-\$1,208,000	-\$706,000	-\$204,000	\$290,000	\$1,594,000
Cumulative Cash Flow (PV)	-\$1,410,000	-\$1,229,643	-\$829,452	-\$472,138	-\$158,192	\$581,733
Decision Models						
Payback - Regular		3.41	Using the simple "eye-ball" formula (see instructions below)			
Payback - Discounted		4.21				
NPV		\$581,733				
IRR		23.91%				
MIRR (@ 12% reinvestment)		20.01%				
Profitability Index		1.41				

Decision Criteria on a Capital Budgeting Project

As noted above, the decision to accept or reject a capital budgeting project should be based on quantifiable data. The six most common methods used to quantify decision criteria for capital budgeting projects are:

<u>Name of Method</u>	<u>Data Produced</u>
Regular Payback	The number of years it takes for <u>regular</u> Net Annual Cash Flow to equal the Initial Investment. The project will be accepted if this number is below a critical threshold and rejected if this number exceeds a critical threshold (e.g. 3 years). See example on the following page.
Discounted Payback	The number of Years it takes for the <u>discounted</u> Net Annual Cash Flow to equal the Initial Investment. The project will be accepted if this number is below a critical threshold and rejected if this number exceeds a critical threshold (e.g. 3 years). See example on the following page.
	Note: As long as the Interest Rate is greater than 0, the Discounted Payback will always be higher (i.e., take more years) than the Regular Payback. Why?
Net Present Value (NPV)	The “net” amount (+\$ or -\$) produced by summing the Present Value of all project Cash Flow, beginning with the Initial Investment (input as a negative number) and followed by all Net Annual Cash Flow (which is either positive or negative based on actual projections). The project will be accepted if NPV is positive, and rejected if NPV is negative.
Internal Rate of Return (IRR)	The percent (%) produced when the Net Annual Cash Flow is measured against the Initial Investment. The project is accepted if the IRR is greater than the Cost of Capital, and rejected if the IRR is less than the Cost of Capital.
	Note: The IRR is the Discount Rate that produces an NPV of 0. Thus, if the IRR exceeds the Cost of Capital, NPV will be positive. Conversely, if the IRR is less than the Cost of Capital, NPV will be negative.
Modified IRR	The same percent (%) produced by the IRR formula, except that the with the modification being that the formula assumes funds are reinvested at the Cost of Capital rather than at the IRR. The method is used to modify the IRR when it is certain that Net Annual Cash Flow will not be reinvested in the project.
Profitability Index (PI)	The number (#) produced when the sum of all Net Annual Cash Flow (in Present Value format) is divided by the Initial Investment. This number will be greater than 1 if NPV is positive, and less than 1 if NPV is negative. This method is often used to choose between mutually exclusive projects that produce similar NPV or IRR calculations. The project with the highest PI will be seen as the one adding the most value.

According to industry data, IRR is most popular with business managers, NPV is most popular with academics, the Payback methods (regular and discounted) are used most often in conjunction with NPV and/or IRR, and the Profitability Index is most often used to differentiate mutually exclusive projects.

Payback (Regular and Discounted) is easy to calculate using Excel (or by hand using a calculator):

INSTRUCTIONS FOR CALCULATING A PRECISE REGULAR PAYBACK						
Year	0	1	2	3	4	5
Project Annual Cash Flow (FV)	-\$1,410,000	\$202,000	\$502,000	\$502,000	\$494,000	\$1,304,000
Cumulative Cash Flow (FV)		-\$1,208,000	-\$706,000	-\$204,000	\$290,000	\$1,594,000
			<u>Data</u>	<u>Step</u>	<u>Instruction</u>	
The year that Payback occurs =			4	A	In which year does the cumulative balance become positive?	
The deficit at beginning of the Payback year =			-\$204,000	B	Note: Remove the negative sign	
The Net Cash Flow (FV) during payback year =			\$494,000	C		
Calculate the fraction of the payback year =			0.41	D	B ÷ C (remove the netative sign)	
Number of years prior to payback year =			3	E	A minus 1	
Payback =			3.41	F	D + E	
INSTRUCTIONS FOR CALCULATING A PRECISE DISCOUNTED PAYBACK						
Year	0	1	2	3	4	5
Project Annual Cash Flow (PV)	-\$1,410,000	\$180,357	\$400,191	\$357,314	\$313,946	\$739,925
Cumulative Cash Flow (PV)		-\$1,229,643	-\$829,452	-\$472,138	-\$158,192	\$581,733
			<u>Data</u>	<u>Step</u>	<u>Instruction</u>	
The year that Payback occurs =			5	A	In which year does the cumulative balance become positive?	
The deficit at beginning of the Payback year =			-\$158,192	B	Note: Remove the negative sign	
The Net Cash Flow (FV) during payback year =			\$739,925	C		
Calculate the fraction of the payback year =			0.21	D	B ÷ C (remove the netative sign)	
Number of years prior to payback year =			4	E	A minus 1	
Payback =			4.21	F	D + E	

NPV is easy to calculate in Excel (or by hand using a calculator) using the following steps:

- Calculate the Present Value of each Net Annual Cash Flow (remember Year 0 is already a Present Value). Note that if Net Annual Cash Flow in any given year is negative, it will be a negative PV number also.
- Input the Initial Investment (Year 0) as a negative number because this is cash required for the project, not cash produced by the project.
- Sum the Present Values (including negative Year 0) to determine the NPV

The Excel formula for NPV is **=NPV(rate, value1, value2, value3,...)** but students must be aware that the Initial Investment (which would be **value0** in the formula) is NOT included in the Excel formula. It must be subtracted (or added as a negative number) to the sum produced by the formula. The **rate** is the cost of capital (or discount factor).

IRR can only be produced in Excel or with special financial calculators. The formula is **=IRR(values, [guess])**. The **values** in this formula is the string of regular Net Annual Cash Flow beginning with Year 0 and ending in the terminal year. Remember, DO NOT use present values in this formula, but rather use the regular (future) values. The **[guess]** is not required, but if used it is simply a percent (%) that the student guesses will be the IRR.

MIRR can only be produced in Excel. The formula is **=MIRR(values, finance_rate, reinvest_rate)**. The **values** is the same as in the IRR formula, the **finance_rate** is the cost of capital, and the **reinvest_rate** is whatever rate is assumed to be the rate at which surplus funds will be reinvested (which is often the same as the finance rate).

The **Profitability Index (PI)** is easy to calculate in Excel or by hand using a calculator:

- Sum the Present Value of all Net Annual Cash Flow (not including the Initial Investment in Year 0).
- Divide this Sum by the Initial Investment (take out the negative sign on the Initial Investment in Year 0).

Apple Inc.

CONSOLIDATED BALANCE SHEETS

(In millions, except number of shares, which are reflected in thousands, and par value)

	September 30, 2023	September 24, 2022
ASSETS:		
Current assets:		
Cash and cash equivalents	\$ 29,965	\$ 23,646
Marketable securities	31,590	24,658
Accounts receivable, net	29,508	28,184
Vendor non-trade receivables	31,477	32,748
Inventories	6,331	4,946
Other current assets	14,695	21,223
Total current assets	143,566	135,405
Non-current assets:		
Marketable securities	100,544	120,805
Property, plant and equipment, net	43,715	42,117
Other non-current assets	64,758	54,428
Total non-current assets	209,017	217,350
Total assets	<u>\$ 352,583</u>	<u>\$ 352,755</u>
LIABILITIES AND SHAREHOLDERS' EQUITY:		
Current liabilities:		
Accounts payable	\$ 62,611	\$ 64,115
Other current liabilities	58,829	60,845
Deferred revenue	8,061	7,912
Commercial paper	5,985	9,982
Term debt	9,822	11,128
Total current liabilities	145,308	153,982
Non-current liabilities:		
Term debt	95,281	98,959
Other non-current liabilities	49,848	49,142
Total non-current liabilities	145,129	148,101
Total liabilities	290,437	302,083
Commitments and contingencies		
Shareholders' equity:		
Common stock and additional paid-in capital, \$0.00001 par value: 50,400,000 shares authorized; 15,550,061 and 15,943,425 shares issued and outstanding, respectively	73,812	64,849
Accumulated deficit	(214)	(3,068)
Accumulated other comprehensive loss	(11,452)	(11,109)
Total shareholders' equity	62,146	50,672
Total liabilities and shareholders' equity	<u>\$ 352,583</u>	<u>\$ 352,755</u>

See accompanying Notes to Consolidated Financial Statements.

Apple Inc.

CONSOLIDATED STATEMENTS OF OPERATIONS

(In millions, except number of shares, which are reflected in thousands, and per-share amounts)

	Years ended		
	September 30, 2023	September 24, 2022	September 25, 2021
Net sales:			
Products	\$ 298,085	\$ 316,199	\$ 297,392
Services	85,200	78,129	68,425
Total net sales	383,285	394,328	365,817
Cost of sales:			
Products	189,282	201,471	192,266
Services	24,855	22,075	20,715
Total cost of sales	214,137	223,546	212,981
Gross margin	169,148	170,782	152,836
Operating expenses:			
Research and development	29,915	26,251	21,914
Selling, general and administrative	24,932	25,094	21,973
Total operating expenses	54,847	51,345	43,887
Operating income	114,301	119,437	108,949
Other income/(expense), net	(565)	(334)	258
Income before provision for income taxes	113,736	119,103	109,207
Provision for income taxes	16,741	19,300	14,527
Net income	\$ 96,995	\$ 99,803	\$ 94,680
Earnings per share:			
Basic	\$ 6.16	\$ 6.15	\$ 5.67
Diluted	\$ 6.13	\$ 6.11	\$ 5.61
Shares used in computing earnings per share:			
Basic	15,744,231	16,215,963	16,701,272
Diluted	15,812,547	16,325,819	16,864,919

See accompanying Notes to Consolidated Financial Statements.

MALIBU BOATS, INC. AND SUBSIDIARIES

Consolidated Balance Sheets (In thousands, except share and per share data)

	June 30, 2023	June 30, 2022
Assets		
Current assets		
Cash	\$ 78,937	\$ 83,744
Trade receivables, net	68,381	51,598
Inventories, net	171,189	157,002
Prepaid expenses and other current assets	7,827	6,155
Total current assets	326,334	298,499
Property, plant and equipment, net	204,792	170,718
Goodwill	100,577	100,804
Other intangible assets, net	221,458	228,304
Deferred tax assets	62,573	42,314
Other assets	10,190	10,687
Total assets	<u>\$925,924</u>	<u>\$851,326</u>
Liabilities		
Current liabilities		
Current maturities of long-term debt	\$ —	\$ 1,563
Accounts payable	40,402	44,368
Accrued expenses	187,078	87,742
Income tax and distribution payable	847	1,670
Payable pursuant to tax receivable agreement, current portion	4,111	3,958
Total current liabilities	232,438	139,301
Deferred tax liabilities	28,453	26,965
Other liabilities	9,926	11,855
Payable pursuant to tax receivable agreement, less current portion	39,354	41,583
Long-term debt	—	118,054
Total liabilities	<u>310,171</u>	<u>337,758</u>
Commitments and contingencies (See Note 17)		
Stockholders' Equity		
Class A Common Stock, par value \$0.01 per share, 100,000,000 shares authorized; 20,603,822 shares issued and outstanding as of June 30, 2023; 20,501,081 shares issued and outstanding as of June 30, 2022	204	203
Class B Common Stock, par value \$0.01 per share, 25,000,000 shares authorized; 12 shares issued and outstanding as of June 30, 2023; 10 shares issued and outstanding as of June 30, 2022	—	—
Preferred Stock, par value \$0.01 per share; 25,000,000 shares authorized; no shares issued and outstanding as of June 30, 2023; no shares issued and outstanding as of June 30, 2022	—	—
Additional paid in capital	86,321	85,294
Accumulated other comprehensive loss, net of tax	(4,340)	(3,507)
Accumulated earnings	525,697	421,184
Total stockholders' equity attributable to Malibu Boats, Inc.	607,882	503,174
Non-controlling interest	7,871	10,394
Total stockholders' equity	<u>615,753</u>	<u>513,568</u>
Total liabilities and stockholders' equity	<u>\$925,924</u>	<u>\$851,326</u>

The accompanying notes are an integral part of these Consolidated Financial Statements.

MALIBU BOATS, INC. AND SUBSIDIARIES

Consolidated Statements of Operations and Comprehensive Income (In thousands, except share and per share data)

	Fiscal Year Ended June 30,		
	2023	2022	2021
Net sales	\$ 1,388,365	\$ 1,214,877	\$ 926,515
Cost of sales	1,037,070	904,826	690,030
Gross profit	351,295	310,051	236,485
Operating expenses:			
Selling and marketing	24,009	22,900	17,540
General and administrative	175,694	66,371	61,915
Amortization	6,808	6,957	7,255
Operating income	144,784	213,823	149,775
Other expense (income), net:			
Other expense (income), net	331	983	(1,015)
Interest expense	2,962	2,875	2,529
Other expense (income), net	3,293	3,858	1,514
Income before provision for income taxes	141,491	209,965	148,261
Provision for income taxes	33,581	46,535	33,979
Net income	107,910	163,430	114,282
Net income attributable to non-controlling interest	3,397	5,798	4,441
Net income attributable to Malibu Boats, Inc.	<u>\$ 104,513</u>	<u>\$ 157,632</u>	<u>\$ 109,841</u>
Comprehensive income:			
Net income	\$ 107,910	\$ 163,430	\$ 114,282
Other comprehensive (loss) income			
Change in cumulative translation adjustment	(833)	(1,868)	1,493
Other comprehensive (loss) income	(833)	(1,868)	1,493
Comprehensive income	107,077	161,562	115,775
Less: comprehensive income attributable to non-controlling interest, net of tax	3,371	5,731	4,507
Comprehensive income attributable to Malibu Boats, Inc., net of tax	<u>\$ 103,706</u>	<u>\$ 155,831</u>	<u>\$ 111,268</u>
Weighted average shares outstanding used in computing net income per share:			
Basic	20,501,844	20,749,237	20,752,652
Diluted	20,641,173	20,986,256	21,011,087
Net income available to Class A Common Stock per share:			
Basic	\$ 5.10	\$ 7.60	\$ 5.29
Diluted	\$ 5.06	\$ 7.51	\$ 5.23

Item 8. Financial Statements and Supplementary Data

CONSOLIDATED BALANCE SHEETS

Yellow Corporation and Subsidiaries

(Amounts in millions except share and per share data)	December 31, 2022	December 31, 2021
Assets		
Current Assets:		
Cash and cash equivalents	\$ 235.1	\$ 310.7
Restricted amounts held in escrow	3.9	4.1
Accounts receivable, less allowances of \$23.7 and \$13.2, respectively	599.7	663.7
Prepaid expenses and other	75.4	65.0
Total current assets	914.1	1,043.5
Property and Equipment:		
Cost	3,109.0	3,164.6
Less – accumulated depreciation	(1,940.0)	(2,032.3)
Net property and equipment	1,169.0	1,132.3
Deferred income taxes, net	0.3	1.4
Pension	34.5	40.5
Operating lease right-of-use assets	139.7	184.8
Other assets	21.7	23.1
Total Assets	\$ 2,279.3	\$ 2,425.6
Liabilities and Shareholders' Deficit		
Current Liabilities:		
Accounts payable	\$ 188.6	\$ 178.4
Wages, vacations and employee benefits	221.4	252.5
Current operating lease liabilities	53.1	76.5
Claims and insurance accruals	116.6	125.9
Other accrued taxes	27.9	72.8
Other current and accrued liabilities	37.6	45.7
Current maturities of long-term debt	71.8	72.3
Total current liabilities	717.0	824.1
Other Liabilities:		
Long-term debt, less current portion	1,466.2	1,482.2
Pension and postretirement	134.0	88.2
Operating lease liabilities	94.6	118.9
Claims and other liabilities	249.0	275.7
Commitments and contingencies		
Shareholders' Deficit:		
Cumulative preferred stock, \$1 par value per share - authorized 5,000,000 shares	—	—
Common stock, \$0.01 par value per share - authorized 95,000,000 shares, issued 51,601,000 and 50,955,000 shares, respectively	0.5	0.5
Capital surplus	2,393.4	2,388.3
Accumulated deficit	(2,453.2)	(2,475.0)
Accumulated other comprehensive loss	(229.5)	(184.6)
Treasury stock, at cost	(92.7)	(92.7)
Total shareholders' deficit	(381.5)	(363.5)
Total Liabilities and Shareholders' Deficit	\$ 2,279.3	\$ 2,425.6

The accompanying notes are an integral part of these statements.

CONSOLIDATED STATEMENTS OF OPERATIONS
Yellow Corporation and Subsidiaries
For the Years Ended December 31

(Amounts in millions except per share data; shares in thousands)	2022	2021	2020
Operating Revenue	\$ 5,244.7	\$ 5,121.8	\$ 4,513.7
Operating Expenses:			
Salaries, wages and employee benefits	2,808.3	2,921.7	2,770.1
Fuel, operating expenses and supplies	1,076.6	858.1	719.1
Purchased transportation	748.5	800.2	638.8
Depreciation and amortization	143.4	143.6	134.9
Other operating expenses	308.1	293.9	239.6
(Gains) losses on property disposals, net	(38.0)	0.7	(45.3)
Total operating expenses	5,046.9	5,018.2	4,457.2
Operating Income	197.8	103.6	56.5
Nonoperating Expenses:			
Interest expense	162.9	150.7	135.9
Non-union pension and postretirement benefits	10.5	59.3	(6.3)
Other, net	(2.1)	(0.4)	—
Nonoperating expenses, net	171.3	209.6	129.6
Income (loss) before income taxes	26.5	(106.0)	(73.1)
Income tax expense (benefit)	4.7	3.1	(19.6)
Net Income (Loss)	\$ 21.8	\$ (109.1)	\$ (53.5)
Average Common Shares Outstanding - Basic	51,346	50,720	41,694
Average Common Shares Outstanding - Diluted	52,233	50,720	41,694
Income (Loss) Per Share - Basic	\$ 0.42	\$ (2.15)	\$ (1.28)
Income (Loss) Per Share - Diluted	\$ 0.42	\$ (2.15)	\$ (1.28)

The accompanying notes are an integral part of these statements.

FERRELLGAS PARTNERS, L.P. AND SUBSIDIARIES
CONSOLIDATED BALANCE SHEETS
(in thousands, except unit data)

	July 31, 2023	July 31, 2022
ASSETS		
Current assets:		
Cash and cash equivalents (including \$11,126 and \$11,208 of restricted cash at July 31, 2023 and 2022, respectively)	\$ 137,347	\$ 158,737
Accounts and notes receivable, net	159,379	150,395
Inventories	98,104	115,187
Price risk management asset	11,966	43,015
Prepaid expenses and other current assets	29,135	30,764
Total current assets	435,931	498,098
Property, plant and equipment, net	615,174	603,148
Goodwill, net	257,006	257,099
Intangible assets (net of accumulated amortization of \$349,614 and \$440,121 at July 31, 2023 and 2022, respectively)	106,615	97,638
Operating lease right-of-use assets	57,839	72,888
Other assets, net	58,838	79,244
Total assets	<u>\$ 1,531,403</u>	<u>\$ 1,608,115</u>
LIABILITIES, MEZZANINE AND EQUITY (DEFICIT)		
Current liabilities:		
Accounts payable	\$ 35,115	\$ 57,586
Current portion of long-term debt	2,597	1,792
Current operating lease liabilities	24,600	25,824
Other current liabilities	197,030	218,610
Total current liabilities	259,342	303,812
Long-term debt	1,456,184	1,450,016
Operating lease liabilities	34,235	47,231
Other liabilities	29,084	43,518
Contingencies and commitments (Note P)		
Mezzanine equity:		
Senior preferred units, net of issue discount and offering costs (700,000 units outstanding at July 31, 2023 and 2022)	651,349	651,349
Equity (Deficit):		
Limited partner unitholders		
Class A (4,857,605 units outstanding at July 31, 2023 and 2022)	(1,205,103)	(1,229,823)
Class B (1,300,000 units outstanding at July 31, 2023 and 2022)	383,012	383,012
General partner unitholder (49,496 units outstanding at July 31, 2023 and 2022)	(70,566)	(71,320)
Accumulated other comprehensive income	1,059	37,907
Total Ferrellgas Partners, L.P. deficit	(891,598)	(880,224)
Noncontrolling interest	(7,193)	(7,587)
Total deficit	(898,791)	(887,811)
Total liabilities, mezzanine and deficit	<u>\$ 1,531,403</u>	<u>\$ 1,608,115</u>

See notes to consolidated financial statements.

FERRELLGAS PARTNERS, L.P. AND SUBSIDIARIES
CONSOLIDATED STATEMENTS OF OPERATIONS
(in thousands, except per unit data)

	For the year ended July 31,		
	2023	2022	2021
Revenues:			
Propane and other gas liquids sales	\$ 1,916,892	\$ 2,017,879	\$ 1,668,852
Other	109,573	96,661	85,458
Total revenues	2,026,465	2,114,540	1,754,310
Costs and expenses:			
Cost of sales - propane and other gas liquids sales	1,003,357	1,174,004	881,936
Cost of sales - other	15,913	12,509	12,728
Operating expense - personnel, vehicle, plant and other	577,520	520,603	465,816
Operating expense - equipment lease expense	23,252	23,094	27,062
Depreciation and amortization expense	93,370	89,897	85,382
General and administrative expense	70,738	52,780	60,065
Non-cash employee stock ownership plan compensation charge	2,935	3,170	3,215
Loss (gain) on asset sales and disposals	5,691	(6,618)	1,831
Operating income	233,689	245,101	216,275
Interest expense	(97,712)	(100,093)	(173,616)
Loss on extinguishment of debt	—	—	(104,834)
Other income, net	2,625	4,833	4,246
Reorganization expense - professional fees	—	—	(10,443)
Earnings (loss) before income taxes	138,602	149,841	(68,372)
Income tax expense	981	981	741
Net earnings (loss)	137,621	148,860	(69,113)
Net earnings (loss) attributable to noncontrolling interest	740	867	(702)
Net earnings (loss) attributable to Ferrellgas Partners, L.P.	\$ 136,881	\$ 147,993	\$ (68,411)
Class A unitholders' interest in net earnings (loss) (Note R)	\$ 10,171	\$ (18,770)	\$ (91,751)
Basic and diluted net earnings (loss) per Class A Unit (Note R)	\$ 2.09	\$ (3.86)	\$ (18.89)

See notes to consolidated financial statements.

COSTCO WHOLESALE CORPORATION
CONSOLIDATED BALANCE SHEETS
(amounts in millions, except par value and share data)

	September 3, 2023	August 28, 2022
ASSETS		
CURRENT ASSETS		
Cash and cash equivalents	\$ 13,700	\$ 10,203
Short-term investments	1,534	846
Receivables, net	2,285	2,241
Merchandise inventories	16,651	17,907
Other current assets	1,709	1,499
Total current assets	35,879	32,696
OTHER ASSETS		
Property and equipment, net	26,684	24,646
Operating lease right-of-use assets	2,713	2,774
Other long-term assets	3,718	4,050
TOTAL ASSETS	\$ 68,994	\$ 64,166
LIABILITIES AND EQUITY		
CURRENT LIABILITIES		
Accounts payable	\$ 17,483	\$ 17,848
Accrued salaries and benefits	4,278	4,381
Accrued member rewards	2,150	1,911
Deferred membership fees	2,337	2,174
Current portion of long-term debt	1,081	73
Other current liabilities	6,254	5,611
Total current liabilities	33,583	31,998
OTHER LIABILITIES		
Long-term debt, excluding current portion	5,377	6,484
Long-term operating lease liabilities	2,426	2,482
Other long-term liabilities	2,550	2,555
TOTAL LIABILITIES	43,936	43,519
COMMITMENTS AND CONTINGENCIES		
EQUITY		
Preferred stock \$0.005 par value; 100,000,000 shares authorized; no shares issued and outstanding	—	—
Common stock \$0.005 par value; 900,000,000 shares authorized; 442,793,000 and 442,664,000 shares issued and outstanding	2	2
Additional paid-in capital	7,340	6,884
Accumulated other comprehensive loss	(1,805)	(1,829)
Retained earnings	19,521	15,585
Total Costco stockholders' equity	25,058	20,642
Noncontrolling interests	—	5
TOTAL EQUITY	25,058	20,647
TOTAL LIABILITIES AND EQUITY	\$ 68,994	\$ 64,166

COSTCO WHOLESALE CORPORATION
CONSOLIDATED STATEMENTS OF INCOME
(amounts in millions, except per share data)

	<u>53 Weeks Ended</u> <u>September 3,</u> <u>2023</u>	<u>52 Weeks Ended</u> <u>August 28,</u> <u>2022</u>	<u>52 Weeks Ended</u> <u>August 29,</u> <u>2021</u>
REVENUE			
Net sales	\$ 237,710	\$ 222,730	\$ 192,052
Membership fees	4,580	4,224	3,877
Total revenue	242,290	226,954	195,929
OPERATING EXPENSES			
Merchandise costs	212,586	199,382	170,684
Selling, general and administrative	21,590	19,779	18,537
Operating income	8,114	7,793	6,708
OTHER INCOME (EXPENSE)			
Interest expense	(160)	(158)	(171)
Interest income and other, net	533	205	143
INCOME BEFORE INCOME TAXES	8,487	7,840	6,680
Provision for income taxes	2,195	1,925	1,601
Net income including noncontrolling interests	6,292	5,915	5,079
Net income attributable to noncontrolling interests	—	(71)	(72)
NET INCOME ATTRIBUTABLE TO COSTCO	<u>\$ 6,292</u>	<u>\$ 5,844</u>	<u>\$ 5,007</u>
NET INCOME PER COMMON SHARE ATTRIBUTABLE TO COSTCO:			
Basic	<u>\$ 14.18</u>	<u>\$ 13.17</u>	<u>\$ 11.30</u>
Diluted	<u>\$ 14.16</u>	<u>\$ 13.14</u>	<u>\$ 11.27</u>
Shares used in calculation (000's)			
Basic	443,854	443,651	443,089
Diluted	444,452	444,757	444,346



KORN FERRY AND SUBSIDIARIES CONSOLIDATED BALANCE SHEETS

	April 30,	
	2024	2023
	(in thousands, except per share data)	
ASSETS		
Cash and cash equivalents	\$ 941,005	\$ 844,024
Marketable securities	42,742	44,837
Receivables due from clients, net of allowance for doubtful accounts of \$44,192 and \$44,377 at April 30, 2024 and 2023, respectively	541,014	569,601
Income taxes and other receivables	40,696	67,512
Unearned compensation	59,247	63,476
Prepaid expenses and other assets	49,456	49,219
Total current assets	1,674,160	1,638,669
Marketable securities, non-current	211,681	179,040
Property and equipment, net	161,849	161,876
Operating lease right-of-use assets, net	160,464	142,690
Cash surrender value of company-owned life insurance policies, net of loans	218,977	197,998
Deferred income taxes	133,564	102,057
Goodwill	908,376	909,491
Intangible assets, net	88,833	114,426
Unearned compensation, non-current	99,913	103,607
Investments and other assets	21,052	24,590
Total assets	\$ 3,678,869	\$ 3,574,444
LIABILITIES AND STOCKHOLDERS' EQUITY		
Accounts payable	\$ 50,112	\$ 53,386
Income taxes payable	24,076	19,969
Compensation and benefits payable	525,466	532,934
Operating lease liability, current	36,073	45,821
Other accrued liabilities	298,792	324,150
Total current liabilities	934,519	976,260
Deferred compensation and other retirement plans	440,396	396,534
Operating lease liability, non-current	143,507	119,220
Long-term debt	396,946	396,194
Deferred tax liabilities	4,540	5,352
Other liabilities	21,636	27,879
Total liabilities	1,941,544	1,921,439
Commitments and contingencies		
Stockholders' equity		
Common stock: \$0.01 par value, 150,000 shares authorized, 77,460 and 76,693 shares issued and 51,983 and 52,269 shares outstanding at April 30, 2024 and 2023, respectively	414,885	429,754
Retained earnings	1,425,844	1,311,081
Accumulated other comprehensive loss, net	(107,671)	(92,764)
Total Korn Ferry stockholders' equity	1,733,058	1,648,071
Noncontrolling interest	4,267	4,934
Total stockholders' equity	1,737,325	1,653,005
Total liabilities and stockholders' equity	\$ 3,678,869	\$ 3,574,444

The accompanying notes are an integral part of these consolidated financial statements.



KORN FERRY AND SUBSIDIARIES CONSOLIDATED STATEMENTS OF INCOME

	Year Ended April 30,		
	2024	2023	2022
	(in thousands, except per share data)		
Fee revenue	\$ 2,762,671	\$ 2,835,408	\$ 2,626,718
Reimbursed out-of-pocket engagement expenses	32,834	28,428	16,737
Total revenue	2,795,505	2,863,836	2,643,455
Compensation and benefits	1,844,164	1,901,203	1,741,452
General and administrative expenses	259,039	268,458	237,272
Reimbursed expenses	32,834	28,428	16,737
Cost of services	300,015	238,499	114,399
Depreciation and amortization	77,966	68,335	63,521
Restructuring charges, net	68,558	42,573	—
Total operating expenses	2,582,576	2,547,496	2,173,381
Operating income	212,929	316,340	470,074
Other income (loss), net	30,681	5,261	(11,880)
Interest expense, net	(20,968)	(25,864)	(25,293)
Income before provision for income taxes	222,642	295,737	432,901
Income tax provision	50,081	82,683	102,056
Net income	172,561	213,054	330,845
Net income attributable to noncontrolling interest	(3,407)	(3,525)	(4,485)
Net income attributable to Korn Ferry	\$ 169,154	\$ 209,529	\$ 326,360
Earnings per common share attributable to Korn Ferry:			
Basic	\$ 3.25	\$ 3.98	\$ 6.04
Diluted	\$ 3.23	\$ 3.95	\$ 5.98
Weighted-average common shares outstanding:			
Basic	51,038	51,482	52,807
Diluted	51,432	51,883	53,401
Cash dividends declared per share:	\$ 1.02	\$ 0.60	\$ 0.48

The accompanying notes are an integral part of these consolidated financial statements.

STELLAR BANCORP, INC.
CONSOLIDATED BALANCE SHEETS

	December 31,	
	2023	2022
	(In thousands, except shares and par value)	
ASSETS		
Cash and due from banks	\$ 121,004	\$ 67,063
Interest-bearing deposits at other financial institutions	278,233	304,642
Total cash and cash equivalents	399,237	371,705
Available for sale securities, at fair value	1,395,680	1,807,586
Loans held for investment	7,925,133	7,754,751
Less: allowance for credit losses on loans	(91,684)	(93,180)
Loans, net	7,833,449	7,661,571
Accrued interest receivable	44,244	44,743
Premises and equipment, net	118,683	126,803
Federal Home Loan Bank stock	25,051	15,058
Bank-owned life insurance	105,084	103,094
Goodwill	497,318	497,260
Core deposit intangibles, net	116,712	143,525
Other assets	111,681	129,092
TOTAL ASSETS	\$ 10,647,139	\$ 10,900,437
LIABILITIES AND SHAREHOLDERS' EQUITY		
LIABILITIES:		
Deposits:		
Noninterest-bearing	\$ 3,546,815	\$ 4,230,169
Interest-bearing		
Demand	1,659,999	1,591,828
Money market and savings	2,136,777	2,575,923
Certificates and other time	1,529,876	869,712
Total interest-bearing deposits	5,326,652	5,037,463
Total deposits	8,873,467	9,267,632
Accrued interest payable	11,288	2,098
Borrowed funds	50,000	63,925
Subordinated debt	109,765	109,367
Other liabilities	81,601	74,239
Total liabilities	9,126,121	9,517,261
COMMITMENTS AND CONTINGENCIES (See Note 15)		
SHAREHOLDERS' EQUITY:		
Preferred stock, \$0.01 par value; 10,000,000 shares authorized; no shares issued or outstanding at both December 31, 2023 and 2022	—	—
Common stock, \$0.01 par value; 140,000,000 shares authorized, 53,291,079 shares issued and outstanding at December 31, 2023 and 52,954,985 shares issued and outstanding at December 31, 2022	533	530
Capital surplus	1,232,627	1,222,761
Retained earnings	405,945	303,146
Accumulated other comprehensive loss	(118,087)	(143,261)
Total shareholders' equity	1,521,018	1,383,176
TOTAL LIABILITIES AND SHAREHOLDERS' EQUITY	\$ 10,647,139	\$ 10,900,437

See notes to consolidated financial statements.

STELLAR BANCORP, INC.
CONSOLIDATED STATEMENTS OF INCOME

	Years Ended December 31,		
	2023	2022	2021
	(Dollars in thousands, except per share data)		
INTEREST INCOME:			
Loans, including fees	\$ 537,722	\$ 280,375	\$ 230,713
Securities:			
Taxable	38,494	27,128	11,889
Tax-exempt	2,553	10,733	9,909
Deposits in other financial institutions	12,048	4,758	673
Total interest income	590,817	322,994	253,184
INTEREST EXPENSE:			
Demand, money market and savings deposits	87,335	19,139	5,365
Certificates and other time deposits	41,286	7,825	11,628
Borrowed funds	17,807	1,216	1,878
Subordinated debt	7,630	5,856	5,749
Total interest expense	154,058	34,036	24,620
NET INTEREST INCOME	436,759	288,958	228,564
Provision for credit losses	8,943	50,712	(2,322)
Net interest income after provision for credit losses	427,816	238,246	230,886
NONINTEREST INCOME:			
Nonsufficient funds fees and overdraft charges	1,298	834	464
Service charges on deposit accounts	4,766	2,856	1,671
Gain (loss) on sale of assets	390	4,050	(272)
Bank-owned life insurance income	2,178	1,125	554
Debit card and ATM income	4,996	4,465	2,996
Other	10,934	7,024	3,149
Total noninterest income	24,562	20,354	8,562
NONINTEREST EXPENSE:			
Salaries and employee benefits	157,034	107,554	90,177
Net occupancy and equipment	16,932	10,335	9,144
Depreciation	7,584	4,951	4,254
Data processing and software amortization	19,526	11,337	8,862
Professional fees	7,955	3,583	3,025
Regulatory assessments and FDIC insurance	11,032	4,914	3,407
Amortization of intangibles	26,883	9,303	3,296
Communications	2,796	1,800	1,406
Advertising	3,627	2,460	1,692
Acquisition and merger-related expenses	15,555	24,138	2,011
Other	21,570	15,701	12,280
Total noninterest expense	290,494	196,076	139,554
INCOME BEFORE INCOME TAXES	161,884	62,524	99,894
Provision for income taxes	31,387	11,092	18,341
NET INCOME	\$ 130,497	\$ 51,432	\$ 81,553
EARNINGS PER SHARE:			
Basic	\$ 2.45	\$ 1.48	\$ 2.85
Diluted	\$ 2.45	\$ 1.47	\$ 2.82
DIVIDENDS PER SHARE	\$ 0.52	\$ 0.43	\$ 0.34

See notes to consolidated financial statements.

Benedictine College School of Business

FINC-3100 Principles of Finance

Fall 2024

MWF 11:00 – 11:50 am in WEST 120

Instructor: Michael King
Office: Ferrell Academic Center 106
Phone: (913) 360-7160
mking@benedictine.edu

Office Hours: MWF: 10.00 am – 11.00 am
MW: 2.00 pm – 3.00 pm
TR: by appointment
[Available if in office all times](#)

Required Materials: *Textbooks* are provided by Prof. King (the \$15 course fee covers this item).

Course Description: This course is a study of the firm's problems of obtaining, allocating, and managing capital funds. Students will examine the sources of capital funds, the costs of obtaining these funds, the management of the firm's capital structure, and financial planning for growth and expansion. Emphasis is placed on composition of capital structure, capital budgeting, and cost of capital theories. *Prerequisite:* ACCT-2090

General Course Objectives:

1. To gain knowledge and understanding of the terminology of corporate finance
2. To gain competence in fundamental financial principles, concepts, and theories
3. To develop the ability to view the business from a financial perspective
4. To apply these skills to successfully analyze and solve business problems related to finance

Specific Course Objectives: Students should learn how to:

1. Read, understand, and interpret financial statements.
2. Use financial and accounting information to perform financial analysis.
3. Produce pro forma financial statements and project financial information into the future.
4. Understand and analyze issues related to working capital management.
5. Understand, analyze, and calculate Time Value of Money principles.
6. Understand, analyze, and interpret principles related to interest rates and required returns.
7. Understand, analyze, and value corporate debt (including publicly traded bonds).
8. Understand, analyze, and value corporate equity (including publicly traded stock).
9. Understand, analyze, and calculate cost of capital.
10. Conduct a complete capital budgeting analysis.

Teaching Method: To achieve these objectives, this course will use a combination of lecture, class discussion, and problem-solving covering the assigned reading material. In addition, students will complete homework problems using Excel spreadsheets that utilize their developing knowledge of finance as the semester proceeds. To succeed in this class, the student should:

- Read the assigned material from the *Textbook* prior to class
- Attend all classes, pay attention, and participate in the class discussions
- Be prepared for the graded work by reading, listening, and completing assignments
- Complete the ten homework problems and thirty quizzes on time
- Prepare for, and perform well on, the three Exams

General Education Requirement: This course satisfies the “Quantitative Analysis” component of the “Skills & Perspectives” requirement. Students will gain the knowledge and ability to correctly use mathematical or quantitative analytical skills. This goal will be evaluated through work completed in class, on exams, and in the Excel homework assignments

Class Attendance: *It is presumed that students attend every session of courses in which they are registered in order that mastery of subject matter may be assisted by the instructor’s leadership, class discussion, and/or class projects, and so that they may contribute to the intellectual exchange that is appropriate to the class. The College’s mission of education within a community of faith and scholarship makes class attendance a particular priority. However, it is recognized that in some cases absence is unavoidable. Sickness excuses are received by the individual instructors. The Director of the Student Success Center will forward information to faculty about long-term absences or absences that may be considered unavoidable. Individual instructors have the right to limit absence in their courses and impose penalties for absenteeism, providing the professor’s policy and attached penalties are identified in the course syllabus. In no case, however, may a student be withdrawn from a course whose absences are less than or equal to two times the number of weekly meeting days assigned to the course. A grade of an “F” will be recorded if a withdrawal occurs after the official withdrawal period.*

See “Class Attendance” at <https://coursecatalog.benedictine.edu/academic-regulations-procedures/>

In this course, the instructor may withdraw any student who misses more than 6 class meetings (and a grade of “F” will be recorded if the withdrawal occurs after the official withdrawal period). **In any case if the instructor chooses not to withdraw a student who misses more than 6 class meetings, the student will receive a final grade no higher than D.**

Schedule: The Semester Schedule of Lecture Topics, Quizzes, Homework Problems, and Exams is printed on the last page of this Syllabus and on the last page of the *Textbook*. This schedule may change as the semester progresses and, if necessary, students will be given an updated schedule.

Weekly Topics: The sequence of weekly Topics is numbered in the *Textbook* and listed on the Semester Schedule. In order to be prepared for the lecture, students should have carefully read the entire Topic prior to any class that covers the material. For each topic, practice problems (similar to what is required on the Exams) will be completed together in class. These problems will be completed using Excel files that are posted on Canvas for student downloading.

Grading: The graded coursework will consist of the following:

43%	300 points - Quizzes
14%	100 points - Homework Assignments (in Excel)
43%	300 points - Exams
100%	700 points – Total

Quizzes: There are 30 timed Quizzes covering the 10 topics that will be taken on Canvas. Each quiz is worth 10 points, will be mostly “conceptual in nature” (i.e., multiple-choice, true/false, matching, etc.). The due dates of all quizzes are on the Semester Schedule. If a student fails to complete a quiz during the allotted time without an approved excuse, no make-up will be allowed.

Homework: There are 10 homework assignments covering the 10 topics. Each assignment is worth 10 points. The instructions and Excel template files are available on Canvas. Due dates and times are on the Semester Schedule. Each homework assignment is worth 10 points. Late assignments will be automatically penalized 3 points up to one week past the due date and at least 5 points thereafter. Copying the formulas/answers of another student or submitting the file of another student is an act of Academic Misconduct.

Exams: Exams will consist primarily of problems similar to those covered in class and on the homework assignments (see Semester Schedule for Exam due dates). There will be an in-class review of each exam prior to its date.

Personal Meeting: Due to the size of the class, each student will be required to personally meet with Professor King in his office at least once during the semester (and students may come in pairs if desired). Any meeting with a student to go over course content will be considered to have satisfied this requirement. Students who fail to satisfy this requirement will NOT benefit from curve points that are added to the final grade count at the end of the semester.

Homework Grading Rubric: The homework assignments will be graded as follows:

Completeness (i.e., answer all the questions, do all the problems):	2 points
Quality of Work (i.e., formatting, organization, order, etc.):	2 points
Use of Formulas (per file uploaded onto Canvas):	2 points
Accuracy of Answers:	4 points
Late submission up to one week following the due date	-3 points
Late submission later than one week following the due date	-5 points

Final Grade: Final grades (out of a maximum 700 points) will be determined as follows:

<u>Grades</u>	<u>Range</u>		<u>Points</u>	
	<u>Lower</u>	<u>Upper</u>	<u>Lower</u>	<u>Upper</u>
A	92.50%	100.00%	648	700
A-	90.00%	92.49%	630	647
B+	87.50%	89.99%	613	630
B	82.50%	87.49%	578	612
B-	80.00%	82.49%	560	577
C+	77.50%	79.99%	543	560
C	72.50%	77.49%	508	542
C-	70.00%	72.49%	490	507
D	60.00%	69.99%	420	490
F	0.00%	60.00%	0	420

Academic Misconduct: *Benedictine College endorses the principles of academic honesty. Any academic misconduct is contradictory to the purposes and welfare of both the student and the College. Academic misconduct includes, but is not limited to, cheating on examinations; plagiarizing; failing to properly document sources either intentionally or unintentionally; turning in work prepared by another source, falsifying data and documents; or gaining or giving unauthorized access to assessment materials. Using artificial intelligence (AI) for coursework in place of a student's own effort to learn constitutes academic misconduct. Therefore, students may use AI for their academics only when in accordance with instructor expectations and direction. Students should refer to the specified guidelines provided by their instructor.*

It is expected that the student will behave in a manner consistent with the values of Benedictine College in interactions in an academic context both inside and outside of the classroom. Should a student engage in inappropriate or aggressive language or behavior in relationship with faculty, or disruptive behavior in or out of the classroom, a faculty member may document the incident with the Vice Provost and the student may be subject to appropriate sanctions.

(See <https://coursecatalog.benedictine.edu/academic-regulations-procedures/> for a description of Consequences and Due Process with respect to Academic Misconduct.)

Reasonable Learning Accommodations: *Benedictine College complies with federal law requiring reasonable accommodations for individuals with documented disabilities that may affect their performance in a class or otherwise affect the individual's ability to enjoy the programs and services offered by the College. If a student requests an accommodation from a faculty member, the faculty member should direct the student to the Director of the Academic Research Center in the Student Success Center. (See pg. 54 of the 2022-23 College Catalog)*

Teaching Quality: Your professor pledges to

- Follow the course schedule.
- Start and end every class on time.
- Keep grades on Canvas updated for each student's progress.
- Be available to students outside of class times.
- Teach with enthusiasm, respect, and confidence in the students.

Classroom Decorum: In order to create and maintain a classroom environment that promotes the highest quality of teaching and learning, students are asked to adhere to the following rules:

- Dress appropriately that reflects a serious work environment
- Fill up all the front rows (nobody in the back rows)
- DO NOT to leave the classroom (and return) unless it is an emergency
- Put away cell phones and personal devices AT ALL TIMES
- DO NOT use your computer for anything other than viewing class material
- DO NOT bring meals into the classroom (snacks, bottled drinks and coffee are allowed)

FINC-3100: Principles of Finance - Semester Schedule - Fall 2024

As of August 28, 2024

Week	Class	Day	Date	Lecture Topics and Readings (Textbook pages)	Quiz # Due	HW # Due	Exam # Due
1	1	Wed	28-Aug	Class Introduction - Review Syllabus and Semester Schedule (pg. 1 of Text)			
	2	Fri	30-Aug	Topic 1: Financial Statements - Balance Sheet (pp. 2-11)			
2	3	Mon	2-Sep	Topic 1: Financial Statements - Balance Sheet (pp. 2-11)			
	4	Wed	4-Sep	Topic 1: Financial Statements - Income Statement (pp. 12-17)	1		
	5	Fri	6-Sep	Topic 1: Financial Statements - I/S (pp. 12-17) and SSE, SCF (pp. 18-23)			
3	6	Mon	9-Sep	Topic 1: Financial Statements - SSE and SCF (pp. 18-23)	2		
	7	Wed	11-Sep	Topic 1: Financial Statements - Practice Problem #1A			
	8	Fri	13-Sep	Topic 1: Financial Statements - Practice Problem #1B	3		
4	9	Mon	16-Sep	Topic 2: Financial Statement Analysis - Ratios and DuPont Analysis (pp. 24-29)	4	1	
	10	Wed	18-Sep	Topic 2: Financial Statement Analysis - Ratios and DuPont Analysis (pp. 24-29)	5		
	11	Fri	20-Sep	Topic 2: Financial Statement Analysis - Practice Problem 2A			
5	12	Mon	23-Sep	Topic 2: Financial Statement Analysis - Common-Size and Growth (pp. 30-33)	6		
	13	Wed	25-Sep	Topic 2: Financial Statement Analysis - Practice Problem 2B	7		
	14	Fri	27-Sep	Topic 2: Financial Statement Analysis - Practice Problem 2B			
6	15	Mon	30-Sep	Topic 3: Working Capital Management - Cash Conversion Cycle (pp. 34-35)	8	2	
	16	Wed	2-Oct	Topic 3: Cash Budget (pp. 38-39) and Practice Problem 3A	9		
	17	Fri	4-Oct	Topic 3: Cash Budget - Practice Problem 3B	10		
7	18	Mon	7-Oct	Topic 4: Short-Term Forecasting and Financial Planning (pp. 40-43)	11	3	
	19	Wed	9-Oct	Topic 4: Short-Term Forecasting - Practice Problem 4A	12		
	20	Fri	11-Oct	Topic 4: Short-Term Forecasting - Practice Problem 4B	13		
8	21	Mon	14-Oct	Topic 5: Time Value of Money (pp. 44-47) Review for Exam 1	14	4	
	22	Wed	16-Oct	Topic 5: Time Value of Money - Practice Problems	15		
	23	Fri	18-Oct	Topic 5: Time Value of Money - Practice Problems	16		
9	24	Mon	21-Oct	Topic 5: Time Value of Money - Practice Problems	17		1
	25	Wed	23-Oct	Topic 5: Time Value of Money - Practice Problems			
		Fri	25-Oct	Fall Break - No Classes			
10	26	Mon	28-Oct	Topic 6: Interest Rates (pp. 48-51)		5	
	27	Wed	30-Oct	Topic 6: Interest Rates, Risk and Return - Practice Problem 6A	18		
	28	Fri	1-Nov	Topic 6: Interest Rates, Risk and Return - Practice Problem 6B	19		
11	29	Mon	4-Nov	Topic 7: Bonds and Bond Pricing (pp. 52-59)	20	6	
	30	Wed	6-Nov	Topic 7: Bonds and Bond Pricing - Practice Problem 7A	21		
	31	Fri	8-Nov	Topic 7: Bonds and Bond Pricing - Practice Problem 7B	22		
12	32	Mon	11-Nov	Topic 8: Stock Valuation (pp. 60-64)	23	7	
	33	Wed	13-Nov	Topic 8: Stock Valuation - Practice Problem 8A	24		
	34	Fri	15-Nov	Topic 8: Stock Valuation - Practice Problem 8B	25		
13	35	Mon	18-Nov	Topic 9: Cost of Capital (pp. 65-67) Review for Exam 2		8	
	36	Wed	20-Nov	Topic 9: Cost of Capital - Practice Problems 9A			
	37	Fri	22-Nov	Topic 9: Cost of Capital - Practice Problems 9B	26		
14	38	Mon	25-Nov	Topic 10: Capital Budgeting (pp. 68-72)	27	9	2
		Wed	27-Nov	Thanksgiving Break - No Classes			
		Fri	29-Nov				
15	39	Mon	2-Dec	Topic 10: Capital Budgeting - Practice Problem (pp. 73-80)			
	40	Wed	4-Dec	Topic 10: Capital Budgeting - Practice Problem (pp. 73-80)	28		
	41	Fri	6-Dec	Topic 10: Capital Budgeting - Practice Problem (pp. 73-80)	29		
16	42	Mon	9-Dec	Review for Exam 3	30	10	
		Wed	11-Dec	Study Day - No Class			
		Mon	16-Dec	Final Exam Time Slot: 8:00 to 9:50 AM			3